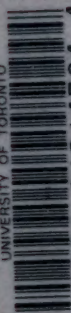


UNIVERSITY OF TORONTO



3 1761 01214734 4

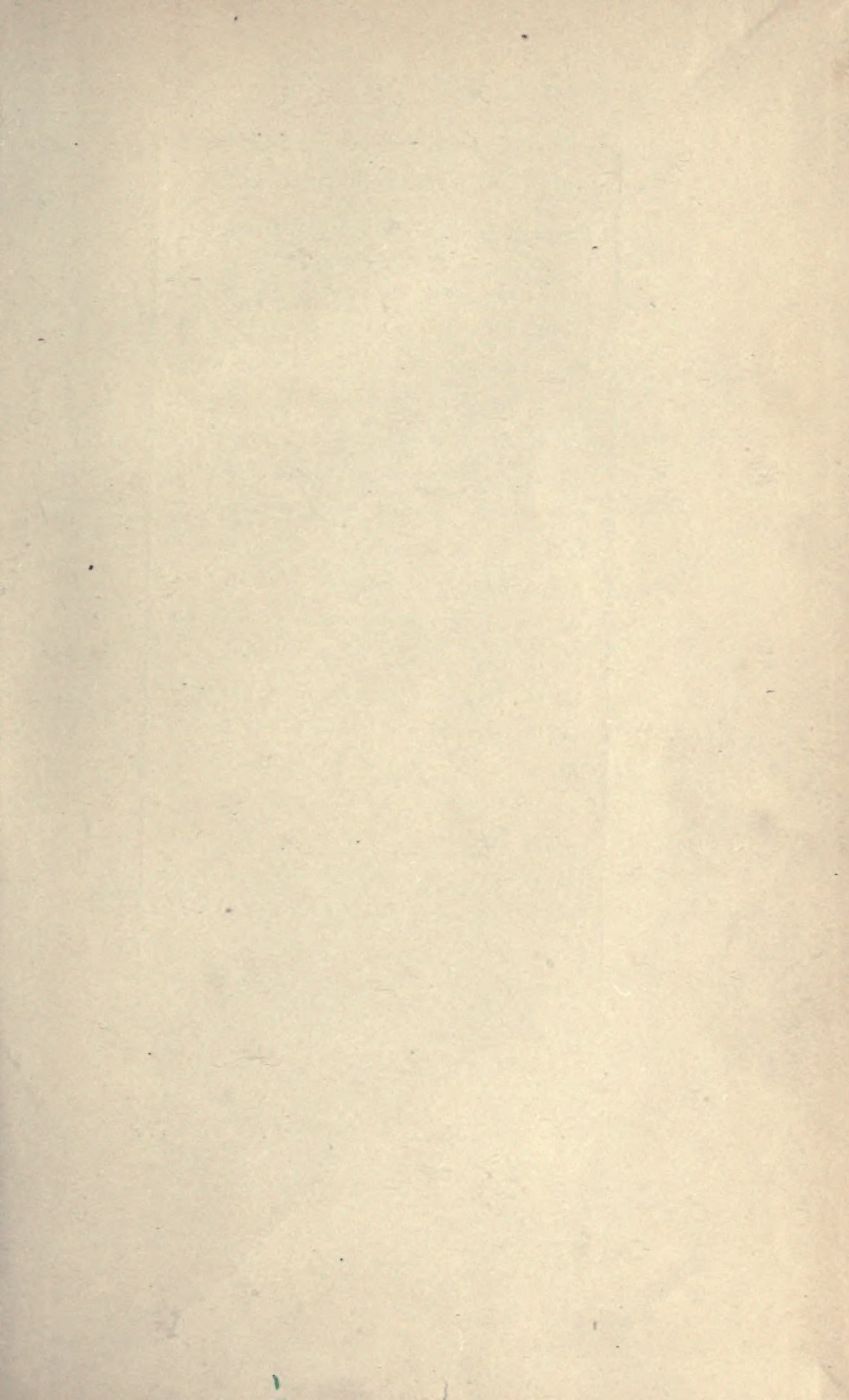




Digitized by the Internet Archive  
in 2007 with funding from  
Microsoft Corporation







## MATHEMATICAL MONOGRAPHS

EDITED BY

Mansfield Merriman and Robert S. Woodward.

Octavo, Cloth.

- No. 1. **History of Modern Mathematics.** By DAVID EUGENE SMITH. \$1.00 *net*.
- No. 2. **Synthetic Projective Geometry.** By GEORGE BRUCE HALSTED. \$1.00 *net*.
- No. 3. **Determinants.** By LAENAS GIFFORD WELD. \$1.00 *net*.
- No. 4. **Hyperbolic Functions.** By JAMES MCMAHON. \$1.00 *net*.
- No. 5. **Harmonic Functions.** By WILLIAM E. BYERLY. \$1.00 *net*.
- No. 6. **Grassmann's Space Analysis.** By EDWARD W. HYDE. \$1.00 *net*.
- No. 7. **Probability and Theory of Errors.** By ROBERT S. WOODWARD. \$1.00 *net*.
- No. 8. **Vector Analysis and Quaternions.** By ALEXANDER MACFARLANE. \$1.00 *net*.
- No. 9. **Differential Equations.** By WILLIAM WOOLSEY JOHNSON. \$1.00 *net*.
- No. 10. **The Solution of Equations.** By MANSFIELD MERRIMAN. \$1.00 *net*.
- No. 11. **Functions of a Complex Variable.** By THOMAS S. FISKE. \$1.00 *net*.
- No. 12. **The Theory of Relativity.** By ROBERT D. CARMICHAEL. \$1.00 *net*.
- No. 13. **The Theory of Numbers.** By ROBERT D. CARMICHAEL. \$1.00 *net*.
- No. 14. **Algebraic Invariants.** By LEONARD E. DICKSON. \$1.25 *net*.
- No. 15. **Mortality Laws and Statistics.** By ROBERT HENDERSON. \$1.25 *net*.
- No. 16. **Diophantine Analysis.** By ROBERT D. CARMICHAEL. \$1.25 *net*.
- No. 17. **Ten British Mathematicians.** By ALEXANDER MACFARLANE. \$1.25 *net*.
- No. 18. **Elliptic Integrals.** By HARRIS HANCOCK. \$1.25 *net*.

PUBLISHED BY

JOHN WILEY & SONS, Inc., NEW YORK.

CHAPMAN & HALL, Limited, LONDON.



MATHEMATICAL MONOGRAPHS

EDITED BY

MANSFIELD MERRIMAN AND ROBERT S. WOODWARD

No. 18

ELLIPTIC INTEGRALS

BY

HARRIS HANCOCK

PROFESSOR OF MATHEMATICS IN THE UNIVERSITY OF CINCINNATI

FIRST EDITION

FIRST THOUSAND

NEW YORK

JOHN WILEY & SONS, INC.

LONDON: CHAPMAN & HALL, LIMITED

1917

144045.  
23/10/17.



COPYRIGHT, 1917  
BY  
HARRIS HANCOCK

QA  
343  
H225

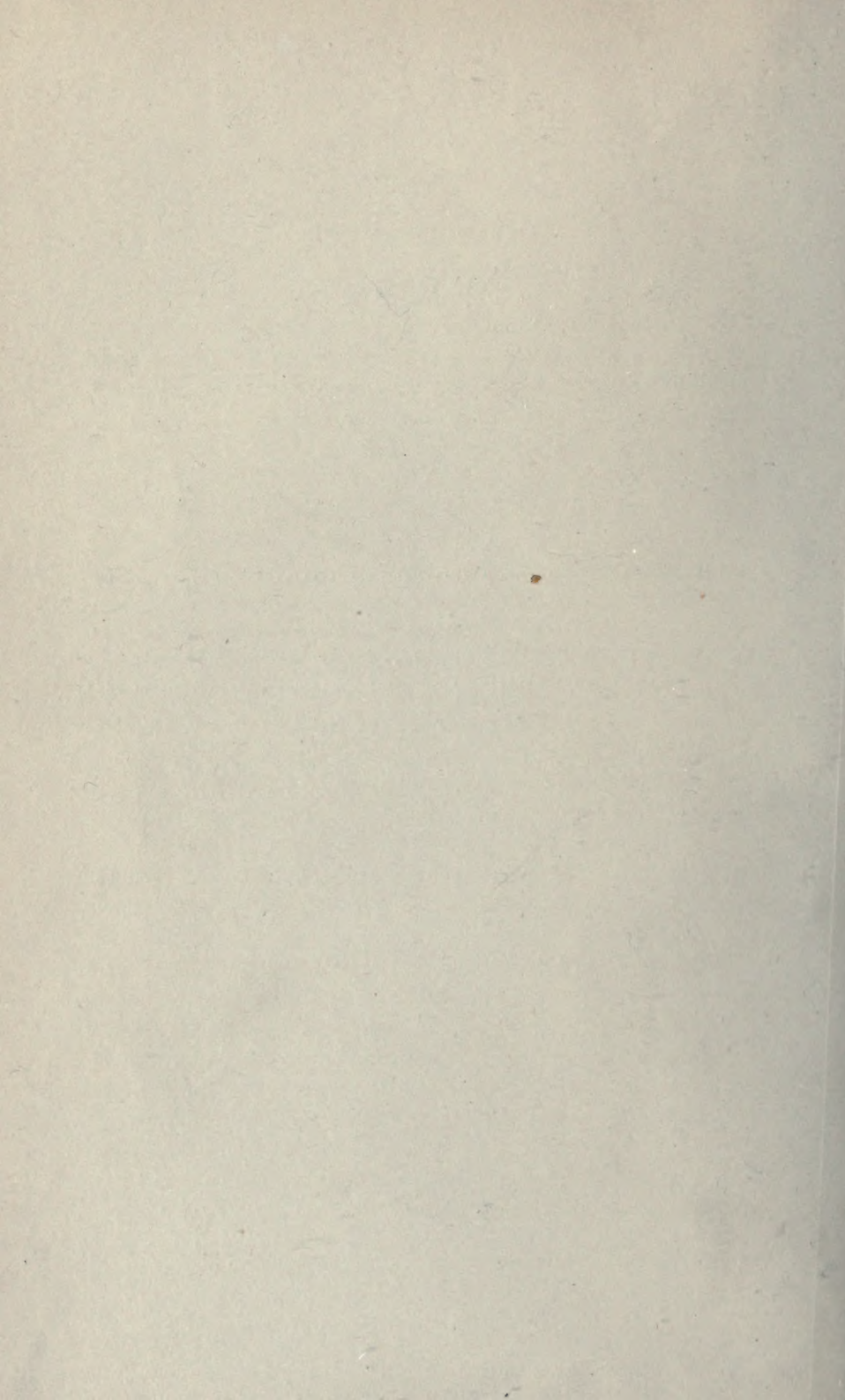
PRESS OF  
BRAUNWORTH & CO.  
BOOK MANUFACTURERS  
BROOKLYN, N. Y.



# CONTENTS

---

|                                                                                                                 | PAGE |
|-----------------------------------------------------------------------------------------------------------------|------|
| INTRODUCTION.....                                                                                               | 5    |
| CHAPTER I                                                                                                       |      |
| ELLIPTIC INTEGRALS OF THE FIRST, SECOND, AND THIRD KINDS. THE<br>LEGENDRE TRANSFORMATIONS.....                  | 9    |
| CHAPTER II                                                                                                      |      |
| THE ELLIPTIC FUNCTIONS.....                                                                                     | 24   |
| CHAPTER III                                                                                                     |      |
| ELLIPTIC INTEGRALS OF THE FIRST KIND REDUCED TO LEGENDRE'S NORMAL<br>FORM.....                                  | 41   |
| CHAPTER IV                                                                                                      |      |
| NUMERICAL COMPUTATION OF THE ELLIPTIC INTEGRALS OF THE FIRST AND<br>SECOND KINDS. LANDEN'S TRANSFORMATIONS..... | 65   |
| CHAPTER V                                                                                                       |      |
| MISCELLANEOUS EXAMPLES AND PROBLEMS.....                                                                        | 88   |
| CHAPTER VI                                                                                                      |      |
| FIVE-PLACE TABLES.....                                                                                          | 92   |
| INDEX.....                                                                                                      | 103  |



## INTRODUCTION

---

THE editors of the present series of mathematical monographs have requested me to write a work on elliptic integrals which "shall relate almost entirely to the three well-known elliptic integrals, with tables and examples showing practical applications, and which shall fill about one hundred octavo pages." In complying with their request, I shall limit the monograph to what is known as the Legendre-Jacobi theory; and to keep the work within the desired number of pages I must confine the discussion almost entirely to what is known as the elliptic integrals of the first and second kinds.

In the elementary calculus are found methods of integrating any rational expression involving under a square root sign a quadratic in one variable; in the present work, which may be regarded as a somewhat more advanced calculus, we have to integrate similar expressions where cubics and quartics in one variable occur under the root sign. Whatever be the nature of these cubics and quartics, it will be seen that the integrals may be transformed into standard normal forms. Tables are given of these normal forms, so that the integral in question may be calculated to any degree of exactness required.

With the trigonometric sine function is associated its inverse function, an integral; and similarly with the normal forms of elliptic integrals there are associated elliptic functions. A short account is given of these functions which emphasizes their doubly periodic properties. By making suitable transformations and using the inverse of these functions, it is found that the integrals in question may be expressed more concisely through the normal forms and in a manner that indicates the transformation employed.



The underlying theory, the philosophy of the subject, I have attempted to give in my larger work on elliptic functions, Vol. I. In the preparation of the present monograph much use has been made of Greenhill's *Application of Elliptic Functions*, a work which cannot be commended too highly; one may also read with great advantage Cayley's *Elliptic Functions*. The standard works of Legendre, Abel and Jacobi are briefly considered in the text. It may also be of interest to note briefly the earlier mathematicians who made possible the writings just mentioned.

The difference of two arcs of an ellipse that do not overlap may be expressed through the difference of two lengths on a straight line; in other words, this difference may be expressed in an *algebraic* manner. This is the geometrical signification of a theorem due to an Italian mathematician, Fagnano, which theorem is published in the twenty-sixth volume of the *Giornale de' letterari d'Italia*, 1716, and later with numerous other mathematical papers in two volumes under the title *Produzioni matematiche del Marchese Giulio Carlo de' Toschi di Fagnano*, 1750.

The great French mathematician Hermite (*Cours*, rédigé par Andoyer, Paris, 1882) writes "*Ce résultat doit être cité avec admiration comme ayant ouvert le premier la voie à la théorie des fonctions elliptiques.*"

Maclaurin in his celebrated work *A Treatise on Fluxions*, Edinburgh, 1742, Vol. II, p. 745, shows "how the *elastic curve* may be constructed in all cases by the rectification of the conic sections." On p. 744 he gives Jacob Bernoulli "as the celebrated author who first resolved this as well as several other curious problems" (see *Acta Eruditorium*, 1694, p. 274). It is thus seen that the elliptic integrals made their appearance in the formative period of the integral calculus.

The results that are given in Maclaurin's work were simplified and extended by d'Alembert in his treatise *Recherches sur le calcul intégral. Histoire de l'Ac. de Berlin, Année 1746*, pp. 182-224. The second part of this work, *Des différentielles qui se rapportent à la rectification de l'ellipse ou de l'hyperbole*,

treats of a number of differentials whose integrals through simple substitutions reduce to the integrals through which the arc of an ellipse or hyperbola may be expressed.

It was also known through the works of Fagnano, Jacob Bernoulli and others that the expressions for  $\sin(\alpha + \beta)$ ,  $\sin(\alpha - \beta)$  etc., gave a means of adding or subtracting the arcs of circles, and that between the limits of two integrals that express lengths of arc of a lemniscate an algebraic relation exists, such that the arc of a lemniscate, although a transcendent of higher order, may be doubled or halved just as the arc of a circle by means of geometric construction.

It was natural to inquire if the ellipse, hyperbola, etc., did not have similar properties. Investigating such properties, Euler made the remarkable discovery of the addition-theorem of elliptic integrals (see *Nov. Comm. Petrop.*, VI, pp. 58-84, 1761; and VII, p. 3; VIII, p. 83). A direct proof of this theorem was later given by Lagrange and in a manner which elicited the great admiration of Euler (see Serret's *Œuvres de Lagrange*, T. II, p. 533).

The addition-theorem for elliptic integrals gave to the elliptic functions a meaning in higher analysis similar to that which the cyclometric and logarithmic functions had enjoyed for a long time.

I regret that space does not permit the derivation of these addition-theorems and that the reader must be referred to a larger work.

The above mathematicians are the ones to whom Legendre refers in the introduction of his *Traité des fonctions elliptiques*, published in three quarto volumes, Paris, 1825. This work must always be regarded as the foundation of the theory of elliptic integrals and their associated functions; and Legendre must be regarded as the founder of this theory, for upon his investigations were established the doubly periodic properties of these functions by Abel and Jacobi and indeed in the very form given by Legendre. Short accounts of these theories are found in the sequel.

For more extended works the reader is referred to Appell

et Lacour, *Fonctions elliptiques*, and to Enneper, *Elliptische Funktionen*, where in particular the historical notes and list of authors cited on pp. 500-598 are valuable. Fricke in the article "*Elliptische Funktionen*," *Encyclopädie der mathematischen Wissenschaften*, Vol. II, gives a fairly complete list of books and monographs that have been written on this subject.

To Dr. Mansfield Merriman I am indebted for suggesting many of the problems of Chapter V and also for valuable assistance in editing this work. I have pleasure also in thanking my colleague, Dr. Edward S. Smith, for drawing the figures carefully to scale.

HARRIS HANCOCK.

2365 Auburn Ave.,  
CINCINNATI, OHIO,  
October 3, 1916.



# ELLIPTIC INTEGRALS

## CHAPTER I

### ELLIPTIC INTEGRALS OF THE FIRST, SECOND AND THIRD KINDS. THE LEGENDRE TRANSFORMATION

**Art. 1.** In the elementary calculus are studied such integrals as  $\int \frac{dx}{s}$ ,  $\int \frac{x dx}{(ax+b)s}$ , etc., where  $s^2 = ax^2 + 2bx + c$ . In general the integral of any rational function of  $x$  and  $s$  can be transformed into other typical integrals, which are readily integrable. Such types of integrals are

$$\int^x \frac{dx}{\sqrt{1-x^2}}, \quad \int_0^1 \frac{dx}{\sqrt{1-x^2}}, \quad \int_0^x \frac{dx}{\sqrt{x^2+1}}, \quad \text{etc.}$$

In the present theory instead of, as above, writing  $s^2$  equal to a quadratic in  $x$ , we shall put  $s^2$  equal to a cubic or quartic in  $x$ . Suppose further that  $F(x, s)$  is any rational function of  $x$  and  $s$  and consider the integral  $\int F(x, s) dx$ . Such an integral may be made to depend upon three types of integral of the form

$$\int \frac{dx}{s}, \quad \int \frac{x^2 dx}{s} \quad \text{and} \quad \int \frac{dx}{(x-b)s}.$$

These three types of integral, in somewhat different notation, were designated by Legendre, the founder of this theory, as elliptic integrals of the *first*, *second*, and *third kinds* respectively, while the general term "elliptic integral" was given by him to any integral of the form  $\int F(x, s) dx$ . The method of expressing the general integral through the three types of inte-

gral as first indicated by Legendre, may be found in my *Elliptic Functions*, Vol. I, p. 180.

**Art. 2.** First consider integrals of the form

$$\int \frac{c'x}{\sqrt{R(x)}}, \quad . . . . . (1)$$

which, as will be shown, reduce to a definite typical normal form,\* when  $R(x)$  is either of the third or fourth degree in  $x$ .

Suppose that  $R(x)$  is of the fourth degree, and write

$$R(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4,$$

where  $a_0, a_1, . . .$ , are real constants. It is seen that (1) may be written

$$\frac{1}{\sqrt{a_0}} \int \frac{dx}{\sqrt{X}}, \quad . . . . . (2)$$

where  $X$ , when decomposed into its factors, is

$$X = \pm(x-\alpha)(x-\beta)(x-\gamma)(x-\delta),$$

and  $\sqrt{a_0}$  is a real quantity. If the roots are all real, suppose that  $\alpha > \beta > \gamma > \delta$ ; if two are complex, take  $\alpha$  and  $\beta$  real and write  $\gamma = \rho + i\sigma$ ,  $\delta = \rho - i\sigma$ , where  $i = \sqrt{-1}$ ; and if all four of the roots are complex, denote them by  $\alpha = \mu + i\nu$ ,  $\beta = \mu - i\nu$ ,  $\gamma = \rho + i\sigma$ ,  $\delta = \rho - i\sigma$ .

In the present work the variable is taken real unless it is stated to the contrary or is otherwise evident.

We shall first so transform the expression  $X$  that only *even* powers of the variable appear. With Legendre (loc. cit., p. 7), write

$$x = \frac{p+qy}{1+y}. \quad . . . . . (3)$$

It follows at once that

$$\frac{dx}{\sqrt{X}} = \frac{(q-p)dy}{\sqrt{\pm Y}}, \quad . . . . . (4)$$

\* See Legendre, *Traité des fonctions elliptiques*, T. I., p. 11, et seq.; Richelot, *Crelle*, Bd. 34, p. 1; Enneper, *Elliptische Functionen*, p. 14.

where

$$Y = [p - \alpha + (q - \alpha)y][p - \beta + (q - \beta)y][p - \gamma + (q - \gamma)y][p - \delta + (q - \delta)y]. \quad (5)$$

As all the results must be real, it will be seen that real values may be given to  $p$  and  $q$  in such a way that only even powers of  $y$  appear on the right-hand side of (5). If in this expression we multiply the first and second factors together, we have

$$(p - \alpha)(p - \beta) + (q - \alpha)(q - \beta)y^2$$

provided

$$(p - \alpha)(q - \beta) + (p - \beta)(q - \alpha) = 0; \quad . \quad . \quad . \quad (6)$$

and similarly if

$$(p - \gamma)(q - \delta) + (p - \delta)(q - \gamma) = 0, \quad . \quad . \quad . \quad (7)$$

the product of the third and fourth factors of (5) is

$$(p - \gamma)(p - \delta) + (q - \gamma)(q - \delta)y^2.$$

From (6) and (7) it follows that

$$pq + \alpha\beta = \frac{p+q}{2}(\alpha + \beta),$$

and

$$pq + \gamma\delta = \frac{p+q}{2}(\gamma + \delta).$$

From the last two equations, it also follows that

$$\frac{p+q}{2} = \frac{\alpha\beta - \gamma\delta}{\alpha + \beta - \gamma - \delta}, \quad pq = \frac{\alpha\beta(\gamma + \delta) - \gamma\delta(\alpha + \beta)}{\alpha + \beta - \gamma - \delta}. \quad (8)$$

From (8) it is seen that the sum and quotient of  $p$  and  $q$  are real quantities whatever the nature of the four roots  $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\delta$  may be; and further from (8) it is seen that

$$\left(\frac{q-p}{2}\right)^2 = \frac{(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)}{(\alpha + \beta - \gamma - \delta)^2}, \quad . \quad . \quad . \quad (9)$$

which is always a positive quantity. It follows that  $q - p$  is a real quantity, and that  $p$  and  $q$  are real.

The equations (8) and (9) cannot be used if  $\alpha + \beta = \gamma + \delta$ .



In this case, as is readily shown, instead of the substitution (3), we may write

$$x = y + \frac{\alpha + \beta}{2} = y + \frac{\gamma + \delta}{2}.$$

It follows that (5) takes the form

$$Y = (\pm m^2 \pm n^2 y^2)(\pm r^2 \pm l^2 y^2),$$

where  $m$ ,  $n$ ,  $r$ , and  $l$  are real quantities.

The expression (4) then becomes

$$\frac{dx}{\sqrt{X}} = \frac{(q-p)dy}{\sqrt{\pm Y}} = \frac{dy}{f\sqrt{\pm(1 \pm g^2 y^2)(1 \pm h^2 y^2)}}, \quad \dots \quad (10)$$

where  $f$ ,  $g$ , and  $h$  are essentially real quantities.

In the expression on the right-hand side, suppose that  $h > g$  and put  $hy = t$ , and  $\frac{g}{h} = c$ , where  $c < 1$ .

It follows that

$$\frac{dx}{\sqrt{X}} = \frac{dt}{fh\sqrt{\pm(1 \pm t^2)(1 \pm c^2 t^2)}}. \quad \dots \quad (11)$$

It is seen that under the radical there are eight combinations of sign. With Legendre, loc. cit., Chap. II, and *Enneper*, p. 17, a table will be given below from which it is seen that the corresponding functions may be expressed by means of trigonometric substitutions in the one normal form

$$\frac{dx}{\sqrt{X}} = \pm \frac{1}{M} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \pm \frac{1}{M} \frac{dv}{\sqrt{(1 - v^2)(1 - k^2 v^2)}}, \quad \dots \quad (12)$$

where  $M$  is a real quantity and  $v = \sin \phi$ .

The quantity  $k$ , called the *modulus*, is also real, and situated within the interval  $0 \leq k \leq 1$ .

Of the expressions under the root sign  $\sqrt{-(1 + t^2)(1 + c^2 t^2)}$  may be neglected, since  $R(x)$ , assumed to be positive for at least some real value of the original  $x$ , cannot be transformed into a function that is always negative by a real substitution.

**Art. 3.** Writing  $\Delta\phi = \sqrt{1 - k^2 \sin^2 \phi}$  and defining the *com-*

plementary modulus  $k'$  by the relation  $k^2 + k'^2 = 1$ , the following table results:

|      |                                                                      |                                                |                           |
|------|----------------------------------------------------------------------|------------------------------------------------|---------------------------|
| I.   | $\frac{dt}{\sqrt{(1+t^2)(1+c^2t^2)}} = \frac{d\phi}{\Delta\phi},$    | $t = \tan \phi,$                               | $k^2 = 1 - c^2$           |
| II.  | $\frac{dt}{\sqrt{(1-t^2)(1+c^2t^2)}} = \frac{-k'd\phi}{\Delta\phi},$ | $t = \cos \phi,$                               | $k^2 = \frac{c^2}{1+c^2}$ |
| III. | $\frac{dt}{\sqrt{(t^2-1)(1+c^2t^2)}} = \frac{k d\phi}{\Delta\phi},$  | $t = \sec \phi,$                               | $k^2 = \frac{1}{1+c^2}$   |
| IV.  | $\frac{dt}{\sqrt{(1+t^2)(1-c^2t^2)}} = \frac{-k d\phi}{\Delta\phi},$ | $t = \frac{\cos \phi}{c},$                     | $k^2 = \frac{1}{1+c^2}$   |
| V.   | $\frac{dt}{\sqrt{(1+t^2)(c^2t^2-1)}} = \frac{k'd\phi}{\Delta\phi},$  | $t = \frac{\sec \phi}{c},$                     | $k^2 = \frac{c^2}{1+c^2}$ |
| VI.  | $\frac{dt}{\sqrt{(1-t^2)(1-c^2t^2)}} = \frac{d\phi}{\Delta\phi},$    | $t = \sin \phi,$                               | $k^2 = c^2$               |
| VIa. | $\frac{dt}{\sqrt{(t^2-1)(c^2t^2-1)}} = -\frac{d\phi}{\Delta\phi},$   | $t = \frac{1}{c \sin \phi},$                   | $k^2 = c^2$               |
| VII. | $\frac{dt}{\sqrt{(t^2-1)(1-c^2t^2)}} = -\frac{d\phi}{\Delta\phi},$   | $t^2 = \sin^2 \phi + \frac{\cos^2 \phi}{c^2},$ | $k^2 = 1 - c^2$           |

The formulas VI and VIa have the same form; in VI it is necessary that  $t \leq 1$ , while in VIa it is required that  $t \geq \frac{1}{c}$ .

Art. 4. It is seen that the eight transformations in the table are all of the form

$$t^2 = \frac{A + B \sin^2 \phi}{C + D \sin^2 \phi}, \quad \dots \dots \dots (i)$$

where  $A$ ,  $B$ ,  $C$ , and  $D$  are real constants; at the same time it is seen that by means of real substitutions the following reduction can always be made:

$$\frac{dx}{\sqrt{R(x)}} = \pm \frac{1}{M} \frac{d\phi}{\Delta\phi} = \pm \frac{1}{M} \frac{dv}{\sqrt{(1-v^2)(1-k^2v^2)}},$$

where  $v = \sin \phi$ .

These substitutions and reductions are given in full in Chap. III.

The radical in  $\frac{dv}{\sqrt{(1-v^2)(1-k^2v^2)}}$  is *real* for real values of  $v$  that are  $1^\circ$  less than unity and  $2^\circ$  greater than  $\frac{1}{k}$ . In the latter case, write  $v = \frac{1}{ks}$ , and then

$$\frac{dv}{\sqrt{(1-v^2)(1-k^2v^2)}} = -\frac{ds}{\sqrt{(1-s^2)(1-k^2s^2)}}.$$

In this substitution as  $v$  passes from  $\frac{1}{k}$  to  $\infty$ , the variable  $s$  passes from 1 to 0.

It is therefore concluded that by making the real substitution (i), the differential expression \*

$$\frac{dt}{\sqrt{\pm(1 \pm g^2t^2)(1 \pm h^2t^2)}}$$

may be reduced to the form

$$\pm \frac{1}{M} \frac{dv}{\sqrt{(1-v^2)(1-k^2v^2)}},$$

where the variable  $v$  lies within the interval 0 . . . 1. Such transformations fail if the expression under the root contains only even powers of  $t$ , the two roots in  $t^2$  being imaginary, i.e., if  $R(x) = Ax^4 + 2Bx^2 + C$ , where  $B^2 - AC < 0$ . This case is considered in Art. 34.

Art. 5. It is also seen that the general elliptic integral

$$\int \frac{Q(t)}{\sqrt{R(t)}} dt,$$

\* For other transformations and tables, see Tannery et Molk, *Fonctions Elliptiques*, Vol. IV, p. 34; Cayley, *Elliptic Functions*, pp. 315-16; Appell et Lacour, *Fonctions Elliptiques*, pp. 240-243.



where  $Q(t)$  is any rational function of  $t$ , and  $R(t)$  is of the fourth degree in  $t$ , may by the real substitutions

$$t = \frac{p+q\tau}{1+\tau}, \quad \tau = \frac{a+bx^2}{c+dx^2},$$

be transformed into

$$\int \frac{f(x)dx}{\sqrt{(1-x^2)(1-k^2x^2)}},$$

where  $f(x)$  is a rational function of  $x$ . The evaluation of this latter integral, see my *Elliptic Functions*, I, p. 186, may be made to depend upon that of three types of integral, viz.:

$$F(k, x) = \int \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}},$$

$$E(k, x) = \int \frac{\sqrt{1-k^2x^2}}{\sqrt{1-x^2}} dx,$$

$$\Pi(n, k, x) = \int \frac{dx}{(1+nx^2)\sqrt{(1-x^2)(1-k^2x^2)}}.$$

Writing  $x = \sin \phi$ , and putting  $\sqrt{1-k^2 \sin^2 \phi} = \Delta(k, \phi)$ , there results the *Legendre notation* as *normal integrals* of the *first kind*

$$F(k, \phi) = \int_0^\phi \frac{d\phi}{\Delta(k, \phi)},$$

of the *second kind*,

$$E(k, \phi) = \int_0^\phi \Delta(k, \phi) d\phi,$$

and of the *third kind*,

$$\Pi(n, k, \phi) = \int_0^\phi \frac{d\phi}{(1+n \sin^2 \phi) \Delta(k, \phi)}.$$

The modulus  $k$  is omitted from the notation when no particular emphasis is put upon it.

The evaluation of these integrals is reserved for Chap. IV. However, the nature of the first two integrals may be studied by observing the graphs in the next article.

**Art. 6.** *Graphs of the integrals  $F(k, \phi)$  and  $E(k, \phi)$ .* In Fig. 1 there are traced the curves  $y = \frac{1}{\Delta(k, \phi)}$  and  $y = \Delta(k, \phi)$ . Let values of  $\phi$  be laid off upon the  $X$ -axis. It is seen that the areas of these curves included between the  $x$ -axis and the ordinates corresponding to the abscissa  $\phi$  will represent the integrals  $F(k, \phi)$  and  $E(k, \phi)$ . See Cayley, *Elliptic Functions*, p. 41.

If  $k=0$ , then  $\Delta\phi=1$ , and the curves  $y=\Delta\phi$ ,  $y=\frac{1}{\Delta\phi}$  each become the straight line  $y=1$ ; while the corresponding integrals

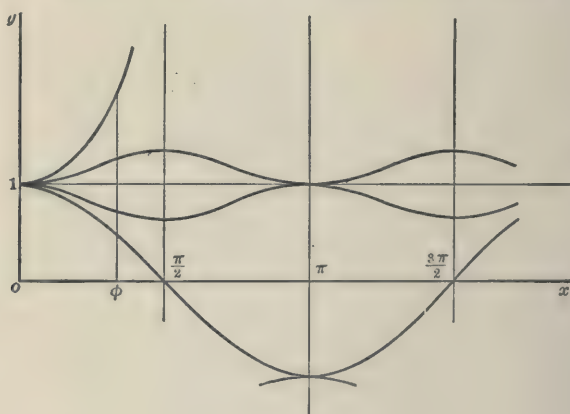


FIG. 1.

$F(\phi)$ ,  $E(\phi)$  are both equal to  $\phi$  and are represented by rectangles upon the sides  $\phi$  and 1. When  $0 < k < 1$ , the curve  $y = \frac{1}{\Delta\phi}$  lies entirely above the line  $y=1$ , while  $y = \Delta\phi$  lies below it. As  $\phi$  increases from zero, the integrals  $F(\phi)$  and  $E(\phi)$  increase from zero in a continuous manner, the integral  $F(\phi)$  being always the larger. Further, for a given value of  $\phi$ , as  $k$  increases the integral  $F(\phi)$  increases and  $E(\phi)$  diminishes; and conversely as  $k$  decreases,  $F(\phi)$  diminishes and  $E(\phi)$  increases.

If  $F\left(k, \frac{\pi}{2}\right)$  be denoted by  $F_1(k)$ , or  $F_1$ , and if we put

$E_1 = E\left(k, \frac{\pi}{2}\right)$ , it is seen that when  $k=0$ ,  $F\left(0, \frac{\pi}{2}\right) = F_1(0) = \frac{\pi}{2} = E_1(0)$ . When  $k$  has a fixed value, it is often omitted in the notation.  $F_1$  and  $E_1$  are called *complete* integrals.

It is evident that both curves are symmetric about the line  $\phi = \frac{1}{2}\pi$ , and that for a fixed value of  $k$ , it is sufficient to

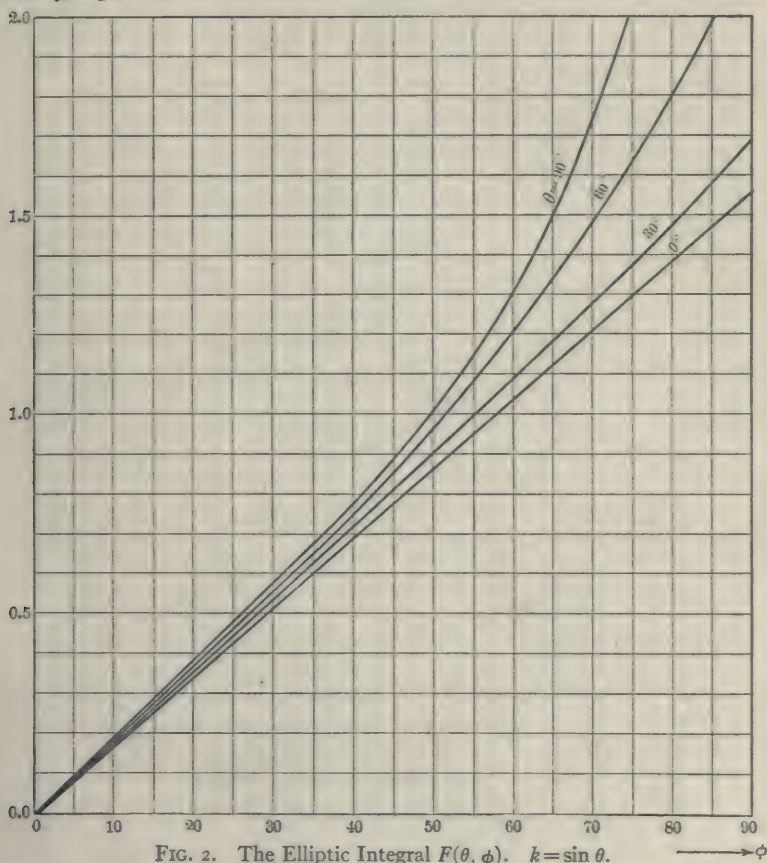


FIG. 2. The Elliptic Integral  $F(\theta, \phi)$ .  $k = \sin \theta$ .

know the values of  $\phi$  from 0 to  $\frac{1}{2}\pi$ . For  $F(\pi) = 2F_1$ , and for any value  $\phi = \alpha$ ,  $F(\alpha) = F(\pi) - F(\pi - \alpha)$ , or  $F(\pi - \alpha) = 2F_1 - F(\alpha)$ .

In the latter formula, as  $\alpha$  diminishes from  $\frac{\pi}{2}$  to 0,  $F(\phi)$  increases

from  $\frac{\pi}{2}$  to  $\pi$ .



Further noting that  $F(-\alpha) = -F(\alpha)$ , the formula

$$\begin{aligned} F(\alpha) &= F(\pi) + F(\alpha - \pi), \\ &= 2F_1 + F(\alpha - \pi), \end{aligned}$$

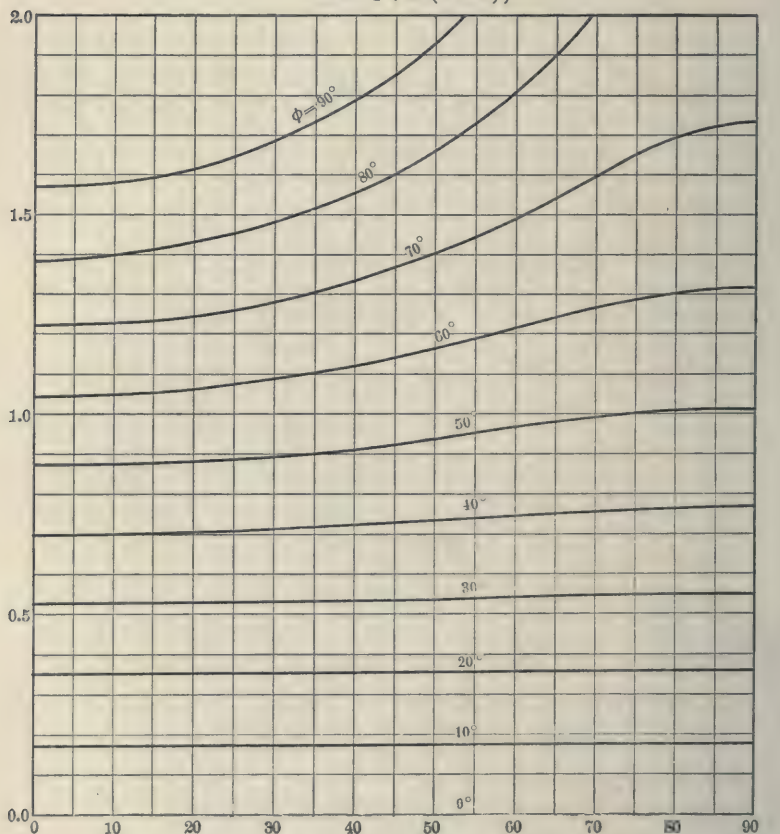


FIG. 3. The Elliptic Integral  $F(\theta, \phi)$ .  $k = \sin \theta$ .

—————→ $\theta$

gives the values of  $F(\phi)$  for values  $\phi = \pi$  to  $\phi = 2\pi$ , etc. In general,

$$F(m\pi \pm \alpha) = 2mF_1 \pm F(\alpha),$$

$$E(m\pi \pm \alpha) = 2mE_1 \pm E(\alpha).$$

**Art. 7.** When  $k = 1$ , the graphs of the two curves in Fig. 1 are entirely changed, the curve  $y = \Delta\phi$  becoming  $y = \cos \phi$ , which as before lies wholly below the line  $y = 1$ . The curve  $y = \frac{1}{\Delta\phi}$

becomes  $y = \sec \phi$ . The ordinate for this latter curve becomes infinite for  $\phi = \frac{1}{2}\pi$ , and between the values  $\frac{1}{2}\pi$  and  $\frac{3}{2}\pi$  there is a branch lying wholly below the line  $y = -1$ , the ordinates for the values  $\phi = \frac{1}{2}\pi$  and  $\phi = \frac{3}{2}\pi$  being  $= -\infty$ .

For the values  $\frac{3}{2}\pi$  and  $\frac{5}{2}\pi$  there is a branch lying wholly

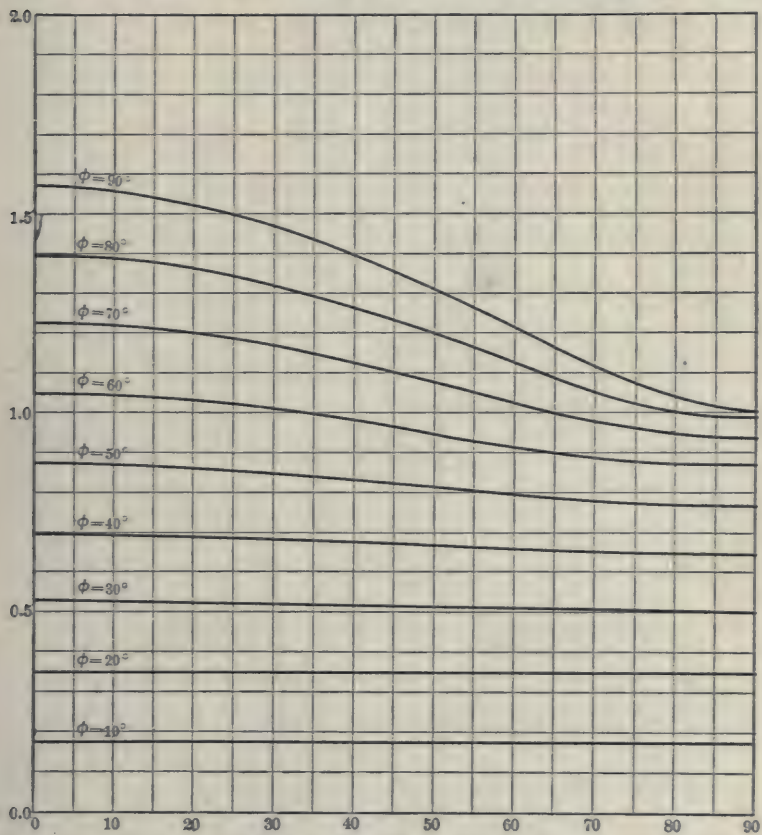


FIG. 4. The Elliptic Integral  $E(\theta, \phi)$ .  $k = \sin \theta$ .

above the line  $y = +1$ , the ordinates for  $\frac{3}{2}\pi$  and  $\frac{5}{2}\pi$  being  $+\infty$  and so on.

Corresponding to the first curve,  $E(\phi) = \int_0^\phi \cos \phi \, d\phi = \sin \phi$  and consequently  $E_1 = 1$ . This, taken in connection with what was given above, shows that as  $k$  increases from 0 to 1,  $E_1$  decreases from  $\frac{1}{2}\pi$  to 1.

For the second curve,  $F(\phi) = \int_0^\phi \sec \phi \, d\phi = \log \tan \left( \frac{\pi}{4} + \frac{\phi}{2} \right)$ , so that  $F_1$  is logarithmically infinite when  $k=1$ ; and this taken in connection with what was given above, shows that

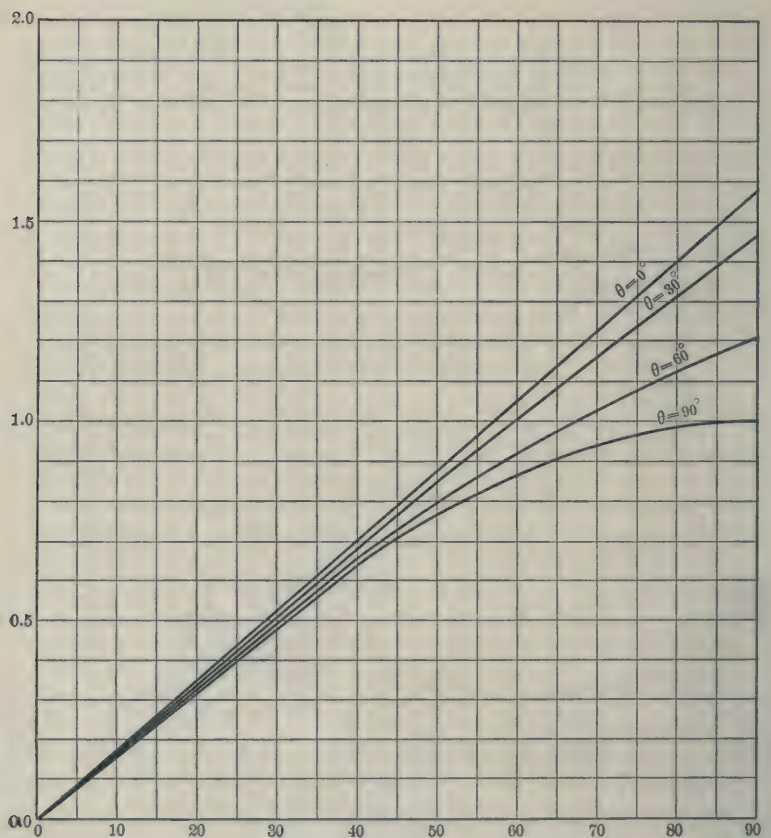


FIG. 5. The Elliptic Integral  $E(\theta, \phi)$ .  $k = \sin \theta$ .

as  $k$  increases from 0 to 1,  $F_1$  increases from  $\frac{1}{2}\pi$  to logarithmic infinity.

**Art. 8.** In Figs. 2-5 are added other graphs of the integrals  $F(k, \phi)$  and  $E(k, \phi)$  which require no further explanation. At the end of the book are found tables which give the values of these integrals for fixed values of  $k$  and  $\phi$ .

## EXAMPLES

1. A quartic function with real coefficients is always equal to the product of two factors  $M=l+2mx+nx^2$ ,  $N=\lambda+2\mu x+\nu x^2$ , where all the coefficients are real. Remove the coefficient of  $x$  in  $M$  and  $N$  in the integral

$$\int \frac{dx}{\sqrt{MN}},$$

and thereby reduce this integral to

$$\int \frac{(q-p)dy}{\sqrt{(ay^2+b)(a'y^2+b')}},$$

by a substitution  $x=\frac{p+qy}{1+y}$ , and show that  $p$  and  $q$  are real. *Legendre*, Vol. I., Chap. II.

2. Show that

$$\int \frac{f(x)dx}{\sqrt{S(x)}}$$

may be reduced to the integral

$$\int \frac{g(z)dz}{\sqrt{4z^3-g_2z-g_3}},$$

where  $f$  and  $g$  are rational functions of their arguments and

$$S(x)=ax^3+3bx^2+3cx+d.$$

The substitution required is  $x=mz+n$ , where  $n=-\frac{b}{a}$ ,  $am^3=4$ .

*Appell et Lacour*, p. 247.

3. Knowing a real root  $\alpha$  of  $R(x)$ , find the form of  $\frac{dx}{\sqrt{R(x)}}$ , when  $x=\alpha+\frac{1}{y}$ .

Write

$$R(x)=(x-\alpha)(cx^3+c_1x^2+c_2x+c_3).$$

*Levy*, p. 77.

4. Show that the substitution

$$\sqrt{cx}=\frac{(1+\sin \phi)+\sqrt{c}(1-\sin \phi)}{(1-\sin \phi)+\sqrt{c}(1+\sin \phi)}$$

transforms

$$\frac{dx}{\sqrt{(x^2-1)(1-c^2x^2)}} \text{ into } \frac{(1+\sqrt{k})^2 d\phi}{2\sqrt{1-k^2 \sin^2 \phi}},$$



where

$$k = \left( \frac{1 - \sqrt{c}}{1 + \sqrt{c}} \right)^2.$$

5. Show that by the substitution  $x = \frac{1-y}{1+y} \sqrt{\frac{\lambda}{\mu}}$ , the integral in which  $R(x)$  has the form  $\lambda^2 + 2\lambda\mu \cos\theta x^2 + \mu^2 x^4$ , is transformed into one which has under the radical an expression of the form  $m^2(1+g^2y^2)(1+h^2y^2)$ .

*Legendre, Vol. I, Chap. XI.*

6. If the four roots of  $X$  are all real, such that  $\alpha > \beta > \gamma > \delta$ , show that the substitution

$$x = \frac{\gamma(\beta - \delta) - \delta(\beta - \gamma) \sin^2 \phi}{(\beta - \delta) - (\beta - \gamma) \sin^2 \phi}$$

transforms

$$\frac{dx}{\sqrt{X}} \text{ into } \frac{2}{\sqrt{(\alpha - \gamma)(\beta - \delta)}} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}},$$

where

$$k^2 = \frac{\beta - \gamma}{\alpha - \gamma} \frac{\alpha - \delta}{\beta - \delta} \quad \text{and} \quad \gamma < x < \beta.$$

7. If  $Y$  is of the third degree and if its roots  $\alpha, \beta, \gamma$  are all real, such that  $\alpha > \beta > \gamma$ , show that the substitution  $y = \gamma + (\beta - \gamma) \sin^2 \phi$  transforms

$$\frac{dy}{\sqrt{Y}} \text{ into } \frac{2}{\sqrt{\alpha - \gamma}} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}},$$

where

$$k^2 = \frac{\beta - \gamma}{\alpha - \gamma} \quad \text{and} \quad \gamma < y < \beta.$$

8. If  $X$  is of the fourth degree with roots  $\alpha, \beta$ , real and  $\gamma, \delta = \rho \pm i\sigma$ , and if  $M^2 = (\rho - \alpha)^2 + \sigma^2$ ,  $N^2 = (\rho - \beta)^2 + \sigma^2$ , show that the substitution

$$\frac{x - \alpha}{x - \beta} = \frac{M}{N} \frac{1 - \cos \phi}{1 + \cos \phi}$$

transforms

$$\frac{dx}{\sqrt{(x - \alpha)(x - \beta)[(x - \rho)^2 + \sigma^2]}} \text{ into } \frac{1}{\sqrt{MN}} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}},$$

where

$$k^2 = \frac{1}{2} \frac{(M + N)^2 - (\alpha - \beta)^2}{2MN}$$

and

$$\infty > x > \alpha \quad \text{or} \quad \beta > x > -\infty.$$

9. Show that the substitution

$$t = e_1 + \frac{(e_2 - e_1)(e_3 - e_1)}{s - e_1}$$

transforms the integral

$$\int \frac{dt}{\sqrt{(t - e_1)(t - e_2)(t - e_3)}}$$

into itself.

10. Show that the substitutions

$$t = \frac{z - a_1}{z - a_2} \cdot \frac{a_2 - a_4}{a_2 - a_1}, \quad k^2 = \frac{a_3 - a_4}{a_3 - a_1} \cdot \frac{a_2 - a_1}{a_2 - a_4};$$

transform

$$\int_0^t \frac{dt}{\sqrt{t(1-t)(1-k^2t)}} \text{ into } \pm \sqrt{(a_4 - a_2)(a_1 - a_3)} \int_{a_1}^z \frac{dz}{\sqrt{(z - a_1)(z - a_2)(z - a_3)(z - a_4)}}.$$

11. Prove that the substitution

$$\frac{z - a_1}{z - a_2} : \frac{a_3 - a_1}{a_3 - a_2} = \frac{t - a_2}{t - a_1} : \frac{a_4 - a_2}{a_4 - a_1}$$

transforms

$$\int \frac{dz}{\sqrt{A(z - a_1)(z - a_2)(z - a_3)(z - a_4)}} \text{ into } \int \frac{dt}{\sqrt{A(t - a_1)(t - a_2)(t - a_3)(t - a_4)}}.$$

## CHAPTER II

### THE ELLIPTIC FUNCTIONS

**Art. 9.** The expressions  $F(k, \phi)$ ,  $E(k, \phi)$ ,  $\Pi(n, k, \phi)$  were called by Legendre *elliptic functions*; these quantities are, however, *elliptic integrals*. It was Abel\* who, about 1823, pointed out that if one studied the integral  $u$  as a function of  $x$  in

$$u = \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \int_0^\phi \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}}, \quad x = \sin \phi, \quad (1)$$

the same difficulty was met, as if he were to study the trigonometric and logarithmic functions by considering  $u$  as a function of  $x$  in

$$u = \int_0^x \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x, \text{ or } u = \int_1^x \frac{dx}{x} = \log x.$$

Abel proposed instead to study the upper limit  $x$  as a function of  $u$ . Jacobi (*Fundamenta Nova*, § 17) introduced the notation  $\phi = \text{amplitude of } u$ , and written  $\phi = \text{am } u$ . Considered as a function of  $u$ , we have  $x = \sin \phi = \sin \text{am } u$ , and associated with this function are the two other elliptic functions  $\cos \phi = \cos \text{am } u$  and  $\Delta \phi = \Delta \text{am } u = \sqrt{1-k^2 \sin^2 \phi}$ . Gudermann (teacher of Weierstrass) in *Crelle's Journal*, Bd. 18, p. 12, proposed to abbreviate this notation and to write

$$\begin{aligned} x &= \sin \phi = \text{sn } u, \\ \sqrt{1-x^2} &= \cos \phi = \text{cn } u, \\ \sqrt{1-k^2x^2} &= \Delta \phi = \text{dn } u, \end{aligned}$$

\* Abel (*Œuvres*, Sylow and Lie edition, T. I., p. 263 and p. 518, 1827-30).

It follows at once that

$$sn^2 u + cn^2 u = 1,$$

$$dn^2 u + k^2 sn^2 u = 1.$$

From (1) results  $\frac{du}{d\phi} = \frac{1}{\Delta\phi}$  or  $\frac{d\phi}{du} = \Delta\phi$ , so that  $\frac{d}{du} am u = \Delta am u = dn u$ .

It is also evident that

$$\frac{d}{du} sn u = \frac{d}{du} \sin \phi = \cos \phi \frac{d\phi}{du} = cn u \, dn u,$$

$$\frac{d}{du} cn u = -sn u \, dn u,$$

$$\frac{d}{du} dn u = -k^2 sn u \, cn u.$$

Further, if  $u=0$ , then the upper limit  $\phi=0$ , so that  $am 0=0$ , and consequently,  $sn 0=0$ ,  $cn 0=1$ ,  $dn 0=1$ .

If  $\phi$  be changed into  $-\phi$ , it is seen that  $u$  changes its sign, so that  $am(-u) = -am u$ , and

$$sn(-u) = -sn u, \quad cn(-u) = cn u, \quad dn(-u) = dn u.$$

**Art. 10.** In the theory of circular functions there is found the numerical transcendent  $\pi$ , a quantity such that  $\sin \frac{\pi}{2} = 1$ ,

$\cos \frac{\pi}{2} = 0$ . Writing

$$u = \int_0^x \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x,$$

we have  $x = \sin u$ . Thus  $\frac{\pi}{2}$  may be defined as the complete integral

$$\frac{\pi}{2} = \int_0^1 \frac{dx}{\sqrt{1-x^2}}.$$



Similarly a real positive quantity  $K$  (Jacobi) may be defined through

$$K = \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}} = F\left(k, \frac{\pi}{2}\right) \quad (\text{Art. 6}).$$

Associated with  $K$  is the transcendental quantity  $K'$ , which is the same function of the complementary modulus  $k'$  as  $K$  is of  $k$ . The transcendental nature of these two functions of  $k$  and  $k'$  may be observed by considering the following infinite series through which they are expressed.

If  $(1-k^2 \sin^2 \phi)^{-\frac{1}{2}}$  be expanded in a series, then

$$\begin{aligned} F(k, \phi) &= \int_0^\phi \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}} \\ &= \phi + \frac{1}{2}k^2 v_2 + \dots + \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} k^{2n} v_{2n} + \dots, \end{aligned}$$

where  $v_{2n} = \int_0^\phi \sin^{2n} \phi \, d\phi$ .

In particular, if  $\phi = \frac{\pi}{2}$ , we have by Wallis's Theorem,

$$\int_0^{\frac{\pi}{2}} \sin^{2n} \phi \, d\phi = \frac{1 \cdot 3 \cdot \dots \cdot 2n-1}{2 \cdot 4 \cdot \dots \cdot 2n} \frac{\pi}{2}.$$

It follows that

$$\frac{2}{\pi} K = 1 + \left(\frac{1}{2}\right)^2 k^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 k^4 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 k^6 + \dots$$

Similarly, it may be proved that

$$\frac{2}{\pi} E\left(k, \frac{\pi}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 k^2 - \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 \frac{k^4}{3} - \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 \frac{k^6}{5} - \dots$$

which confirm the results of Arts. 6 and 7.

**Art. 11.** If in the integral  $\int_{\frac{2n-1}{2}\pi}^{n\pi} \frac{d\phi}{\Delta\phi}$  there be put  $\phi = n\pi - \theta$ ,

then it becomes

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{\Delta\theta} = K; \text{ and if in the integral } \int_{n\pi}^{\frac{2n+1}{2}\pi} \frac{d\phi}{\Delta\phi}$$

we put  $\phi = n\pi + \theta$ , then this integral is

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{\Delta\theta} = K.$$

It follows that

$$\int_0^{\frac{\pi}{2}} \frac{d\phi}{\Delta\phi} = \int_0^{\frac{\pi}{2}} \frac{d\phi}{\Delta\phi} + \int_{\frac{\pi}{2}}^{\pi} \frac{d\phi}{\Delta\phi} + \dots + \int_{(n-1)\frac{\pi}{2}}^{\frac{n\pi}{2}} \frac{d\phi}{\Delta\phi} = nK,$$

so that  $\frac{n\pi}{2} = am\ nK$ ; or, since  $\frac{\pi}{2} = am\ K$ , we have  $am\ nK = n\ am\ K$ .

Note that

$$\int_0^{n\pi+\beta} \frac{d\phi}{\Delta\phi} = \int_0^{n\pi} \frac{d\phi}{\Delta\phi} + \int_{n\pi}^{n\pi+\beta} \frac{d\phi}{\Delta\phi} = 2nK + u,$$

where

$$u = \int_{n\pi}^{n\pi+\beta} \frac{d\phi}{\Delta\phi} = \int_0^{\beta} \frac{d\theta}{\Delta\theta};$$

further, since any arc  $\alpha$  may be put  $= n\pi \pm \beta$ , where  $\beta$  is an arc between 0 and  $\frac{\pi}{2}$ , we may always write

$$\alpha = n\pi \pm \beta = am(2nK \pm u),$$

or

$$2n\ am\ K \pm am\ u = am(2nK \pm u).$$

Art. 12. Making use of the formula just written, it is seen that  $am\ K = \frac{\pi}{2}$ ,

$$sn\ K = 1, \quad cn\ K = 0, \quad dn\ K = k'.$$

$$sn(u \pm 2K) = -sn\ u, \quad cn(u \pm 2K) = -cn\ u, \quad dn(u \pm 2K) = dn\ u;$$

$$sn(u \pm 4K) = sn\ u, \quad cn(u \pm 4K) = cn\ u, \quad dn(u \pm 4K) = dn\ u.$$

Note that  $4K$  is a *period* of the three elliptic transcendents  $sn\ u$ ,  $cn\ u$  and  $dn\ u$ ; in fact, it is seen that  $2K$  is a period of  $dn\ u$  and of  $\frac{sn\ u}{cn\ u} = tn\ u$ . Also note that

$$\operatorname{sn} 2K = 0, \quad \operatorname{cn} 2K = -1, \quad \operatorname{dn} 2K = 1,$$

$$\operatorname{sn} 4K = 0, \quad \operatorname{cn} 4K = 1, \quad \operatorname{dn} 4K = 1.$$

Of course, the modulus of the above functions is  $k$ ; and, since  $K'$  is the same function of  $k'$  as  $K$  is of  $k$ , we also have

$$\operatorname{sn}(u \pm 2K', k') = -\operatorname{sn}(u, k'),$$

$$\operatorname{sn}(u \pm 4K', k') = \operatorname{sn}(u, k'), \text{ etc.}$$

**Art. 13.** *The Gudermannian.* As introductory to the Jacobi imaginary transformation of the following article, there is a particular case \* where  $k=1$ . Then

$$u = F(1, \phi) = \int_0^\phi \frac{d\phi}{\sqrt{1 - \sin^2 \phi}} = \log \tan \left( \frac{\pi}{4} + \frac{\phi}{2} \right). \quad (\text{Cf. Art. 7.})$$

Here  $\phi$ , considered as a function of  $u$ , may be called the Gudermannian and written  $\phi = \operatorname{gd} u$ , the functions corresponding to  $\operatorname{sn} u$  and  $\operatorname{cn} u$  being  $\operatorname{sg} u$  and  $\operatorname{cg} u$ . Then

$$e^u = \tan \left( \frac{\pi}{4} + \frac{\phi}{2} \right) = \frac{1 + \tan \phi/2}{1 - \tan \phi/2} = \frac{1 + \sin \phi}{\cos \phi} = \frac{\cos \phi}{1 - \sin \phi};$$

or,

$$e^u = \frac{1 + \operatorname{sg} u}{\operatorname{cg} u}, \quad e^{-u} = \frac{\operatorname{cg} u}{1 + \operatorname{sg} u} = \frac{1 - \operatorname{sg} u}{\operatorname{cg} u}.$$

It follows that

$$\operatorname{cg} u = \frac{2}{e^u + e^{-u}} = \frac{1}{\cos iu} = \frac{1}{\cosh u} = \operatorname{sech} u,$$

and

$$\operatorname{sg} u = \frac{e^u - e^{-u}}{e^u + e^{-u}} = -i \frac{\sin iu}{\cos iu} = \frac{\sinh u}{\cosh u} = \tanh u.$$

These formulas may be written

$$\begin{array}{l|l} \operatorname{sg} u = -i \tan iu, & \sin iu = i \operatorname{tg} u, \\ \operatorname{cg} u = 1/\cos iu, & \cos iu = 1/\operatorname{cg} u, \\ \operatorname{tg} u = -i \sin iu; & \tan iu = i \operatorname{sg} u. \end{array}$$

\* See Gudermann, *Crelle*, Bd. 18, pp. 1, et seq.; see also Cayley, loc. cit. p. 56; Weierstrass, *Math. Werke* I, pp. 1-49 and the remark p. 50.

The above relations may also be derived by considering two angles  $\theta$  and  $\phi$  connected by the equation  $\cos \theta \cos \phi = 1$ . For there follows at once

$$\begin{array}{l|l} \sin \theta = i \tan \phi, & \sin \phi = -i \tan \theta, \\ \cos \theta = 1/\cos \phi, & \cos \phi = 1/\cos \theta, \\ \tan \theta = i \sin \phi, & \tan \phi = -i \sin \theta. \end{array}$$

Further, there results,

$$\cos \theta d\theta = i \sec^2 \phi d\phi, \quad \text{or} \quad d\theta = i \frac{d\phi}{\cos \phi}.$$

It follows that

$$\theta = i \log \tan \left( \frac{\pi}{4} + \frac{\phi}{2} \right).$$

Then, by assuming that  $\phi = gdu$ , we have  $\theta = iu$ , and consequently the foregoing relations.

**Art. 14.** *Jacobi's Imaginary Transformations.\** Writing  $\sin \theta = i \tan \phi$ ,  $\cos \theta = \frac{1}{\cos \phi}$ ,  $\sin \phi = -i \tan \theta$ ,  $\Delta(\theta, k) = \frac{\Delta(\phi, k')}{\cos \phi}$ ,

we have  $d\theta = i \frac{d\phi}{\cos \phi}$  and  $\int_0^\theta \frac{d\theta}{\Delta(\theta, k)} = i \int_0^\phi \frac{d\phi}{\Delta(\phi, k')}$ .

If, then,  $\int_0^\phi \frac{d\phi}{\Delta(\phi, k')} = u$ , so that  $\phi = am(u, k')$ , there results  $\int_0^\theta \frac{d\theta}{\Delta(\theta, k)} = iu$ , and  $\theta = am iu$ .

These expressions, substituted in the above relations, give

$$sn(iu, k) = i \, tn(u, k'),$$

$$cn(iu, k) = \frac{1}{cn(u, k')},$$

$$dn(iu, k) = \frac{dn(u, k')}{cn(u, k')}.$$

From this it is evident that the two functions  $cn$  and  $dn$  have real values for imaginary values of the argument, while  $sn(iu)$  is an imaginary quantity.

\* Jacobi, *Fundamenta Nova*, § 19. See also Abel, *Œuvres*, T. I., p. 272.



Among the trigonometric and exponential functions, we have, for example, the relation

$$\cos iu = \frac{e^u + e^{-u}}{2},$$

where the argument of the trigonometric function is real while that of the exponential function is real. We note that an elliptic function with imaginary argument may be expressed through an elliptic function with real argument, whose modulus is the complement of the original modulus.

**Art. 15.** From the formulas of the preceding article it follows at once

$$sn[i(u+4K'), k] = i \, tn(u+4K', k') = sn(iu, k),$$

and also

$$cn(iu+4iK', k) = cn(iu, k),$$

$$dn(iu+4iK', k) = dn(iu, k).$$

If in these formulas  $iu$  be changed into  $u$ , we have

$$sn(u \pm 4iK', k) = sn(u, k),$$

$$cn(u \pm 4iK', k) = cn(u, k),$$

$$dn(u \pm 4iK', k) = dn(u, k).$$

It also follows that  $sn(u \pm 4iK, k') = sn(u, k')$ , etc. If in the formula  $sn(iu) = i \, tn(u, k')$ , we put  $u+2K'$  in the place of  $u$ , then

$$sn(iu+2iK', k) = i \, tn(u+2K', k') = i \, tn(u, k') = sn \, iu.$$

Changing  $iu$  to  $u$ , we have

$$sn(u \pm 2iK') = sn \, u, \quad cn(u \pm 2iK') = -cn \, u, \quad dn(u \pm 2iK') = -dn \, u,$$

and

$$sn(2iK') = 0, \quad cn(2iK') = -1, \quad dn(2iK') = -1.$$

The modulus  $k$  is always understood, unless another modulus is indicated.

It follows at once that

$$sn(u \pm 4iK') = sn u, \quad cn(u \pm 4iK') = cn u, \quad dn(u \pm 4iK') = dn u,$$

and

$$sn(4iK') = 0, \quad cn(4iK') = 1, \quad dn(4iK') = 1.$$

It is also seen that

$$\begin{aligned} sn(u \pm 2K \pm 2iK') &= -sn u, \\ sn(u \pm 4K \pm 4iK') &= sn u, \text{ etc.} \end{aligned}$$

In particular, notice that

$$\begin{aligned} \text{the periods of } snu &\text{ are } 4K \text{ and } 2iK', \\ \text{the periods of } cnu &\text{ are } 4K \text{ and } 2K + 2iK', \\ \text{the periods of } dnu &\text{ are } 2K \text{ and } 4iK'. \end{aligned}$$

**Art. 16. Periodic Functions.** Consider the simple case of the exponential function  $e^u$  and suppose that  $u = x + iy$ . It may be shown that  $e^{u+2\pi i} = e^u$  for all values of  $u$ ; for it is seen that  $e^u = e^{x+iy} = e^x(\cos y + i \sin y)$ . If we increase  $u$  by  $2\pi i$ , then  $y$  is increased by  $2\pi$  and consequently

$$e^{u+2\pi i} = e^x[\cos(y+2\pi) + i \sin(y+2\pi)] = e^x(\cos y + i \sin y) = e^u.$$

It follows that if it is desired to examine the function  $e^u$ , then clearly this function need not be studied in the whole  $u$ -plane, but only within a strip which lies above the  $X$ -axis and has the breadth  $2\pi$ ; for we see at once that to every point  $u_0$  which lies without this *period-strip* there corresponds a point  $u_1$  within the strip and in such a way that the function has the same value and the same properties at  $u_0$  and  $u_1$ .

Similarly it is seen that the two functions  $\sin u$  and  $\cos u$  have the real period  $2\pi$ , and consequently it is necessary to study these functions only within a period-strip which lies adjacent to the  $Y$ -axis with a breadth  $2\pi$ . As already noted, Abel and Jacobi found that the elliptic functions had two periods. In the preceding article it was seen that  $sn u$  had the real period  $4K$  and the imaginary period  $2iK'$ .

On the  $X$ -axis lay off a distance  $4K$  and on the  $Y$ -axis a distance  $2K'$  and construct the rectangle on these two sides. Further suppose that the whole plane is filled out with such rectangles.

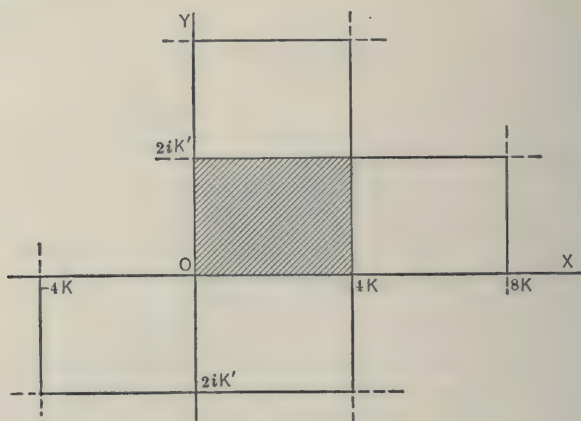


FIG. 6.

Then it will be seen that the function  $\operatorname{sn} u$  behaves in every rectangle precisely as it does in the initial rectangle. Similar parallelograms may be constructed for the functions  $\operatorname{cn} u$  and  $\operatorname{dn} u$ . See Art. 21.

**Art. 17.** Next write  $\sin \phi = \frac{\cos \theta}{\Delta \theta}$ , so that  $\cos \phi = \frac{k' \sin \theta}{\Delta \theta}$ , and  $\Delta \phi = \frac{k'}{\Delta \theta}$ . It follows that  $\frac{d\phi}{\Delta \phi} = -\frac{d\theta}{\Delta \theta}$  and consequently

$$\int_0^\phi \frac{d\phi}{\Delta \phi} = \int_\theta^{\frac{\pi}{2}} \frac{d\theta}{\Delta \theta} = \int_0^{\frac{\pi}{2}} \frac{d\theta}{\Delta \theta} - \int_0^\theta \frac{d\theta}{\Delta \theta} = K - u,$$

if we put  $u = \int_0^\theta \frac{d\theta}{\Delta \theta}$ , or  $\theta = \operatorname{am} u$ . It follows that  $\phi = \operatorname{am}(K - u)$ ,

and from the above formulas

$$\operatorname{sn}(K - u) = \frac{\operatorname{cn} u}{\operatorname{dn} u}, \quad \operatorname{cn}(K - u) = \frac{k' \operatorname{sn} u}{\operatorname{dn} u}, \quad \operatorname{dn}(K - u) = \frac{k'}{\operatorname{dn} u}.$$

In these formulas change  $-u$  to  $u$  and note that  $\operatorname{sn}(-u) = -\operatorname{sn} u$ , etc.

It is seen that

$$\operatorname{sn}(u \pm K) = \pm \frac{\operatorname{cn} u}{\operatorname{dn} u} \quad \operatorname{sn} K = 1,$$

$$\operatorname{cn}(u \pm K) = \mp \frac{k' \operatorname{sn} u}{\operatorname{dn} u}, \quad \operatorname{cn} K = 0,$$

$$\operatorname{dn}(u \pm K) = + \frac{k'}{\operatorname{dn} u}, \quad \operatorname{dn} K = k'.$$

For the calculation of the elliptic functions, the above relations permit the reduction of the argument so that it is always comprised between 0 and  $\frac{1}{2}K$ , just as in trigonometry the angle may be reduced so as to lie between 0 and  $45^\circ$  for the calculation of the circular functions.

Art. 18. In the above formulas put  $iu$  in the place of  $u$ , and it is seen that

$$\operatorname{sn}(iu \pm K) = \pm \frac{\operatorname{cn} iu}{\operatorname{dn} iu} = \pm \frac{1}{\operatorname{dn}(u, k')},$$

$$\operatorname{cn}(iu \pm K) = \mp \frac{ik' \operatorname{sn}(u, k')}{\operatorname{dn}(u, k')},$$

$$\operatorname{dn}(iu \pm K) = \frac{k' \operatorname{cn}(u, k')}{\operatorname{dn}(u, k')}.$$

Further, in the formulas  $\operatorname{sn} iu = i \operatorname{tn}(u, k')$ , etc., write  $u \pm iK$  for  $u$  and it is seen that

$$\operatorname{sn}(iu \pm iK', k) = i \operatorname{tg} \operatorname{am}(u \pm K', k') = -\frac{i \operatorname{cn}(u, k')}{k \operatorname{sn}(u, k')},$$

$$\operatorname{cn}(iu \pm iK', k) = \mp \frac{\operatorname{dn}(u, k')}{\operatorname{sn}(u, k')},$$

$$\operatorname{dn}(iu \pm iK', k) = \mp \frac{1}{\operatorname{sn}(u, k')}.$$

In the above formulas change  $iu$  to  $u$ . We then have

$$\operatorname{sn}(u \pm iK') = \frac{1}{k} \frac{1}{\operatorname{sn} u},$$

$$\operatorname{cn}(u \pm iK') = \mp \frac{i}{k} \frac{\operatorname{dn} u}{\operatorname{sn} u},$$

$$\operatorname{dn}(u \pm iK') = \mp i \operatorname{cot} \operatorname{am} u.$$



If in these formulas  $u=0$ , then

$$\operatorname{sn}(\pm iK') = \infty, \quad \operatorname{cn}(\pm iK') = \infty, \quad \operatorname{dn}(\pm iK') = \infty.$$

Further, if in the preceding formulas  $u+K$  be put in the place of  $u$ , then

$$\operatorname{sn}(u+K \pm iK') = \frac{1}{k} \frac{1}{\operatorname{sn}(u+K)} = \frac{1}{k} \frac{\operatorname{dn} u}{\operatorname{cn} u}$$

$$\operatorname{cn}(u+K \pm iK') = \mp \frac{ik'}{k \operatorname{cn} u},$$

$$\operatorname{dn}(u+K \pm iK') = \pm ik' \operatorname{tg} am u;$$

and from these formulas, writing,  $u=0$ , there results

$$\operatorname{sn}(K \pm iK') = \frac{1}{k}, \quad \operatorname{cn}(K \pm iK') = \mp \frac{ik'}{k}, \quad \operatorname{dn}(K \pm iK') = 0.$$

**Art. 19.** Note the analogy of the transcendent  $K$  of the elliptic functions to  $\frac{\pi}{2}$  of the circular functions. Due to the relation  $am(K-u) = \frac{\pi}{2} - am u$  (Art. 11) Jacobi called the amplitude of  $K-u$  the *co-amplitude* of  $u$  and wrote  $am(K-u) = coam u$ .

It follows at once from the above formulas that

$$\sin coam u = \frac{\operatorname{cn} u}{\operatorname{dn} u},$$

$$\cos coam u = \frac{k' \operatorname{sn} u}{\operatorname{dn} u},$$

$$\Delta coam u = \frac{k'}{\operatorname{dn} u}.$$

$$\sin coam(iu, k) = \frac{1}{\operatorname{dn}(u, k')}, \text{ etc.}$$

**Art. 20. Remark.** The results obtained for the imaginary argument have been derived by making use of Jacobi's imaginary transformation; and by changing  $iu$  into  $u$  we have implicitly made the assumption (proved in my *Elliptic Functions*, Vol. I,

Chaps X and XI) that the elliptic functions have the same properties for real and imaginary arguments.

Art. 21. By a *zero* of a function,  $sn\,u$  for example, we mean that value of  $u$  which, when substituted for  $u$  in  $sn\,u$ , causes this function to be zero, while an *infinity* of a function is a value of  $u$  which causes the function to become infinite.

In studying the following graphs note that on the boundaries of the period parallelogram of  $sn\,u$ , there are six points at which this function becomes zero; but if the adjacent period parallelograms be constructed, it will be seen that only *two* zeros belong to each parallelogram. In fact, in each period-parallelogram there are *two* values of  $u$  which cause the function to take any fixed value; that is, any value being fixed, there are always two values of  $u$  which cause the function to take this value. From the following graphs it is seen that any *real* value situated within the interval  $-\infty$  to  $+\infty$  is taken twice by each of the three functions  $sn\,u$ ,  $cn\,u$ ,  $dn\,u$ .

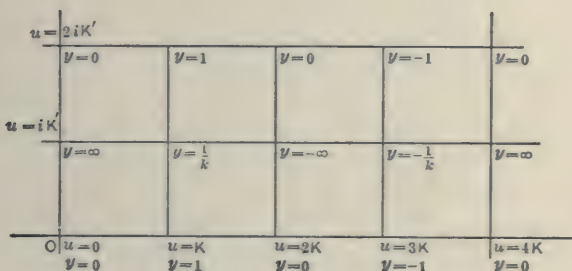


FIG. 7.  $y = sn(u, k)$ .

ZEROS

$$2mK + 2niK'$$

INFINITIES

$$2mK + (2n+1)K'$$

where  $m$  and  $n$  are any integers.

PERIODS

$$4K, 2iK'$$

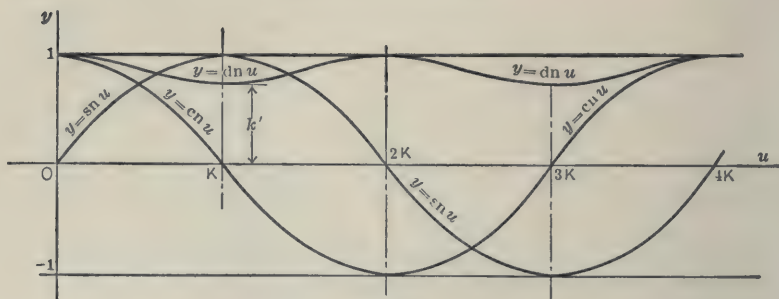
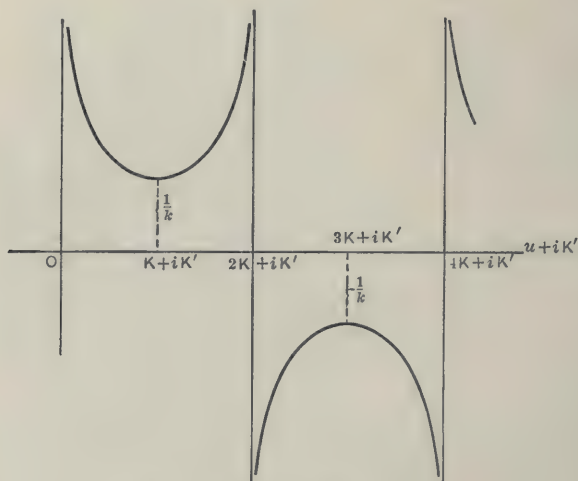
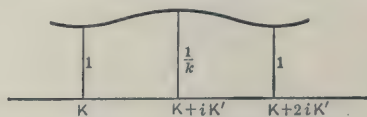
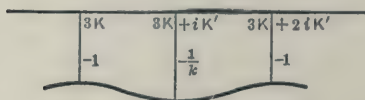
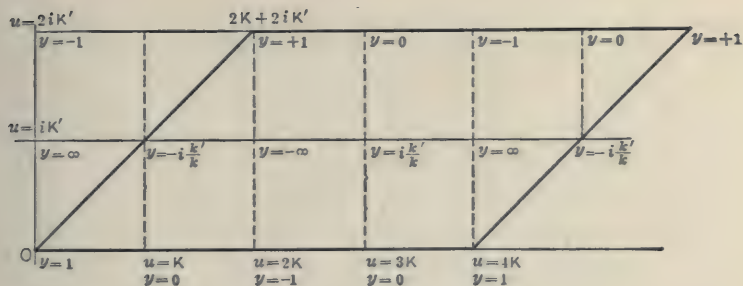


FIG. 8.

FIG. 9.  $y = \operatorname{sn}(u + iK')$ .

In Fig. 9, the value  $iK'$  coincides with the origin.

FIG. 10a.  $y = \operatorname{sn}(iu + K)$ .FIG. 10b.  $y = \operatorname{sn}(iu + 3K)$ .


 FIG. 11.  $y = \text{cn}(u)$ .

ZEROS

$$(2m+1)K + 2niK'$$

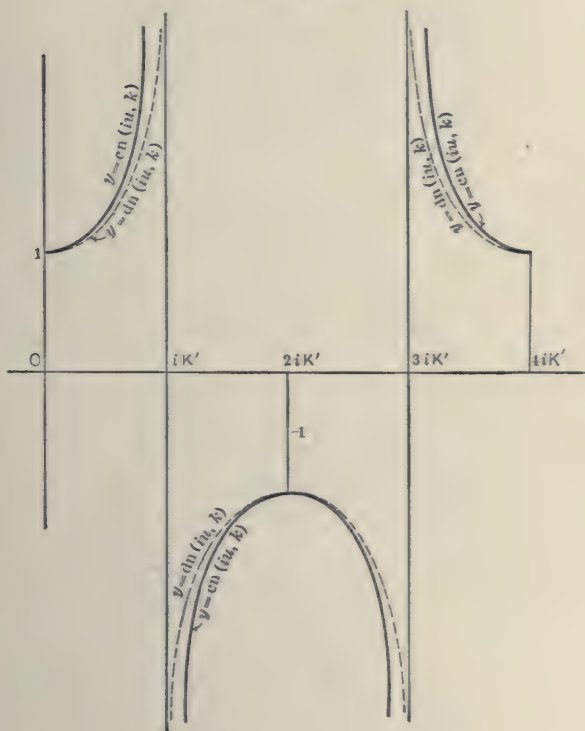
INFINITIES

$$2mK + (2n+1)iK'$$

 where  $m$  and  $n$  are any integers.

PERIODS

$$4K, 2K + 2iK'$$


 FIG. 12.  $y = \text{cn}(iu)$ ;  $y = \text{dn}(iu)$ .



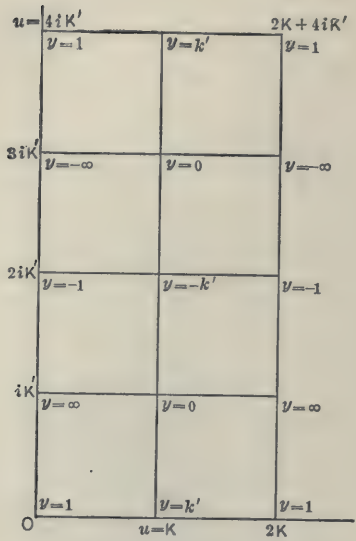


FIG. 13.  $y = \text{dn}(u)$ .

ZEROS

$$(2m+1)K + (2n+1)iK'$$

where  $m$  and  $n$  are integers.

INFINITIES

$$2mK + (2n+1)iK'$$

PERIODS

$$2K, 4iK'$$

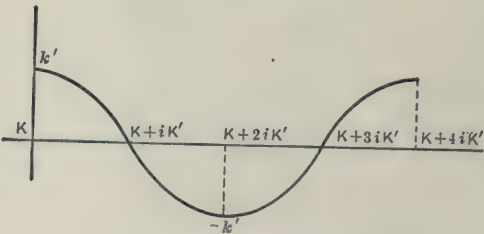


FIG. 14.  $y = \text{dn}(K+iu)$ .

## EXAMPLES

1. In the formulas of Art. 17 put  $u = \frac{K}{2}$ , and show first that  $dn \frac{K}{2} = \sqrt{k'}$  and then  $sn^2 \frac{K}{2} = \frac{1-k'}{k^2} = \frac{1}{1+k'}$ ,  $cn^2 \frac{K}{2} = \frac{k'}{1+k'}$ ,  $am \frac{K}{2} = \tan^{-1} \sqrt{\frac{1}{k'}}$ .

2. Prove that

$$sn \frac{3}{2}K = \frac{1}{\sqrt{1+k'}}, \quad cn \frac{3}{2}K = -\frac{\sqrt{k'}}{\sqrt{1+k'}}, \quad dn \frac{3}{2}K = \sqrt{k'}.$$

3. Prove that

$$sn \frac{iK'}{2} = \frac{i}{\sqrt{k}}, \quad cn \frac{iK'}{2} = \frac{\sqrt{1+k}}{\sqrt{k}}, \quad dn \frac{iK'}{2} = \sqrt{1+k}.$$

4. Show that

$$sn(K + \frac{1}{2}iK') = \frac{1}{\sqrt{k}}, \quad cn(K + \frac{1}{2}iK') = -i \frac{\sqrt{1-k}}{\sqrt{k}}, \quad dn(K + \frac{1}{2}iK') = \sqrt{1-k}.$$

5. Show that

$$sn(\frac{1}{2}K + \frac{1}{2}iK') = \frac{1}{\sqrt{2k}} [\sqrt{1+k} + i\sqrt{1-k}],$$

$$cn(\frac{3}{2}K + \frac{1}{2}iK') = -\frac{1+i\sqrt{k'}}{\sqrt{2k}},$$

$$dn(\frac{1}{2}K + \frac{3}{2}iK') = -\frac{\sqrt{k'}}{2} (\sqrt{1+k'} + i\sqrt{1-k'}).$$

6. Show that

$$sn(u + K + 3iK') = \frac{dn u}{k cn u},$$

$$cn(u + 3K + iK') = \frac{ik'}{k cn u},$$

$$dn(u + 3K + 3iK') = \frac{-k' sn u}{cn u}.$$

7. Making the linear transformation  $x = kz$ , we have

$$\int_0^x \frac{dx}{\sqrt{(1-x^2) \left(1 - \frac{x^2}{k^2}\right)}} = k \int_0^{\frac{x}{k}} \frac{dz}{\sqrt{(1-z^2) (1 - k^2 z^2)}}.$$

Further, put

$$u = \int_0^z \frac{dz}{\sqrt{(1-z^2) (1 - k^2 z^2)}}, \quad ku = \int_0^x \frac{dx}{\sqrt{(1-x^2) \left(1 - \frac{x^2}{k^2}\right)}},$$

and show that

$$\operatorname{sn}\left(ku, \frac{1}{k}\right) = k \operatorname{sn}(u, k),$$

$$\operatorname{cn}\left(ku, \frac{1}{k}\right) = \operatorname{dn}(u, k),$$

$$\operatorname{dn}\left(ku, \frac{1}{k}\right) = \operatorname{cn}(u, k);$$

$$\operatorname{sn}\left(ku, \frac{ik'}{k}\right) = \cos \operatorname{coam}(u, k'),$$

$$\operatorname{cn}\left(ku, \frac{ik'}{k}\right) = \sin \operatorname{coam}(u, k'),$$

$$\operatorname{dn}\left(ku, \frac{ik'}{k}\right) = \frac{1}{\Delta \operatorname{am}(u, k')}.$$

8. The quadratic substitution  $t = \frac{(1+k)z}{1+kz^2}$  transforms  $\frac{dz}{\sqrt{(1-z^2)(1-k^2z^2)}}$

into  $\frac{Mdt}{\sqrt{(1-t^2)(1-l^2t^2)}}$ , where  $l = \frac{2\sqrt{k}}{1+k}$  and  $M = \frac{1}{1+k}$ .

9. Show that

$$\operatorname{sn}\left[(1+k)u, \frac{2\sqrt{k}}{1+k}\right] = \frac{(1+k)\operatorname{sn}(u, k)}{1+k\operatorname{sn}^2(u, k)},$$

$$\operatorname{cn}\left[(1+k)u, \frac{2\sqrt{k}}{1+k}\right] = \frac{\operatorname{cn}(u, k)\operatorname{dn}(u, k)}{1+k\operatorname{sn}^2(u, k)},$$

$$\operatorname{dn}\left[(1+k)u, \frac{2\sqrt{k}}{1+k}\right] = \frac{1-k\operatorname{sn}^2(u, k)}{1+k\operatorname{sn}^2(u, k)}.$$

## CHAPTER III

### ELLIPTIC INTEGRALS OF THE FIRST KIND REDUCED TO LEGENDRE'S NORMAL FORM

**Art. 22.** In the elementary calculus such integrals as the following have been studied

$$\int_0^x \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x = \cos^{-1} \sqrt{1-x^2},$$

$$\int_x^\infty \frac{dx}{x^2+1} = \cot^{-1} x = \tan^{-1} \frac{1}{x},$$

$$\int_1^x \frac{dx}{\sqrt{x^2-1}} = \cosh^{-1} x = \sinh^{-1} \sqrt{x^2-1} = \log \{x + \sqrt{x^2-1}\}.$$

Following Clifford\* an analogous notation for the elliptic integrals will be introduced. Write (see Art. 9),

$$x = sn\, u, \quad \sqrt{1-x^2} = cn\, u, \quad \sqrt{1-k^2x^2} = dn\, u.$$

Since (see Art. 9),  $\frac{dx}{du} = cn\, u\, dn\, u$ , it follows that

$$\frac{dx}{du} = \sqrt{(1-x^2)(1-k^2x^2)};$$

or

$$\begin{aligned} \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} &= u = sn^{-1} x = cn^{-1} \sqrt{1-x^2} = dn^{-1} \sqrt{1-k^2x^2} \\ &= F(k, \phi) = F(k, \sin^{-1} x). \quad . \quad . \quad . \quad (1) \end{aligned}$$

In particular, it is seen from this formula that the substitution  $x = \sin \phi$  transforms the integral  $\int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}$

\* Clifford, *Mathematical Papers*, p. 207.



into the normal form  $\int_0^\phi \frac{dx}{\sqrt{1-k^2 \sin^2 \phi}} = F(k, \phi)$ . Further, from the tables given at the end of the book, which we shall learn later to construct and use, the integral is known as soon as  $x$  is fixed.

Similarly, if there be put  $x = cn u$ ,  $\sqrt{1-x^2} = sn u$ ,  $\sqrt{k'^2 + k^2 x^2} = dn u$ ,  $\frac{dx}{du} = \frac{d cn u}{du} = -sn u \, dn u = -\sqrt{(1-x^2)(k'^2 + k^2 x^2)}$ , it follows that

$$\begin{aligned} \int_x^1 \frac{dx}{\sqrt{(1-x^2)(k'^2 + k^2 x^2)}} &= u = cn^{-1} x = sn^{-1} \sqrt{1-x^2} = dn^{-1} \sqrt{k'^2 + k^2 x^2} \\ &= F(k, \phi) = F(k, \cos^{-1} x) \\ &= F(k, \sin^{-1} \sqrt{1-x^2}) \dots \dots \dots (2) \end{aligned}$$

It is seen also that the substitution  $x = \cos \phi$  transforms the integral on the right-hand side into the normal form.

If  $x = dn u$ ,  $\frac{\sqrt{1-x^2}}{k} = sn u$ ,  $\frac{\sqrt{x^2 - k'^2}}{k} = cn u$ ,  $\frac{dx}{du} = -k^2 sn u \, cn u = -\sqrt{(1-x^2)(x^2 - k'^2)}$ , we have

$$\begin{aligned} \int_x^1 \frac{dx}{\sqrt{(1-x^2)(x^2 - k'^2)}} &= u = dn^{-1} x = sn^{-1} \left( \frac{\sqrt{1-x^2}}{k} \right) \\ &= cn^{-1} \left( \frac{\sqrt{x^2 - k'^2}}{k} \right) = F(k, \phi) \\ &= F \left[ k, \sin^{-1} \left( \frac{\sqrt{1-x^2}}{k} \right) \right] \dots \dots \dots (3) \end{aligned}$$

Further, writing  $x = \tan am u$ , it follows that  $sn u = \frac{x}{\sqrt{1+x^2}}$ ,

$$cn u = \frac{1}{\sqrt{1+x^2}}, \quad dn u = \frac{\sqrt{1+k'^2 x^2}}{\sqrt{1+x^2}}, \quad \frac{dx}{du} = \frac{dn u}{cn^2 u} = \sqrt{(1+x^2)(1+k'^2 x^2)},$$

and

$$\begin{aligned} \int_0^x \frac{dx}{\sqrt{(1+x^2)(1+k'^2 x^2)}} &= u = tn^{-1} x = sn^{-1} \left( \frac{x}{\sqrt{1+x^2}} \right) \\ &= F(k, \tan^{-1} x). \quad (4) \end{aligned}$$

**Art. 23.** 1. If  $a > b > x > 0$ , write  $x = b \sin \phi$  in the integral,

$$v = \int_0^x \frac{dx}{\sqrt{(a^2 - x^2)(b^2 - x^2)}},$$

and we have, if  $k^2 = \frac{b^2}{a^2}$ ,

$$v = \frac{1}{a} \int_0^\phi \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \frac{1}{a} \operatorname{sn}^{-1} \left[ \frac{x}{b}, \frac{b}{a} \right] \dots \dots (5a)$$

2. If  $\infty > x > a$ , write  $x = \frac{a}{\sin \phi}$ , and it is seen that

$$\int_x^\infty \frac{dx}{\sqrt{(x^2 - a^2)(x^2 - b^2)}} = \frac{1}{a} \operatorname{sn}^{-1} \left[ \frac{a}{x}, \frac{b}{a} \right] \dots \dots (5b)$$

If  $a > b > x > 0$ ,

$$\int_x^b \frac{dx}{\sqrt{(a^2 + x^2)(b^2 - x^2)}} = \frac{1}{\sqrt{a^2 + b^2}} \operatorname{cn}^{-1} \left[ \frac{x}{b}, \frac{b}{\sqrt{a^2 + b^2}} \right]; \quad (6a)$$

(see IV, in Art. 3), and also

$$\int_b^x \frac{dx}{\sqrt{(a^2 + x^2)(x^2 - b^2)}} = \frac{1}{\sqrt{a^2 + b^2}} \operatorname{cn}^{-1} \left[ \frac{b}{x}, \frac{a}{\sqrt{a^2 + b^2}} \right], \quad (6b)$$

(see V in Art. 3).

It is almost superfluous to add that for example in (6a) the substitution  $\frac{x}{b} = \cos \phi$  transforms the integral

$$\int_x^b \frac{dx}{\sqrt{(a^2 + x^2)(b^2 - x^2)}}$$

into

$$\frac{1}{\sqrt{a^2 + b^2}} \int_0^\phi \frac{d\phi}{\sqrt{1 - \frac{b^2}{a^2 + b^2} \sin^2 \phi}} = \frac{1}{\sqrt{a^2 + b^2}} F \left[ \frac{b}{\sqrt{a^2 + b^2}}, \cos^{-1} \frac{x}{b} \right].$$

It is also seen that if  $a > x > b > 0$ ,

$$\int_x^a \frac{dx}{\sqrt{(a^2 - x^2)(x^2 - b^2)}} = \frac{1}{a} \operatorname{dn}^{-1} \left[ \frac{x}{a}, \frac{\sqrt{a^2 - b^2}}{a} \right]; \dots \dots (7)$$

that is, the integral on the left-hand side becomes

$$\frac{1}{a} \int_0^\phi \frac{d\phi}{\sqrt{1 - \frac{a^2 - b^2}{a^2} \sin^2 \phi}},$$

for the substitution

$$\frac{x}{a} = \sqrt{1 - \frac{a^2 - b^2}{a^2} \sin^2 \phi}.$$

Further if  $a > b$

$$\int_0^x \frac{dx}{\sqrt{(x^2 + a^2)(x^2 + b^2)}} = \frac{1}{a} \operatorname{tn}^{-1} \left[ \frac{x}{b}, \sqrt{\frac{a^2 - b^2}{a^2}} \right] \quad . \quad . \quad (8)$$

(See I in Art. 3.)

**Art. 24.** In the formulas (1), (2), (3) and (4) above, substitute  $x$  for  $x^2$ , and it is seen that

$$\begin{aligned} \int_0^x \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} &= 2sn^{-1}(\sqrt{x}, k) = 2cn^{-1}(\sqrt{1-x}, k) \\ &= 2dn^{-1}(\sqrt{1-k^2x}, k), \quad . \quad . \quad . \quad (9) \end{aligned}$$

$$\int_x^1 \frac{dx}{\sqrt{x(1-x)(k'^2 + k^2x)}} = 2cn^{-1}(\sqrt{x}, k), \quad . \quad . \quad . \quad (10)$$

$$\int_x^1 \frac{dx}{\sqrt{x(1-x)(x - k'^2)}} = 2dn^{-1}(\sqrt{x}, k), \quad . \quad . \quad . \quad (11)$$

$$\int_0^x \frac{dx}{\sqrt{x(1+x)(1+k'^2x)}} = 2tn^{-1}(\sqrt{x}, k) \quad . \quad . \quad . \quad (12)$$

**Art. 25.** Suppose that  $\alpha$ ,  $\beta$ , and  $\gamma$  are real quantities such that  $\alpha > \beta > \gamma$ ; further write  $M = \frac{\sqrt{\alpha - \gamma}}{2}$ ,  $k_1^2 = \frac{\beta - \gamma}{\alpha - \gamma}$  and  $k_2^2 = \frac{\alpha - \beta}{\alpha - \gamma}$ , where  $k_1^2 + k_2^2 = 1$ , so that the one is the complementary modulus of the other. Put  $X = (x - \alpha)(x - \beta)(x - \gamma)$ .

If  $\infty > x > \alpha > \beta > \gamma$ , write  $x - \gamma = (\alpha - \gamma) \operatorname{cosec}^2 \phi$  and we have

$$M \int_x^\infty \frac{dx}{\sqrt{X}} = sn^{-1} \left( \sqrt{\frac{\alpha - \gamma}{x - \gamma}}, k_1 \right) = cn^{-1} \left( \sqrt{\frac{x - \alpha}{x - \gamma}}, k_1 \right). \quad (13)$$

When  $\infty > x > \alpha > \beta > \gamma$ , it is seen that

$$M \int_{\alpha}^x \frac{dx}{\sqrt{X}} = sn^{-1} \left( \sqrt{\frac{x-\alpha}{x-\beta}}, k_1 \right) = cn^{-1} \left( \sqrt{\frac{\alpha-\beta}{x-\beta}}, k_1 \right), \quad (14)$$

and when  $\beta > x > \gamma$ , we have

$$\begin{aligned} M \int_x^{\beta} \frac{dx}{\sqrt{X}} &= sn^{-1} \left[ \sqrt{\frac{(\alpha-\gamma)(\beta-x)}{(\beta-\gamma)(\alpha-x)}}, k_1 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\alpha-\beta)(x-\gamma)}{(\beta-\gamma)(\alpha-x)}}, k_1 \right]. \end{aligned} \quad (15)$$

Further if  $\beta > x > \gamma$ , then

$$\begin{aligned} M \int_{\gamma}^x \frac{dx}{\sqrt{X}} &= sn^{-1} \left( \sqrt{\frac{x-\gamma}{\beta-\gamma}}, k_1 \right) = cn^{-1} \left( \sqrt{\frac{\beta-x}{\beta-\gamma}}, k_1 \right) \\ &= dn^{-1} \left( \sqrt{\frac{\alpha-x}{\alpha-\gamma}}, k_1 \right). \end{aligned} \quad (16)$$

Art. 26. As above write

$$M = \frac{\sqrt{\alpha-\gamma}}{2}, \quad k_2^2 = \frac{\alpha-\beta}{\alpha-\gamma}, \quad X = (x-\alpha)(x-\beta)(x-\gamma).$$

For the interval  $\alpha > x > \beta > \gamma$ , it is seen that

$$M \int_x^{\alpha} \frac{dx}{\sqrt{-X}} = sn^{-1} \left[ \sqrt{\frac{\alpha-x}{\alpha-\beta}}, k_2 \right] = cn^{-1} \left[ \sqrt{\frac{x-\beta}{\alpha-\beta}}, k_2 \right], \quad (17)$$

and for the same interval

$$\begin{aligned} M \int_{\beta}^x \frac{dx}{\sqrt{-X}} &= sn^{-1} \left[ \sqrt{\frac{(\alpha-\gamma)(x-\beta)}{(\alpha-\beta)(x-\gamma)}}, k_2 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\beta-\gamma)(\alpha-x)}{(\alpha-\beta)(x-\gamma)}}, k_2 \right]. \end{aligned} \quad (18)$$

Further, if  $\gamma > x > -\infty$ , then

$$M \int_x^{\gamma} \frac{dx}{\sqrt{-X}} = sn^{-1} \left[ \sqrt{\frac{\gamma-x}{\beta-x}}, k_2 \right] = cn^{-1} \left[ \sqrt{\frac{\beta-\gamma}{\beta-x}}, k_2 \right], \quad (19)$$

and for the same interval

$$M \int_{-\infty}^x \frac{dx}{\sqrt{-X}} = sn^{-1} \left( \sqrt{\frac{\alpha-\gamma}{\alpha-x}}, k_2 \right) = cn^{-1} \left( \sqrt{\frac{\gamma-x}{\alpha-x}}, k_2 \right). \quad (20)$$

Art. 27. From formula (14) it is seen that, if  $\infty > x > \frac{1}{k^2}$ ,

$$\begin{aligned} \int_{\frac{1}{k^2}}^x \frac{dx}{\sqrt{x(x-1)(k^2x-1)}} &= \frac{1}{k} \int_{\frac{1}{k^2}}^x \frac{dx}{\sqrt{x(x-1)(x-1/k^2)}} \\ &= 2sn^{-1}\left(\sqrt{\frac{x-1/k^2}{x-1}}, k\right) = 2cn^{-1}\left(\sqrt{\frac{1-k^2}{k^2(x-1)}}, k\right), \quad (21) \end{aligned}$$

and from formula (13) for the same interval,

$$\int_x^\infty \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} = 2sn^{-1}\left(\sqrt{\frac{1}{k^2x}}, k\right) = 2cn^{-1}\left(\sqrt{\frac{k^2x-1}{k^2x}}, k\right) \quad (22).$$

Using formula (17), it follows that, if  $\frac{1}{k^2} > x > 1$ ,

$$\begin{aligned} \int_x^{\frac{1}{k^2}} \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} &= 2i sn^{-1}\left(\sqrt{\frac{1-k^2x}{1-k^2}}, k'\right) \\ &= 2i cn^{-1}\left(\sqrt{\frac{k^2(x-1)}{1-k^2}}, k'\right), \quad (23) \end{aligned}$$

and for the same interval (see formula (18)),

$$\begin{aligned} \int_1^x \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} &= 2i sn^{-1}\left(\sqrt{\frac{x-1}{x(1-k^2)}}, k'\right) \\ &= 2i cn^{-1}\left(\sqrt{\frac{1-k^2x}{x(1-k^2)}}, k'\right). \quad (24) \end{aligned}$$

If  $0 > x > -\infty$ , the formula (19) offers

$$\begin{aligned} \int_x^0 \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} &= 2i sn^{-1}\left(\sqrt{\frac{-x}{1-x}}, k'\right) \\ &= 2i cn^{-1}\left(\sqrt{\frac{1}{1-x}}, k'\right); \quad (25) \end{aligned}$$

while for the same interval it follows from formula (20) that

$$\begin{aligned} \int_{-\infty}^x \frac{dx}{\sqrt{x(1-x)(1-k^2x)}} &= 2i sn^{-1}\left(\sqrt{\frac{1}{1-k^2x}}, k'\right) \\ &= 2i cn^{-1}\left(\sqrt{\frac{-k^2x}{1-k^2x}}, k'\right). \quad (26) \end{aligned}$$



**Art. 28.** Next let  $X = (x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$  and further put

$$N = \frac{\sqrt{(\alpha - \gamma)(\beta - \delta)}}{2}, \quad k_3^2 = \frac{(\beta - \gamma)(\alpha - \delta)}{(\alpha - \gamma)(\beta - \delta)}, \quad k_4^2 = \frac{(\alpha - \beta)(\gamma - \delta)}{(\alpha - \gamma)(\beta - \delta)},$$

and note that  $k_3^2 + k_4^2 = 1$ .

If then  $\infty > x > \alpha$ , there results, supposing always that  $\alpha > \beta > \gamma > \delta$ ,

$$\begin{aligned} N \int_{\alpha}^x \frac{dx}{\sqrt{X}} &= sn^{-1} \left[ \sqrt{\frac{(\beta - \delta)(x - \alpha)}{(\alpha - \delta)(x - \beta)}}, k_3 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\alpha - \beta)(x - \delta)}{(\alpha - \delta)(x - \beta)}}, k_3 \right]; \quad (27) \end{aligned}$$

and if  $\alpha > x > \beta$

$$\begin{aligned} N \int_x^{\alpha} \frac{dx}{\sqrt{-X}} &= sn^{-1} \left[ \sqrt{\frac{(\beta - \delta)(\alpha - x)}{(\alpha - \beta)(x - \delta)}}, k_4 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\alpha - \delta)(x - \beta)}{(\alpha - \beta)(x - \delta)}}, k_4 \right]. \quad (28) \end{aligned}$$

If  $\alpha > x > \beta$ ,

$$\begin{aligned} N \int_{\beta}^x \frac{dx}{\sqrt{-X}} &= sn^{-1} \left[ \sqrt{\frac{(\alpha - \gamma)(x - \beta)}{(\alpha - \beta)(x - \gamma)}}, k_4 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\beta - \gamma)(\alpha - x)}{(\alpha - \beta)(x - \gamma)}}, k_4 \right]; \quad (29) \end{aligned}$$

while if  $\beta > x > \gamma$ ,

$$\begin{aligned} N \int_x^{\beta} \frac{dx}{\sqrt{X}} &= sn^{-1} \left[ \sqrt{\frac{(\alpha - \gamma)(\beta - x)}{(\beta - \gamma)(\alpha - x)}}, k_3 \right] \\ &= cn^{-1} \left[ \sqrt{\frac{(\alpha - \beta)(x - \gamma)}{(\beta - \gamma)(\alpha - x)}}, k_3 \right]. \quad (30) \end{aligned}$$

When  $x$  lies within the interval  $\beta > x > \gamma$ ,

$$\begin{aligned} N \int_{\gamma}^x \frac{dx}{\sqrt{X}} &= sn^{-1} \left( \sqrt{\frac{(\beta - \delta)(x - \gamma)}{(\beta - \gamma)(x - \delta)}}, k_3 \right) \\ &= cn^{-1} \left( \sqrt{\frac{(\gamma - \delta)(\beta - x)}{(\beta - \gamma)(x - \delta)}}, k_3 \right); \quad (31) \end{aligned}$$

and when  $\gamma > x > \delta$ , it is seen that

$$\begin{aligned} N \int_x^\gamma \frac{dx}{\sqrt{(\alpha-x)(\beta-x)(\gamma-x)(x-\delta)}} &= sn^{-1} \left( \sqrt{\frac{(\beta-\delta)(\gamma-x)}{(\gamma-\delta)(\beta-x)}}, k_4 \right) \\ &= cn^{-1} \left( \sqrt{\frac{(\beta-\gamma)(x-\delta)}{(\gamma-\delta)(\beta-x)}}, k_4 \right). \quad (32) \end{aligned}$$

If  $\gamma > x > \delta$ ,

$$\begin{aligned} N \int_\delta^x \frac{dx}{\sqrt{-X}} &= sn^{-1} \left( \sqrt{\frac{(\alpha-\gamma)(x-\delta)}{(\gamma-\delta)(\alpha-x)}}, k_4 \right) \\ &= cn^{-1} \left( \sqrt{\frac{(\alpha-\delta)(\gamma-x)}{(\gamma-\delta)(\alpha-x)}}, k_4 \right), \quad (33) \end{aligned}$$

and if  $\delta > x > -\infty$

$$\begin{aligned} N \int_0^\delta \frac{dx}{\sqrt{X}} &= sn^{-1} \left( \sqrt{\frac{(\alpha-\gamma)(\delta-x)}{(\alpha-\delta)(\gamma-x)}}, k_3 \right) \\ &= cn^{-1} \left( \sqrt{\frac{(\gamma-\delta)(\alpha-x)}{(\alpha-\delta)(\gamma-x)}}, k_3 \right). \quad (34) \end{aligned}$$

**Art. 29.** By means of the above formulas it is possible to integrate the reciprocal of the square root of any cubic or biquadratic which has real roots; for example (see Byerly, *Integral Calculus*, 1902, p. 276),

$$\begin{aligned} \int_{\frac{a}{2}}^a \frac{dx}{\sqrt{(2ax-x^2)(a^2-x^2)}} &= \int_0^a \frac{dx}{\sqrt{(2a-x)(a-x)x(a+x)}} \\ &\quad - \int_{\frac{a}{2}}^a \frac{dx}{\sqrt{(2a-x)(a-x)x(a+x)}} = \frac{1}{a} \left[ sn^{-1} \left( 1, \frac{\sqrt{3}}{2} \right) \right. \\ &\quad \left. - sn^{-1} \left( \frac{\sqrt{6}}{3}, \frac{\sqrt{3}}{2} \right) \right] \quad [\text{cf. (30)}] \\ &= \frac{1}{a} F \left( \frac{\sqrt{3}}{2}, \sin^{-1} 1 \right) - \frac{1}{a} F \left( \frac{\sqrt{3}}{2}, \sin^{-1} \frac{\sqrt{6}}{3} \right). \end{aligned}$$

Remark.—In the above integrals it is well to note that (34), for example, may be written

$$N \int_x^\delta \frac{dx}{\sqrt{(\alpha-x)(\beta-x)(\gamma-x)(\delta-x)}},$$

showing that each factor under the root sign is positive for the interval in question.

Art. 30. It is seen that the substitution

$$\frac{\alpha-\gamma}{x-\gamma} = \frac{y-\gamma}{\beta-\gamma}, \quad \text{or} \quad \frac{x-\alpha}{x-\gamma} = \frac{\beta-\gamma}{\beta-\gamma} \quad \text{or} \quad \frac{x-\beta}{x-\gamma} = \frac{\alpha-\gamma}{\alpha-\gamma}$$

changes

$$\int_x^\infty \frac{dx}{\sqrt{(x-\alpha)(x-\beta)(x-\gamma)}} \quad \text{into} \quad \int_\gamma^y \frac{dy}{\sqrt{(y-\alpha)(y-\beta)(y-\gamma)}},$$

or (13) into (16). For example,

$$\begin{aligned} \int_\alpha^\infty \frac{dx}{\sqrt{(x-\alpha)(x-\beta)(x-\gamma)}} &= \int_\gamma^\beta \frac{dy}{\sqrt{(y-\alpha)(y-\beta)(y-\gamma)}} \\ &= \frac{2K}{\sqrt{\alpha-\gamma}}, \quad (35) \end{aligned}$$

where  $k^2 = \frac{\beta-\gamma}{\alpha-\gamma}$ , see (16).

By the same substitution (14) becomes (15).

Similarly the substitution

$$\frac{\alpha-x}{\alpha-\beta} = \frac{\alpha-\gamma}{\alpha-\gamma}, \quad \text{or} \quad \frac{x-\beta}{\alpha-\beta} = \frac{\gamma-\gamma}{\alpha-\gamma}, \quad \text{or} \quad \frac{x-\gamma}{\alpha-\gamma} = \frac{\beta-\gamma}{\alpha-\gamma}$$

changes (17) into (20) and shows that

$$\begin{aligned} \int_\beta^\alpha \frac{dx}{\sqrt{(\alpha-x)(x-\beta)(x-\gamma)}} &= \int_{-\infty}^\gamma \frac{dy}{\sqrt{(\alpha-y)(\beta-y)(\gamma-y)}} \\ &= \frac{2K'}{\sqrt{\alpha-\gamma}} \quad (36) \end{aligned}$$

where  $\frac{\alpha-\beta}{\alpha-\gamma} = k^2$ .

By the same substitution (18) becomes (19).

**Art. 31.** Let the roots of the cubic be one real and two imaginary, so that  $X$  has the form  $(x-\alpha)[(x-\rho)^2+\sigma^2]$ .

Make the substitution

$$y = \frac{X}{(x-\alpha)^2} = \frac{(x-\rho)^2 + \sigma^2}{x-\alpha}, \quad \text{or}$$

(1)  $(x-\rho)^2 + \sigma^2 - y(x-\alpha) = 0$ , which is an hyperbola.

The condition that this quadratic in  $x$  have equal roots, is

(2)  $y^2 + 4(\rho-\alpha)y - 4\sigma^2 = 0$ .

The roots of this equation are, say,

$$(y_1, y_2) = -2(\rho-\alpha) \pm 2\sqrt{(\rho-\alpha)^2 + \sigma^2}.$$

It is evident that  $y_1$  is positive and  $y_2$  is negative.

If we eliminate  $y$  from (1) and (2), we have the biquadratic

$$[(x-\rho)^2 + \sigma^2]^2 + 4(\rho-\alpha)(x-\alpha)[(x-\rho)^2 + \sigma^2] - 4\sigma^2(x-\alpha)^2 = 0,$$

the left hand side being, as we know *a priori*, a perfect square.

Equating to zero one of these double factors, we have

$$(3) \quad x^2 - 2\alpha x + 2\alpha\rho - \rho^2 - \sigma^2 = 0.$$

Further let  $x_1, x_2$  denote the values of  $x$  which correspond to the values  $y_1, y_2$  of  $y$ .

From (3) it follows that

$$(x_1, x_2) = \alpha \pm \sqrt{(\alpha-\rho)^2 + \sigma^2},$$

or

$$x_1 = \rho + \frac{1}{2}y_1, \quad x_2 = \rho + \frac{1}{2}y_2.$$

Further there results

$$y - y_1 = \frac{(x-x_1)^2}{x-\alpha}, \quad y - y_2 = \frac{(x-x_2)^2}{x-\alpha},$$

and

$$\frac{dy}{dx} = \frac{(x-x_1)(x-x_2)}{(x-\alpha)^2}.$$

It follows at once that

$$\begin{aligned}\int_x^\infty \frac{dx}{\sqrt{X}} &= \int_x^\infty \frac{dx}{(x-\alpha)\sqrt{y}} = \int_x^\infty \frac{(x-\alpha)dy}{(x-x_1)(x-x_2)\sqrt{y}} \\ &= \int_y^\infty \frac{dy}{\sqrt{y(y-y_1)(y-y_2)}} = \frac{2}{\sqrt{y_1-y_2}} cn^{-1} \left( \sqrt{\frac{y-y_1}{y-y_2}}, \sqrt{\frac{-y_2}{y_1-y_2}} \right) \\ &\quad (\text{cf. (13)}) = \frac{\sqrt{2}}{\sqrt{x_1-x_2}} cn^{-1} \left( \frac{x-x_1}{x-x_2}, k \right), \quad \dots \quad (37)\end{aligned}$$

where  $k^2 = \frac{-y_2}{y_1-y_2}$  and  $k'^2 = \frac{y_1}{y_1-y_2}$ .

In the same way, with the same substitutions, it may be proved that

$$\begin{aligned}\int_{-\infty}^x \frac{dx}{\sqrt{(\alpha-x)[(x-\rho)^2+\sigma^2]}} &= \int_{-\infty}^y \frac{dy}{\sqrt{-y(y_1-y)(y_2-y)}} \\ &= \frac{2}{\sqrt{y_1-y_2}} cn^{-1} \left( \sqrt{\frac{y_2-y}{y_1-y}}, k' \right)\end{aligned}$$

[cf. (20), where  $k'^2 = \frac{y_1}{y_1-y_2}$  is the complementary modulus of the preceding integral], or

$$\int_{-\infty}^x \frac{dx}{\sqrt{-X}} = \frac{\sqrt{2}}{\sqrt{x_1-x_2}} cn^{-1} \left( \frac{x_2-x}{x_1-x}, k' \right), \quad \dots \quad (38)$$

Further write  $M^2 = (\rho-\alpha)^2 + \sigma^2$ , so that  $x_1 = \alpha + M$  and  $x_2 = \alpha - M$ . It is evident that

$$\begin{aligned}\int_\alpha^x \frac{dx}{\sqrt{(x-\alpha)[(x-\rho)^2+\sigma^2]}} &= \int_\infty^y \frac{dy}{\sqrt{y(y-y_1)(y-y_2)}} \\ &= \frac{\sqrt{2}}{\sqrt{x_1-x_2}} cn^{-1} \left( \frac{x_1-x}{x-x_2}, k \right), \text{ cf. (37),} \\ &= \frac{1}{\sqrt{M}} cn^{-1} \left[ \frac{M-(x-\alpha)}{M+(x-\alpha)}, k \right], k^2 = \frac{1}{2} - \frac{1}{2} \frac{\alpha-\rho}{M}. \quad \dots \quad (39)\end{aligned}$$



Similarly, it may be shown that

$$\int_x^\alpha \frac{dx}{\sqrt{(\alpha-x)[(x-\rho)^2+\sigma^2]}} = \frac{1}{\sqrt{M}} cn^{-1} \left( \frac{M-(\alpha-x)}{M+(\alpha-x)}, k' \right), \quad (40)$$

where

$$k'^2 = \frac{1}{2} + \frac{1}{2} \frac{\alpha-\rho}{M}.$$

Note that the modulus here is the complementary modulus of the one in (39) and that the product of the two moduli is, say,

$$2kk' = \frac{\sigma}{M}.$$

As numerical examples, prove that

$$\begin{aligned} \int_x^\infty \frac{dx}{\sqrt{x^3-1}} &= \frac{1}{\sqrt[4]{3}} cn^{-1} \left( \frac{x-1-\sqrt{3}}{x-1+\sqrt{3}}, k_1 \right), \\ \int_1^x \frac{dx}{\sqrt{x^3-1}} &= \frac{1}{\sqrt[4]{3}} cn^{-1} \left( \frac{\sqrt{3}+1-x}{\sqrt{3}-1+x}, k_1 \right), \\ \int_x^1 \frac{dx}{\sqrt{1-x^3}} &= \frac{1}{\sqrt[4]{3}} cn^{-1} \left( \frac{\sqrt{3}-1+x}{\sqrt{3}+1-x}, k_2 \right), \\ \int_{-\infty}^x \frac{dx}{\sqrt{1-x^3}} &= \frac{1}{\sqrt[4]{3}} cn^{-1} \left( \frac{1-x-\sqrt{3}}{1-x+\sqrt{3}}, k_2 \right), \end{aligned}$$

where  $2k_1k_2 = \frac{1}{2} = \sin 30^\circ$ ,  $k_1 = \sin 15^\circ$ ,  $k_2 = \sin 75^\circ$ .

(Greenhill, loc. cit., p. 40.)

**Art. 32.** Suppose next that we have a quartic with two imaginary roots. It is always possible to write

$$X = (ax^2 + 2bx + c)(Ax^2 + 2Bx + C),$$

where the real roots constitute the first factor, and the imaginary roots the second so that  $b^2 - ac$  is positive and  $B^2 - AC$  is negative.

Make the substitution

$$y = \frac{ax^2 + 2bx + c}{Ax^2 + 2Bx + C} = \frac{N}{D}, \text{ say, } \dots \dots \dots (i)$$

or

$$(a) \quad x^2(Ay-a) + 2x(By-b) + Cy-c = 0.$$

This equation has equal roots in  $x$ , if

$$(b) \quad (By-b)^2 - (Ay-a)(Cy-c) = 0.$$

Let the roots of this equation be  $y_1$  and  $y_2$ .

From (a) it is seen that

$$[2x(By-b)]^2 = x^4(Ay-a)^2 + 2x^2(Ay-a)(Cy-c) + (Cy-c)^2,$$

which combined with (b), gives

$$(c) \quad -x = \frac{Cy-c}{By-b} = \frac{By-b}{Ay-a}, \quad Ax+B = \frac{Ab-aB}{Ay-a},$$

$$(d) \quad y = \frac{ax+b}{Ax+B} = \frac{bx+c}{Bx+C}, \quad Ay-a = \frac{(Ab-aB)x + Ac-aC}{Bx+C}.$$

From (i) it follows, if  $D$  is put for  $Ax^2 + 2Bx + C$ , and since

$Ax_1^2 + 2Bx_1 + C \equiv x_1(Ax_1+B) + Bx_1 + C$ , that

$$y_1 - y = \frac{x - x_1}{D} \frac{2(Ab - Ba)xx_1 + (Ac - aC)(x + x_1) + 2(Bc - bC)}{x_1(Ax_1+B) + (Bx_1+C)},$$

which, see (c) and (d),

$$= \frac{x - x_1}{D} (Ay_1 - a) \frac{x \{ 2(Ab - aB)x_1 + Ac - aC \} + x_1(Ac - aC) + 2(Bc - bC)}{x_1(Ab - aB) + x_1(Ab - aB) + Ac - aC},$$

so that

$$y_1 - y = \frac{x - x_1}{D} (Ay_1 - a)(x - x_1);$$

and similarly

$$y - y_2 = \frac{(a - Ay_2)(x - x_2)^2}{D}$$

and

$$\frac{dy}{dx} = \frac{2(Ab - aB)(x_1 - x)(x - x_2)}{D^2}.$$

It follows that

$$\begin{aligned} & \frac{dx}{\sqrt{(ax^2+2bx+c)(Ax^2+2Bx+C)}} \\ &= \frac{dX}{D\sqrt{y}} = \frac{Ddy}{2(Ab-Ba)(x_1-x)(x-x_2)\sqrt{y}} \\ &= \frac{\sqrt{(Ay_1-a)(a-Ay_2)}}{2(Ab-aB)} \frac{dy}{\sqrt{y(y_1-y)(y-y_2)}}. \end{aligned}$$

Noting that

$$(Ay_1-a)(a-Ay_2) = -A^2y_1y_2 + Aa(y_1+y_2) - a^2 = \frac{(Ab-aB)^2}{AC-B^2},$$

it follows that

$$(e) \quad \frac{dx}{\sqrt{X}} = \frac{1}{\sqrt{AC-B^2}} \frac{dy}{\sqrt{4y(y_1-y)(y-y_2)}}.$$

From (b) it is seen that  $y_1 > 0$  and  $y_2 < 0$ , and from (e) it is evident that  $y$  varies from 0 to  $y_1$  for real values of  $\sqrt{X}$ . Hence, see (17),

$$\int_x^{x_1} \frac{dx}{\sqrt{X}} = \frac{1}{2\sqrt{AC-B^2}} \int_y^{y_1} \frac{dy}{\sqrt{y(y_1-y)(y-y_2)}},$$

or,

$$\sqrt{y_1-y_2} \sqrt{AC-B^2} \int_x^{x_1} \frac{dx}{\sqrt{X}} = sn^{-1} \left( \sqrt{\frac{y_1-y}{y_1}}, k \right) = cn^{-1} \left( \sqrt{\frac{y}{y_1}}, k \right) \quad (41)$$

$$\text{where } k^2 = \frac{y_1}{y_1-y_2} \text{ and } k'^2 = \frac{-y_2}{y_1-y_2}.$$

**Art. 33.** Suppose next in the quartic

$$X = (ax^2+2bx+c)(Ax^2+2Bx+C),$$

that all the roots are imaginary so that  $b^2-ac < 0$  and  $B^2-AC < 0$ . In this case the roots  $y_1$  and  $y_2$  of the equation of the preceding article

$$(AC-B^2)y^2 - (Ac+aC-2Bb)y + ac-b^2 = 0$$

are both positive.

Hence the integral of the equation (e) may be written [cf. (17)] in the form

$$\begin{aligned}\sqrt{AC-B^2} \int_x^{x_1} \frac{dx}{\sqrt{X}} &= \frac{1}{2} \int_v^{v_1} \frac{dy}{\sqrt{-y(y-y_1)(y-y_2)}} \\ &= \frac{1}{\sqrt{y_1}} sn^{-1} \left( \sqrt{\frac{y_1-y}{y_1-y_2}}, k \right) = \frac{1}{\sqrt{y_1}} cn^{-1} \left( \sqrt{\frac{y-y_2}{y_1-y_2}}, k \right) \\ &= \frac{1}{\sqrt{y_1}} dn^{-1} \left( \sqrt{\frac{y}{y_1}}, k \right) \quad \dots \dots \dots (42)\end{aligned}$$

where

$$k^2 = 1 - \frac{y_2}{y_1}, \quad k'^2 = \frac{y_2}{y_1}$$

and where  $y$  oscillates between the two positive values  $y_1$  and  $y_2$ .

**Art. 34.** As an example of the preceding article, let

$$X = x^4 + 2v^2x^2 \cos 2\omega + v^4 = (x^2 + 2vx \sin \omega + v^2)(x^2 - 2vx \sin \omega + v^2).$$

If we put

$$y = \frac{x^2 + 2vx \sin \omega + v^2}{x^2 - 2vx \sin \omega + v^2},$$

it is seen that

$$y_1 = \tan^2 \left( \frac{\pi}{4} + \frac{\omega}{2} \right), \quad y_2 = \tan^2 \left( \frac{\pi}{4} - \frac{\omega}{2} \right), \quad x_1 = v, \quad x_2 = -v,$$

$$k = \frac{1 - \sin \omega}{1 + \sin \omega} = \tan^2 \left( \frac{\pi}{4} - \frac{\omega}{2} \right),$$

and

$$\begin{aligned}\int_x^v \frac{dx}{\sqrt{x^4 + 2v^2x^2 \cos 2\omega + v^4}} \\ = \frac{1}{v(1 + \sin \omega)} dn^{-1} \sqrt{\frac{1 - \sin \omega}{1 + \sin \omega} \cdot \frac{x^2 + 2vx \sin \omega + v^2}{x^2 - 2vx \sin \omega + v^2}} \quad (43)\end{aligned}$$

When  $\omega = \frac{\pi}{4}$ ,  $v = 1$ , the preceding equation becomes

$$\int_x^1 \frac{dx}{\sqrt{1+x^4}} = (2 - \sqrt{2}) dn^{-1} \left\{ (\sqrt{2} - 1) \sqrt{\frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1}}, k \right\} \quad (44)$$

where  $k = (\sqrt{2} - 1)^2$ .

For the substitution  $\frac{x^2}{v^2} = \frac{1+z}{1-z}$ , there results

$$\int_x^\infty \frac{dx}{\sqrt{x^4 + 2v^2x^2 \cos 2\omega + v^4}} = \frac{1}{2v} \int_z^1 \frac{dz}{\sqrt{(1-z^2)(\cos^2 \omega + z^2 \sin^2 \omega)}},$$

which, see (2),

$$= \frac{1}{2v} cn^{-1}(z, \sin \omega) = \frac{1}{2v} cn^{-1}\left(\frac{x^2 - v^2}{x^2 + v^2}, \sin \omega\right).$$

If in this formula we put  $\omega = \frac{1}{4}\pi$  and  $v = 1$ , we have

$$\int_x^\infty \frac{dx}{\sqrt{x^4 + 1}} = \frac{1}{2} cn^{-1}\left(\frac{x^2 - 1}{x^2 + 1}, \frac{1}{2}\sqrt{2}\right),$$

$$\int_0^x \frac{dx}{\sqrt{1 + x^4}} = \frac{1}{2} cn^{-1}\left(\frac{1 - x^2}{1 + x^2}, \frac{1}{2}\sqrt{2}\right).$$

**Art. 35.** It was shown above that the substitution

$$\sin^2 \phi = \frac{1 - x^2}{1 - k^2 x^2}$$

transforms the integral

$$(A) \quad \int_x^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \text{ into } sn^{-1}\left(\sqrt{\frac{1-x^2}{1-k^2x^2}}, k\right).$$

On the other hand

$$(B) \quad \int_x^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} - \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = K - u,$$

where

$$u = \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}.$$

It follows that

$$sn^{-1}\sqrt{\frac{1-x^2}{1-k^2x^2}} = K - sn^{-1}x,$$



a relation among the integrals. It is also at once evident that

$$\sqrt{\frac{1-x^2}{1-k^2x^2}} = \operatorname{sn}(K-u), \quad \text{or} \quad \frac{cn u}{dn u} = \operatorname{sn}(K-u),$$

which is a relation among the functions.

In (A) make  $k=0$ , and then

$$\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} \sqrt{1-x^2},$$

and from (B) it is seen that

$$\int_x^1 \frac{dx}{\sqrt{1-x^2}} = \int_0^1 \frac{dx}{\sqrt{1-x^2}} - \int_0^x \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2} - u,$$

if 
$$u = \int_0^x \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x.$$

Hence

$$\sin^{-1} \sqrt{1-x^2} = \frac{\pi}{2} - \sin^{-1} x.$$

a relation among the integrals; and on the other hand it is seen that

$$\sqrt{1-\sin^2 u} = \sin\left(\frac{\pi}{2} - u\right),$$

a relation among the functions.

It is thus made evident that we may study the nature of the elliptic functions and their characteristic properties directly from their associated integrals just as we may study the properties of the circular, hyperbolic, logarithmic and exponential functions from their associated integrals. This should be emphasized both in the study of the elementary calculus and in the theory of elliptic integrals and elliptic functions.

**Art. 36.** In the applications of the elementary calculus it was often necessary to evaluate such integrals as  $\int \sin u \, du$ ; so here we must study the integrals of the most usual elliptic functions. From the integral  $u = \int_0^\phi \frac{d\phi}{\Delta\phi}$ , it is seen at once that

$du = \frac{d\phi}{\Delta\phi}$ , or  $dn u du = d\phi$ , so that  $d am u = dn u du$ ,  $d sn u = cn u dn u du$ ,  $d cn u = -sn u dn u du$ ,  $d dn u = -k^2 cn u sn u du$ . We further note that

$$sn^2 u + cn^2 u = 1, \quad dn^2 u - k'^2 = k^2 cn^2 u, \quad dn^2 u + k^2 sn^2 u = 1.$$

We have without difficulty

$$\int sn u du = -\frac{1}{k^2} \int \frac{-k^2 sn u cn u du}{cn u} = -\frac{1}{k} \int \frac{dv}{\sqrt{v^2 - k'^2}},$$

(if  $v = dn u$ ). The last integral is

$$-\frac{1}{k} \log (v + \sqrt{v^2 - k'^2}) = -\frac{1}{k} \cosh^{-1} \frac{v}{k'} = -\frac{1}{k} \cosh^{-1} \left( \frac{dn u}{k'} \right).$$

Further since  $dn K = k'$ , Art. 17, we have

$$k \int_u^K sn u du = \cosh^{-1} \left( \frac{dn u}{k'} \right) = \sinh^{-1} \left( k \frac{cn u}{k'} \right) = \log \frac{dn u + k cn u}{k'}.$$

Similarly it may be proved that

$$k \int_0^u cn u du = \cos^{-1}(dn u) = \sin^{-1}(k sn u),$$

and

$$\int_0^u dn u du = \phi = am u = \sin^{-1} sn u = \cos^{-1} cn u.$$

**Art. 37.** The following integrals should be noted:

$$\int \frac{du}{sn u} = \int \frac{sn u cn u dn u du}{sn^2 u cn u dn u} = \frac{1}{2} \int \frac{dv}{v \sqrt{(1-v)(1-k^2v)}} \quad (\text{if } v = sn^2 u).$$

Further writing  $\sqrt{(1-v)(1-k^2v)} = (1-v)z$ , the last integral becomes

$$\begin{aligned} & -\frac{1}{2} \log \left[ \frac{\sqrt{(1-v)(1-k^2v)} + 1}{v} - \frac{1+k^2}{2} \right] - \frac{1}{2} \log \frac{1-k^2}{2} \\ & = -\frac{1}{2} \log \left[ \frac{cn u dn u + 1}{sn^2 u} - \frac{1+k^2}{2} \right] + C \\ & = -\frac{1}{2} \log \left[ \frac{2cn u dn u + cn^2 u + dn^2 u}{2sn^2 u} \right] + C, \end{aligned}$$

so that, omitting  $C$ ,

$$\int \frac{du}{sn u} = \log \left[ \frac{sn u}{cn u + dn u} \right],$$

where the arbitrary constant is omitted. Similarly it may be shown that

$$\int \frac{du}{cn u} = \frac{1}{k'} \log \left[ \frac{k' sn u + dn u}{cn u} \right],$$

and that

$$\int \frac{du}{dn u} = \frac{1}{2k'} \sin^{-1} \left[ \frac{k'^2 sn^2 u - cn^2 u}{dn^2 u} \right].$$

Further by definition  $E(k, \phi) = \int_0^\phi \Delta \phi d\phi$  (cf. Art. 5), or since  $\phi = am u$  and  $d am u = dn u du$ ,

$$E(am u) = \int_0^u dn^2 u du.$$

It follows that

$$\int_0^u sn^2 u du = \frac{1}{k^2} [u - E(am u, k)],$$

and

$$\int_0^u cn^2 u du = \frac{1}{k^2} [E(am u, k) - k'^2 u].$$

**Art. 38. Reduction formulas.** The following is a very useful and a very general reduction formula.\* Consider the identity

$$\begin{aligned} (m + \sin^2 \phi)^\mu \sin \phi \cos \phi \Delta \phi &= \int_0^\phi \frac{d}{d\phi} \{ (m + \sin^2 \phi)^\mu \sin \phi \cos \phi \Delta \phi \} d\phi \\ &= \int_0^\phi \{ 2\mu (m + \sin^2 \phi)^{\mu-1} \sin^2 \phi \cos^2 \phi \Delta^2 \phi \\ &\quad + (m + \sin^2 \phi)^\mu [\cos^2 \phi \Delta^2 \phi - \sin^2 \phi \Delta^2 \phi - k^2 \sin^2 \phi \cos^2 \phi] \} \frac{d\phi}{\Delta \phi}. \end{aligned}$$

In this expression put  $m + \sin^2 \phi = v$ , so that  $\sin^2 \phi = v - m$ ,  $\cos^2 \phi = 1 - v + m$ ,  $\Delta^2 \phi = 1 - k^2 v + k^2 m$ , and writing

$$V_\mu = \int_0^\phi \frac{v^\mu d\phi}{\Delta \phi} = \int_0^\phi \frac{(m + \sin^2 \phi)^\mu d\phi}{\Delta \phi},$$

\* See, for example, Durège, *Elliptische Funktionen*, § 4, Second edition.

then there is found

$$(m + \sin^2 \phi)^\mu \sin \phi \cos \phi \Delta \phi = -2\mu A V_{\mu-1} + (2\mu + 1) B V_\mu - (2\mu + 2) C V_{\mu+1} + (2\mu + 3) k^2 V_{\mu+2}, \quad (i)$$

where  $A = m(1+m)(1+k^2m)$ ;  $B = 1 + 2m + 2k^2m + 3k^2m^2$ ;  $C = 1 + k^2 + 3k^2m$ .

From this formula it is evident that every integral  $V_\mu$  may be expressed through the three integrals  $V_0$ ,  $V_1$ ,  $V_{-1}$ , the latter being forms of integrals which in Chapter I have been called *elliptic integrals of the first, second and third kinds* respectively.

The following formulas may be derived immediately from the formula above, by writing

$$S_m(u) = \int sn^m u \, du, \quad C_m(u) = \int cn^m u \, du, \quad D_m(u) = \int dn^m u \, du,$$

$$(n+1)k^2 S_{n+2}(u) - n(1+k^2)S_n(u) + (n-1)S_{n-2}(u) = sn \, cn \, u \, dn \, u, \quad (ii)$$

$$(n+1)k^2 C_{n+2}(u) + n(k'^2 - k^2)C_n(u) - (n-1)k'^2 C_{n-2}(u) = cn \, sn \, u \, dn \, u, \quad (iii)$$

$$(n+1)D_{n+2}(u) - n(1+k'^2)D_n(u) + (n-1)k'^2 D_{n-2}(u) = k^2 dn \, sn \, u \, cn \, u. \quad (iv)$$

In particular, if  $u = K$  say in (ii), there results

$$(n+1)k^2 S_{n+2}(K) - n(1+k^2)S_n(K) + (n-1)S_{n-2}(K) = 0,$$

which is the analogue of Wallis's formula for  $\int_0^{\frac{\pi}{2}} \sin^n \theta \, d\theta$ .

**Art. 39.** It may be noted that any of the quantities  $F(\sin^2 \phi)$ ,  $F(\cos^2 \phi)$ ,  $F(\tan^2 \phi)$ , where  $F$  is a rational function of its argument, may be expressed through an aggregate of terms of the form  $M(m + \sin^2 \phi)^\mu$ , where  $\mu$  is a positive or negative integer or zero and where  $M$  and  $m$  are real or imaginary constants.

Further by writing  $x = \frac{a+bz}{c+dz}$ , where  $z = \sin \phi$ , or  $z = \cos \phi$ ,

or  $z = \tan \phi$ , it is seen that the general elliptic integral of Art. 5, namely,  $\int \frac{Q(x)dx}{\sqrt{R(x)}}$  may be put in the form  $\int \frac{F(\sin^2 \phi)d\phi}{\Delta\phi}$ , which in turn may be expressed through integrals that correspond to the integrals  $V_0$ ,  $V_1$  and  $V_{-1}$  of the preceding article.

**Art. 40.** Returning to formula (i) above, make  $\mu = -1$ , and note that if  $m=0$ , we have  $A=0$ ,  $B=1$ ; the formula becomes

$$(a) \quad \cot \phi \Delta\phi = - \int \frac{1}{\sin^2 \phi} \frac{d\phi}{\Delta\phi} + k^2 \int \frac{\sin^2 \phi d\phi}{\Delta\phi}.$$

Next let  $m = -1$ , so that  $A=0$ ,  $B=-k'^2$ , and we have

$$(b) \quad -\tan \phi \Delta\phi = -k'^2 \int \frac{1}{\cos^2 \phi} \frac{d\phi}{\Delta\phi} - k^2 \int \frac{\cos^2 \phi d\phi}{\Delta\phi};$$

finally let  $m = -\frac{1}{k^2}$ , so that  $A=0$ ,  $B=\frac{k'^2}{k^2}$ , and the reduction formula is

$$(c) \quad -\frac{k^2 \sin \phi \cos \phi}{\Delta\phi} = k'^2 \int \frac{1}{\Delta^2 \phi} \frac{d\phi}{\Delta\phi} - \int \Delta\phi d\phi.$$

**Art. 41.** Legendre, *Traité, etc.*, I, p. 256, offers the following integrals "which are often met with in the application of the elliptic integrals." These may for the most part be derived at once from the formulas given above.

$$\int_0^\phi \frac{d\phi}{\Delta\phi} = F(k, \phi), \quad \text{where } \Delta\phi = \sqrt{1 - k^2 \sin^2 \phi} = \Delta,$$

$$\int_0^\phi \Delta d\phi = E(k, \phi), \text{ or } \int_0^u dn^2 u du = E(u), \text{ since } d\phi = dn u du.$$

$$\int_0^\phi \frac{d\phi}{\Delta^3} = \frac{1}{k'^2} E(k, \phi) - \frac{k^2 \sin \phi \cos \phi}{k'^2 \Delta}, \text{ or}$$

$$\int_0^u \frac{du}{dn^2 u} = \frac{E(u)}{k'^2} - \frac{k^2 sn u cn u}{k'^2 dn u},$$

$$\int_0^\phi \frac{d\phi \sin^2 \phi}{\Delta} = \frac{1}{k^2} [F(k, \phi) - E(k, \phi)], \text{ or}$$



$$\int_0^u sn^2 u \, du = \frac{u - E(u)}{k^2},$$

$$\int_0^\phi \frac{d\phi \cos^2 \phi}{\Delta} = \frac{1}{k^2} [E(k, \phi) - k'^2 F(k, \phi)], \text{ or}$$

$$\int_0^u cn^2 u \, du = \frac{-k'^2 u + E(u)}{k^2}.$$

$$\int_0^\phi \frac{d\phi}{\Delta \cos^2 \phi} = \frac{1}{k'^2} [\Delta \tan \phi + k'^2 F(k, \phi) - E(k, \phi)], \text{ or}$$

$$\int_0^u \frac{du}{cn^2 u} = \frac{\ln u \, dn u + k'^2 u - E(u)}{k'^2},$$

$$\int_0^\phi \frac{d\phi \tan^2 \phi}{\Delta} = \frac{\Delta \tan \phi - E(k, \phi)}{k'^2}, \text{ or}$$

$$\int_0^u tn^2 u \, du = \frac{dn u \, \ln u - E(u)}{k'^2}.$$

$$\int_0^\phi \frac{d\phi \cos^2 \phi}{\Delta^3} = \frac{1}{k^2} [F(k, \phi) - E(k, \phi)] + \frac{\sin \phi \cos \phi}{\Delta},$$

$$\int_0^\phi \frac{d\phi \sin^2 \phi}{\Delta^3} = \frac{1}{k^2 k'^2} [E(k, \phi) - k'^2 F(k, \phi)] - \frac{\sin \phi \cos \phi}{k'^2 \Delta},$$

$$\int_0^\phi \frac{\Delta d\phi}{\cos^2 \phi} = \Delta \tan \phi + F(k, \phi) - E(k, \phi),$$

$$\int_0^\phi \Delta \tan^2 \phi \, d\phi = \Delta \tan \phi + F(k, \phi) - 2E(k, \phi),$$

$$\int_0^\phi \Delta^3 d\phi = \frac{k^2}{3} \Delta \sin \phi \cos \phi + \frac{2 + 2k'^2}{3} E(k, \phi) - \frac{k'^2}{3} F(k, \phi),$$

$$\int_0^\phi \Delta \sin^2 \phi \, d\phi = \frac{-1}{3} \Delta \sin \phi \cos \phi + \frac{2k^2 - 1}{3k^2} E(k, \phi) + \frac{k'^2}{3k^2} F(k, \phi),$$

$$\int_0^\phi \Delta \cos^2 \phi \, d\phi = \frac{1}{3} \Delta \sin \phi \cos \phi + \frac{1 + k^2}{3k^2} E(k, \phi) - \frac{k'^2}{3k^2} F(k, \phi).$$

To these may be added

$$\int_\phi^{\frac{\pi}{2}} \frac{d\phi}{\sin^2 \phi \Delta} = \cot \phi \Delta \phi + K - E_1 - F(k, \phi) + E(k, \phi), \text{ or}$$

$$\int_u^K \frac{du}{sn^2 u} = \cot am u dn u + K - E_1 - u + E(u),$$

$$\int_u^K \frac{du}{tn^2 u} = \cot am u dnu - E_1 + E(u), \text{ or}$$

$$\int_\phi^{\frac{\pi}{2}} \frac{d\phi}{\tan^2 \phi \Delta} = \int_\phi^{\frac{\pi}{2}} \frac{1 - \sin^2 \phi}{\sin^2 \phi} \frac{d\phi}{\Delta \phi}.$$

## EXAMPLES

1. Show that

$$\int_x^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{2} \sqrt{2} cn^{-1}(x, \frac{1}{2} \sqrt{2}),$$

$$\int_1^x \frac{dx}{\sqrt{x^4-1}} = \frac{1}{2} \sqrt{2} cn^{-1}\left(\frac{1}{x}, \frac{1}{2} \sqrt{2}\right).$$

2. Show that

$$\int_0^1 \sqrt{1-x^4} dx = 2\sqrt{2} \int_0^K (dn^2 x - dn^4 x) dx = \frac{\sqrt{2}}{3} K \left( \text{mod } \frac{\sqrt{2}}{2} \right) = 0.87401 \dots$$

3. Show that  $\int_0^b \sqrt{\frac{a^2-bx}{bx-x^2}} dx = 2a \int_0^K dn^2 x dx = 2aE\left(\frac{b}{a}, \frac{\pi}{2}\right).$

4. Show that  $\int_0^K \frac{sn u du}{dn u + k'} = \frac{1}{k'(1+k')}.$

5. If  $u = \int_0^b \sqrt{(a^2-x^2)(b^2-x^2)} dx$ , write  $y = sn^{-1}\left(\frac{x}{b}, \frac{b}{a}\right)$ , cf. formula (5a),  $b \neq \pm a$  and show that

$$u = ab^2 \int_0^K cn^2 y dn^2 y dy = \frac{1}{3} a \left[ (a^2+b^2)E\left(\frac{b}{a}, \frac{\pi}{2}\right) - (a^2-b^2)K \right], \left( \text{mod. } \frac{b}{a} \right).$$

Byerly.

6. Show that for the inverse functions,

(i)  $\int sn^{-1} u du = u sn^{-1} u + \frac{1}{k^2} \cosh^{-1} \left( \frac{\sqrt{1-k^2 u^2}}{k'} \right);$

(ii)  $\int cn^{-1} u du = u cn^{-1} u - \frac{1}{k} \cos^{-1} (\sqrt{k'^2 + k^2 u^2});$

(iii)  $\int dn^{-1} u du = u dn^{-1} u - \sin^{-1} \left( \frac{\sqrt{1-u^2}}{k} \right).$

7. Note that if  $X = ax^2 + 2bx + c$ ,

$$d[x^p \sqrt{X}] = \frac{a(p+1)x^{p+1} + b(2p+1)x^p + cpx^{p-1}}{\sqrt{X}} dx;$$

or, if we put  $v_p = \int \frac{x^p dx}{\sqrt{X}}$ , we have

$$x^p \sqrt{X} = a(p+1)v_{p+1} + b(2p+1)v_p + cpx_{p-1}.$$

Further, if  $t = sn^2 u$ , it is seen that

$$\int sn^m u \, du = \frac{1}{2} \int \frac{t^{\frac{m-1}{2}} dt}{\sqrt{(1-t)(1-k^2 t)}}.$$

Derive the reduction formulas (ii), (iii), (iv) of Art. 38.

8. Prove that 
$$\int \frac{dS}{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}} = \frac{4\pi abc}{\sqrt{a^2 - c^2}} cn^{-1} \left( \frac{c}{a}, \sqrt{\frac{a^2 - b^2}{a^2 - c^2}} \right),$$

where the integration is taken over the surface  $S$  of a sphere  $x^2 + y^2 + z^2 = r^2$ .

*Burnside, Math. Tripos, 1881*

9. Show that

$$\int \frac{sn \, u}{cn \, u} \, du = \frac{1}{k'} \log \frac{dn \, u + k'}{cn \, u},$$

$$\int \frac{cn \, u}{sn \, u} \, du = \log \frac{1 - dn \, u}{sn \, u},$$

$$\int \frac{sn \, u}{cn \, u \, dn \, u} \, du = \frac{1}{k'^2} \log \frac{dn \, u}{cn \, u},$$

$$\int \frac{cn \, u}{sn^2 \, u} \, du = -\frac{dn \, u}{sn \, u},$$

$$\int \frac{sn \, u}{cn^2 \, u} \, du = -\frac{1}{k'^2} \frac{dn \, u}{cn \, u}.$$

## CHAPTER IV

# THE NUMERICAL COMPUTATION OF THE ELLIPTIC INTEGRALS OF THE FIRST AND SECOND KINDS. LANDEN'S TRANSFORMATIONS

**Art. 42.** With Jacobi\* consider two fixed circles as in Fig. 15 and suppose that  $R$  is the radius of the larger circle and  $r$  the radius of the smaller circle. Let the distance  $OQ = l$ . From any point  $B$  on the large circle draw a tangent to the small

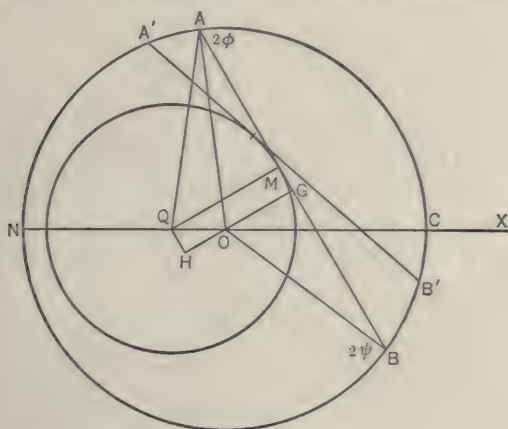


FIG. 15.

circle which again cuts the large circle in  $A$ . Denote the azimuth angle  $BOX$  by  $2\psi$  and  $AOX$  by  $2\phi$ .  $OG$  is drawn perpendicular to  $AB$  and its length is denoted by  $p$ . Note that the angle  $GOX = \phi - \psi$  and  $GOB = \phi + \psi$ ,  $p = R \cos (\phi + \psi)$  and  $QM = r = p + OH = R \cos (\phi + \psi) + l \cos (\phi - \psi)$ , or

$$r = (R+l) \cos \phi \cos \psi - (R-l) \sin \phi \sin \psi.$$

\* Jacobi, *Crelle's Journal*, Vol. III, p. 376, 1828; see also Cayley's *Elliptic Functions*, p. 28.

When  $\psi = 0$ , let the corresponding value of  $\phi$  be  $\mu$ , so that

$$r = (R+l) \cos \mu, \text{ or } \cos \mu = \frac{r}{R+l}, \sin \mu = \frac{\sqrt{(R+l)^2 - r^2}}{R+l}.$$

Denote the ratio  $\frac{QN}{QC}$  by  $\Delta\mu$ , so that  $\Delta\mu = \frac{R-l}{R+l}$ ; then since

$$\Delta\mu^2 = 1 - k^2 \sin^2 \mu, \text{ it is seen that } k^2 = \frac{4lR}{(R+l)^2 - r^2}.$$

Returning to the figure, it is seen that

$$\begin{aligned} \overline{AM}^2 &= \overline{AQ}^2 - \overline{MQ}^2 = R^2 + l^2 + 2Rl \cos 2\phi - r^2 \\ &= (R+l)^2 - r^2 - 4lR \sin^2 \phi; \end{aligned}$$

or

$$\overline{AM}^2 = \{(R+l)^2 - r^2\} \Delta^2 \phi;$$

and similarly

$$\overline{BM}^2 = \{(R+l)^2 - r^2\} \Delta^2 \psi.$$

If the tangent is varied, its new position becoming  $A'B'$ , consecutive to the initial position, then clearly we have

$$AA' : BB' = AM : BM;$$

or

$$\frac{d\phi}{AM} + \frac{d\psi}{BM} = 0;$$

and if for  $AM$  and  $BM$  their values be substituted, it follows that

$$\frac{d\phi}{\Delta\phi} + \frac{d\psi}{\Delta\psi} = 0.$$

Suppose that the smaller circle is varied, the centre moving along the  $X$ -axis while  $r$  and  $l$  are subjected to the condition

$$k^2 = \frac{4lR}{(R+l)^2 - r^2}, \text{ } k \text{ being constant.}$$



In particular when the smaller circle reduces to the point circle at  $L$ , as in Fig. 16, then

$$r=0, \quad OL=l \text{ and } k^2 = \frac{4lR}{(R+l)^2}.$$

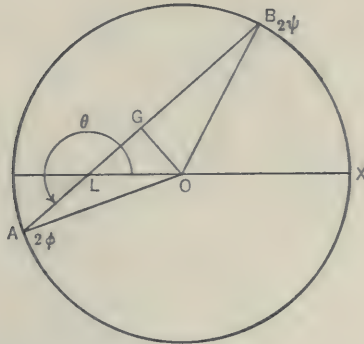


FIG. 16.

Let  $\theta$  represent the angle  $XLA$ . It is seen that

$$\theta = \frac{\pi}{2} + \phi + \psi,$$

and consequently  $d\theta = d\phi + d\psi$ .

It is also seen that the angle  $LAO = \theta - 2\phi$  and  $GOX = \phi + \psi$ . From the triangle  $ALO$  it follows at once that

$$l \sin \theta = R \sin (2\phi - \theta). \quad . \quad . \quad . \quad (1)$$

The relation  $\frac{d\phi}{AM} + \frac{d\psi}{BM} = 0$ , becomes here

$$\frac{d\phi}{AM} = \frac{d\psi}{BM} = \frac{d\theta}{2AG};$$

or, since

$$\overline{AG}^2 = R^2 - l^2 \cos^2 (180 - \phi - \psi) = R^2 - l^2 \sin^2 \theta,$$

it follows that

$$\frac{d\phi}{\Delta\phi} = \frac{d\theta (R+l)}{2\sqrt{R^2 - l^2 \sin^2 \theta}}. \quad . \quad . \quad . \quad (2)$$

Formula (1) may be regarded as the algebraic integral\* of (2), or (2) may be considered as being produced by the transformation (1).

Write  $k_1 = \frac{l}{R}$  and put  $\phi_1$  in the place of  $\theta$ .

It is seen that

$$k = \frac{2\sqrt{lR}}{R+l} = \frac{2\sqrt{k_1}}{1+k_1}, \quad k' = \frac{1-k_1}{1+k_1}, \quad k_1 = \frac{1-k'}{1+k'}, \quad \dots \quad (3)$$

and

$$\frac{d\phi}{\Delta(k, \phi)} = \frac{1}{2}(1+k_1) \frac{d\phi_1}{\Delta(k_1, \phi_1)}, \quad \dots \quad (2')$$

$$k_1' \sin \phi_1 = \sin (2\phi - \phi_1). \quad \dots \quad (1')$$

The last expression may be written

$$k_1 \sin (\phi_1 - \phi + \phi) = \sin (\phi - \phi_1 + \phi),$$

from which we have at once

$$\tan (\phi_1 - \phi) = \frac{1-k_1}{1+k_1} \tan \phi = k' \tan \phi, \quad \dots \quad (3)$$

or

$$\tan \phi_1 = \frac{(1+k') \tan \phi}{1-k' \tan^2 \phi}, \quad \sin \phi_1 = \frac{(1+k') \sin \phi \cos \phi}{\Delta(k, \phi)}.$$

Art. 43. It is seen that  $k_1 = \frac{l}{r} < 1$  and since  $\frac{2\sqrt{k_1}}{1+k_1} > k_1$ , it

follows that  $k > k_1$ . From (1') it is seen that  $0 < \phi < \phi_1$ , if  $\phi \leq \frac{\pi}{2}$ .

From (2') it is seen that

$$\begin{aligned} F(k, \phi) &= \frac{1}{2}(1+k_1)F(k_1, \phi_1) \\ &= (1+k_1)(1+k_2) \dots (1+k_n) \frac{F(k_n, \phi_n)}{2^n}, \quad \dots \quad (A) \end{aligned}$$

\* John Landen, An investigation of a general theorem for finding the length of an arc of any conic, etc., Phil. Trans. 65 (1775), pp. 283, et. seq.; or *Mathematical Memoirs* I, p. 32 of John Landen (London, 1780). An article by Cayley on John Landen is given in the Encyc. Brit., Eleventh Edition, Vol. XVI, p. 153. See also Lagrange, *Œuvres*, II, p. 253; Legendre, *Traité*, etc., I, p. 89.

where the moduli are decreasing and the amplitudes are increasing.

It is also seen that

$$k_v = \frac{1 - \sqrt{1 - k_{v-1}^2}}{1 + \sqrt{1 - k_{v-1}^2}}, \quad \left( \begin{array}{l} v = 1, 2, \dots, n \\ k_0 = k \end{array} \right),$$

$$\tan(\phi_v - \phi_{v-1}) = \sqrt{1 - k_{v-1}^2} \tan \phi_{v-1}. \quad (i)$$

It is further evident that  $F(k_n, \phi_n)$  approaches the limit  $\int_0^\Phi d\phi = \Phi$ , where  $\Phi$  is the limiting value of  $\phi$  as  $n$  increases.

If  $\phi = \frac{\pi}{2}$ , it follows at once from (i), see also Art. 49, that

$$\phi_1 = \pi, \phi_2 = 2\pi, \dots, \phi_n = 2^{n-1}\pi,$$

and consequently

$$K = F\left(k, \frac{\pi}{2}\right) = \frac{\pi}{2}(1+k_1)(1+k_2)(1+k_3)\dots$$

**Art. 44.** Suppose, for example, that it is required to find  $F(\frac{1}{2}, 40^\circ)$ . Using the seven-place logarithm tables of Vega, it is found that for

$$\phi = 40, \sin \theta = k = \frac{1}{2}, \text{ or } \theta = 30,$$

$$\sqrt{1 - k^2} = k' = 0.86603$$

$$1 - k' = 0.13397$$

$$1 + k' = 1.86603$$

$$\log(1 - k') = 9.1270076$$

$$\text{colog}(1 + k') = 9.7290814$$

$$k_1 = 0.071794$$

$$\log k_1 = 8.8560890$$

$$1 - k_1 = 0.928206$$

$$1 + k_1 = 1.071794$$

$$\log(1 - k_1) = 9.9676444$$

$$\log(1 + k_1) = 0.0301098$$

$$\log k_1^2 = 9.9977542$$

$$k'_1 = 0.997418$$

$$\log k'_1 = 9.9988771$$

$$1 - k'_1 = 0.002582$$

$$1 + k'_1 = 1.997418$$

$$\log(1 - k'_1) = 7.4121244$$

$$\text{colog}(1 + k'_1) = 9.6995263$$

$$k_2 = 0.001293$$

$$\log k_2 = 7.1116507$$

$$\begin{aligned} 1 - k_2 &= 0.998707 & \log(1 - k_2) &= 9.9994381 \\ 1 + k_2 &= 1.001293 & \log(1 + k_2) &= 0.0005599 \end{aligned}$$

$$\log k'^2 = 9.9999980$$

$$k'_2 = 1$$

$$\log k'_2 = 9.9999990$$

$$k_3 = 0$$

$$\log k' = 9.9375329$$

$$\log \tan \phi = 9.9238135$$

$$\log \tan(\phi_1 - \phi) = 9.8613464$$

$$\phi_1 - \phi = 36^\circ \quad 0' \quad 20''$$

$$\phi_1 = 76^\circ \quad 0' \quad 20''$$

$$\log k'_1 = 9.9988771$$

$$\log \tan \phi_1 = 0.6034084$$

$$\log \tan(\phi_2 - \phi_1) = 0.6022855$$

$$\phi_2 - \phi_1 = 75^\circ \quad 58' \quad 15''$$

$$\phi_2 = 151^\circ \quad 58' \quad 35''$$

$$\tan(\phi_3 - \phi_2) = \tan \phi_2$$

$$\Phi = \phi_3 = 2\phi_2 = 303^\circ \quad 57' \quad 10''$$

$$\frac{1}{2^3}\Phi = 37^\circ \quad 59' \quad 39''$$

$$= 136779''$$

$$\pi = 648000''$$

$$\log\left(\frac{1}{2^3}\Phi\right)'' = 5.1360194$$

$$\text{colog } \pi'' = 4.1884250$$

$$\log \pi = 0.4971499$$

$$\log\left(\frac{1}{2^3}\Phi\right) = 9.8215943$$

$$\log(1 + k_1) = 0.0301098$$

$$\log(1 + k_2) = 0.0005599$$

$$\log\left(\frac{1}{2^3}\Phi\right) = 9.8215943$$

$$\log F\left(\frac{1}{2}, 40^\circ\right) = 9.8522640$$

$$F\left(\frac{1}{2}, 40^\circ\right) = .711646$$

The value given in Legendre's tables is

$$.7116472757$$

**Art. 45.** The formulas of Art. 42 may be used to increase the modulus and decrease the amplitude; for if the subscripts be interchanged, it is seen that

$$F(k, \phi) = \frac{2}{1+k} F(k_1, \phi_1), \quad . . . . . (i)$$

$$k_1 = \frac{2\sqrt{k}}{1+k},$$

$$\sin(2\phi_1 - \phi) = k \sin \phi,$$

where  $k_1 > k$  and  $\phi_1 < \phi$ .

Applying the formula (i)  $n$  times, there results

$$F(k, \phi) = \frac{2}{1+k} \cdot \frac{2}{1+k_1} \cdot . . . . \frac{2}{1+k_{n-1}} F(k_n, \phi_n);$$

or, since

$$\frac{2}{1+k} = \frac{k_1}{\sqrt{k}}, \quad \frac{2}{1+k_1} = \frac{k_2}{\sqrt{k_1}}, \text{ etc.,}$$

it is seen that

$$F(k, \phi) = k_n \sqrt{\frac{k_1 k_2 \cdot . . . \cdot k_{n-1}}{k}} F(k_n, \phi_n),$$

where

$$k_v = \frac{2\sqrt{k_{v-1}}}{1+k_{v-1}}, \quad \sin(2\phi_v - \phi_{v-1})$$

$$= k_{v-1} \sin \phi_{v-1} (v = 1, 2, . . . ; k_0 = k, \phi_0 = \phi).$$

It follows also that

$$F(k_n, \phi_n) = F(1, \Phi)$$

$$= \int_0^\Phi \frac{d\phi}{\sqrt{1 - \sin^2 \phi}} = \int_0^\Phi \sec \phi d\phi = \log_e \tan\left(\frac{\pi}{4} + \frac{\Phi}{2}\right)$$



and

$$F(k, \phi) = \sqrt{\frac{k_1 k_2 \dots k_{n-1}}{k}} \log_e \tan\left(\frac{\pi}{4} + \frac{\Phi}{2}\right).$$

**Art. 46.** The method of the preceding articles may also be used to evaluate  $F(30^\circ, 40^\circ)$ , thus

$$\begin{array}{ll} k = .5 & \log k = 9.6989700 \\ 1+k = 1.5 & \log (1+k) = 0.1760913 \end{array}$$

$$\begin{array}{r} \log \sqrt{k} = 9.8494850 \\ \log 2 = 0.3010300 \\ \text{colog } (1+k) = 9.8239087 \\ \hline \log k_1 = 9.9744237 \end{array}$$

$$\begin{array}{ll} k_1 = .942809 & \log k_1 = 9.9744237 \\ 1+k_1 = 1.942809 & \log (1+k_1) = 0.2884301 \end{array}$$

$$\begin{array}{r} \log \sqrt{k_1} = 9.9872118 \\ \log 2 = 0.3010300 \\ \text{colog } (1+k_1) = 9.7115699 \\ \hline \log k_2 = 9.9998117 \end{array}$$

$$\begin{array}{ll} k_2 = .999567 & \log k_2 = 9.9998117 \\ 1+k_2 = 1.999567 & \log (1+k_2) = 0.3009359 \end{array}$$

$$\begin{array}{r} \log \sqrt{k_2} = 9.9999059 \\ \log 2 = 0.3010300 \\ \text{colog } (1+k_2) = 9.6990641 \\ \hline \log k_3 = 0.0000000 \end{array}$$

$$k_3 = 1.$$

$$\begin{array}{r} \log k = 9.6989700 \\ \log \sin \phi = 9.8080675 \\ \hline \end{array}$$

$$\log \sin (2\phi_1 - \phi) = 9.5070375$$

$$\begin{array}{lll} 2\phi_1 - \phi = 18^\circ & 44' & 50.''05 \\ 2\phi_1 = 58^\circ & 44' & 50.''10 \\ \phi_1 = 29^\circ & 22' & 25.''05 \end{array}$$

$$\log k_1 = 9.99744237$$

$$\log \sin \phi_1 = 9.6906403$$

$$\log \sin (2\phi_2 - \phi_1) = 9.6650640$$

|                               |       |           |
|-------------------------------|-------|-----------|
| $2\phi_2 - \phi_1 = 27^\circ$ | $32'$ | $43.''08$ |
| $2\phi_2 = 56^\circ$          | $54'$ | $68.''13$ |
| $\phi_2 = 28^\circ$           | $27'$ | $34.''06$ |

$$\log k_2 = 9.9998117$$

$$\log \sin \phi_2 = 9.6780866$$

$$\log \sin (2\phi_3 - \phi_2) = 9.6778983$$

|                               |       |           |
|-------------------------------|-------|-----------|
| $2\phi_3 - \phi_2 = 28^\circ$ | $26'$ | $45.''53$ |
| $2\phi_3 = 56^\circ$          | $54'$ | $19.''59$ |
| $\phi_3 = 28^\circ$           | $27'$ | $9.''78$  |

When  $k_3 = 1$ , then  $\sin (2\phi_4 - \phi_3) = \sin \phi_3$ , or  $\phi_4 = \phi_3$ .

$$\therefore \phi_4 = 28^\circ \quad 27' \quad 9.''78$$

$$\frac{\phi_4}{2} = 14^\circ \quad 13' \quad 34.''89$$

$$\Phi = \frac{\phi_4}{2} + \frac{\pi}{4} = 59^\circ \quad 13' \quad 34.''89$$

$$\Phi = 59^\circ \quad 13' \quad 34.''89$$

$$\log_{10} \tan \Phi = .2251208$$

$$\log \log \tan \Phi = 9.3524156$$

$$\text{colog } M = 0.3622157 \text{ (*see below)}$$

$$\log \sqrt{k_1} = 9.9872118$$

$$\log \sqrt{k_2} = 9.9999059$$

$$\text{colog } \sqrt{k} = 0.1505150$$

$$\log F(30^\circ, 40^\circ) = 9.8522640$$

$$F(30^\circ, 40^\circ) = .711647. \dots$$

Art. 47. Cayley, *Elliptic Functions*, p. 324, introduced instead of the standard form of the radical, a new form

$$\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi} \quad (a > b);$$

\* Division is made by the modulus  $M$  to change from the natural to the common logarithm, where  $M = .43429448$ .

and he further wrote

$$F(a, b, \phi) = \int_0^\phi \frac{d\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}, \quad \dots \quad (1)$$

$$E(a, b, \phi) = \int_0^\phi \sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi} \, d\phi. \quad \dots \quad (2)$$

It is clear that

$$\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi} = a \sqrt{1 - k^2 \sin^2 \phi},$$

where

$$k^2 = 1 - \frac{b^2}{a^2}, \quad k' = \frac{b}{a}.$$

The functions (1) and (2) are consequently  $\frac{1}{a}F(k, \phi)$  and  $aE(k, \phi)$ .

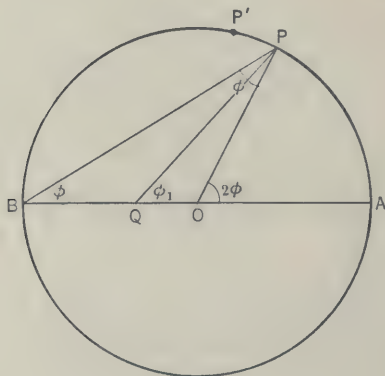


FIG. 17.

In the figure let  $P$  be a point on the circle, whose centre is  $O$  and let  $Q$  be any point on the diameter  $AB$ .

Further let

$$QA = a, \quad QB = b, \quad \angle AQP = \phi_1, \quad \angle AOP = 2\phi, \quad \angle ABP = \phi.$$

$$\text{Write } a_1 = \frac{1}{2}(a+b), \quad b_1 = \sqrt{ab}, \quad c_1 = \frac{1}{2}(a-b).$$

It follows at once that

$$OA = OB = OP = a_1, \quad OQ = a_1 - b = \frac{1}{2}(a-b) = c_1,$$

$$QP \sin \phi_1 = a_1 \sin 2\phi,$$

$$QP \cos \phi_1 = c_1 + a_1 \cos 2\phi.$$

On the other hand

$$\begin{aligned}\overline{QP^2} &= c_1^2 + 2c_1a_1 \cos 2\phi + a_1^2 = \frac{1}{2}(a^2 + b^2) + \frac{1}{2}(a^2 - b^2) \cos 2\phi \\ &= \frac{1}{2}(a^2 + b^2)(\cos^2 \phi + \sin^2 \phi) + \frac{1}{2}(a^2 - b^2)(\cos^2 \phi - \sin^2 \phi) \\ &= a^2 \cos^2 \phi + b^2 \sin^2 \phi.\end{aligned}$$

Therefore it follows that

$$\sin \phi_1 = \frac{a_1 \sin 2\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}, \quad \cos \phi_1 = \frac{c_1 + a_1 \cos 2\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}},$$

and consequently

$$a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1 = \frac{a_1^2 (a \cos^2 \phi + b \sin^2 \phi)^2}{a^2 \cos^2 \phi + b^2 \sin^2 \phi}. \quad (1)$$

It is seen at once that

$$\begin{aligned}\sin (2\phi - \phi_1) &= \frac{\frac{1}{2}(a - b) \sin 2\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}, \\ \cos (2\phi - \phi_1) &= \frac{a \cos^2 \phi + b \sin^2 \phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}; \text{ or, from (1),} \\ \cos (2\phi - \phi_1) &= \frac{1}{a_1} \sqrt{a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1}.\end{aligned}$$

If in the figure we consider the point  $P'$  consecutive to  $P$ , then,

$$PQ d\phi_1 = PP' \sin PP'Q = 2a_1 \cos (2\phi - \phi_1) d\phi;$$

or, writing for  $PQ$  its value from above, there results

$$\frac{2d\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}} = \frac{d\phi_1}{\sqrt{a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1}}.$$

Integrating, this expression becomes

$$F(a, b, \phi) = \frac{1}{2} F(a_1, b_1, \phi_1),$$

or

$$F(k, \phi) = \frac{1}{2} \frac{a}{a_1} F(k', \phi') = \frac{1}{1+k'} F(k_1, \phi_1),$$

where

$$\sin \phi_1 = \frac{\frac{1}{2}(1+k') \sin 2\phi}{\sqrt{1-k^2 \sin^2 \phi}}.$$

Note that  $k^2 = 1 - \frac{b^2}{a^2}$ ,  $k' = \frac{b}{a}$ ;  $k_1^2 = 1 - \frac{b_1^2}{a_1^2} = \left(\frac{a-b}{a+b}\right)^2 = \left(\frac{1-k'}{1+k'}\right)^2$ ;

or,  $k_1 = \frac{1-k'}{1+k'}$ , and  $k' = \frac{1-k_1}{1+k_1}$ , as given at the end of Art. 42.

**Art. 48.** Cayley derives a similar formula for the integrals of the second kind as follows, his work being here in places considerably simplified. From the relation of Art. 42, we have

$$\sin (2\phi - \phi_1) = k_1 \sin \phi_1, \text{ or}$$

$$\sin 2\phi \cos \phi_1 - \cos 2\phi \sin \phi_1 = k_1 \sin \phi_1;$$

it follows that

$$\cos 2\phi = -k_1 \sin^2 \phi_1 + \cos \phi_1 \Delta \phi_1,$$

and consequently

$$2 \cos^2 \phi = 1 - k_1 \sin^2 \phi_1 + \cos \phi_1 \Delta \phi_1,$$

$$2 \sin^2 \phi = 1 + k_1 \sin^2 \phi_1 - \cos \phi_1 \Delta \phi_1.$$

From these two relations it is seen at once that

$$\begin{aligned} 2(a^2 \cos^2 \phi + b^2 \sin^2 \phi) &= a^2 + b^2 - (a^2 - b^2)k_1 \sin^2 \phi_1 \\ &\quad + (a^2 - b^2)\cos \phi_1 \Delta \phi_1 = (a^2 + b^2)(\cos^2 \phi_1 + \sin^2 \phi_1) \\ &\quad - (a^2 - b^2)k_1 \sin^2 \phi_1 + (a^2 - b^2)\cos \phi_1 \Delta \phi_1 \\ &= 4(a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1) \\ &\quad - 2b_1^2 + 4c_1 \cos \phi_1 \sqrt{a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1}. \end{aligned}$$

Multiply this expression by the differential relation given above, viz.,

$$\frac{2d\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}} = \frac{d\phi_1}{\sqrt{a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1}},$$



and integrating, there results

$$E(a, b, \phi) = E(a_1, b_1, \phi_1) - \frac{1}{2} b_1^2 F(a_1, b_1, \phi_1) + c_1 \sin \phi_1,$$

where

$$\sin \phi_1 = \frac{a_1 \sin 2\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}.$$

It follows at once that

$$E(k, \phi) = \frac{a_1}{a} E(k_1, \phi_1) - \frac{1}{2} \frac{b_1^2}{a a_1} F(k_1, \phi_1) + \frac{c_1}{a} \sin \phi_1,$$

or

$$E(k, \phi) = \frac{1}{2} (1 + k') E(k_1, \phi_1) - \frac{k'}{1 + k'} F(k_1, \phi_1) + \frac{1}{2} (1 - k') \sin \phi_1,$$

with the initial relation

$$\sin (2\phi - \phi_1) = k_1 \sin \phi_1.$$

**Art. 49.** From the formula connecting  $\phi$  and  $\phi_1$ , which may be written in the form (see end of Art. 42)

$$\tan \phi_1 = \frac{(1 + k') \tan \phi}{1 - k' \tan^2 \phi}, \quad \dots \dots \dots (1)$$

it is seen that  $\phi$  and  $\phi_1$  vanish at the same time; and further since

$$\frac{d\phi'}{d\phi} = (1 + k') \frac{1 + k' \tan^2 \phi}{(1 - k' \tan^2 \phi)^2} \frac{\cos^2 \phi_1}{\cos^2 \phi},$$

a positive quantity, it appears that  $\phi_1$  increases with  $\phi$ . It is further evident that  $\tan \phi_1 = 0$  when  $\tan \phi = \infty$ . It is clear from

(1) that when  $\phi = 0$ ,  $\phi_1 = 0$  and when  $\tan \phi = \sqrt{\frac{1}{k'}} = \sqrt{\frac{a}{b}}$ , then

$\phi_1 = \frac{1}{2}\pi$ ; and in general to the values  $\frac{\pi}{2}, \pi, 2\pi, \dots$  of  $\phi$ , there correspond the values  $\pi, 2\pi, 4\pi, \dots$  of  $\phi_1$ .

**Art. 50.** Denote the complete functions  $F\left(a, b, \frac{\pi}{2}\right), E\left(a, b, \frac{\pi}{2}\right)$

by  $F(a, b)$ ,  $E(a, b)$ , then

$$F(a, b) = \frac{1}{2}F(a_1, b_1, \pi) = F\left(a_1, b_1, \frac{\pi}{2}\right) = F(a_1, b_1);$$

and similarly

$$E(a, b) = 2E(a_1, b_1) - b_1^2 F(a_1, b_1).$$

**Art. 51.** *Continued repetition of the above transformations.* In the same manner as  $a_1, b_1, c_1$  were derived from  $a, b$ , we may derive  $a_2, b_2, c_2$  from  $a_1, b_1$ , etc., and thus form the following table:

$$\begin{array}{lll} a_1 = \frac{1}{2}(a+b), & b_1 = \sqrt{ab}, & c_1 = \frac{1}{2}(a-b), \\ a_2 = \frac{1}{2}(a_1+b_1), & b_2 = \sqrt{a_1b_1}, & c_2 = \frac{1}{2}(a_1-b_1), \\ a_3 = \frac{1}{2}(a_2+b_2), & b_3 = \sqrt{a_2b_2}, & c_3 = \frac{1}{2}(a_2-b_2), \\ \cdot & \cdot & \cdot \end{array}$$

Note that  $a_1 - b_1 = \frac{(\sqrt{a} - \sqrt{b})^2}{2}$  and that

$$a_2 - b_2 = \frac{a_1 + b_1}{2} - \sqrt{a_1b_1} = \frac{a_1 - b_1}{2} - [\sqrt{a_1} - \sqrt{b_1}]\sqrt{b_1},$$

so that

$$a_2 - b_2 < \frac{a_1 - b_1}{2} \text{ or } a_2 - b_2 < \frac{(\sqrt{a} - \sqrt{b})^2}{2^2}.$$

Similarly it is seen that  $a_3 - b_3 < \frac{a_2 - b_2}{2} < \frac{(\sqrt{a} - \sqrt{b})^3}{2^3}$ ; and in

general  $a_n - b_n < \frac{(\sqrt{a} - \sqrt{b})^n}{2^n}$ , or  $\lim (a_n - b_n) = 0$ . It is clear that

as  $n$  increases  $a_n$  and  $b_n$  approach (very rapidly) one and the same limit, which is called\* by Gauss the *arithmetico-geometrical mean* and denoted by him with the symbol  $M(a, b) = \mu$ . However, when  $a_n = b_n$ , then

$$F(a_n, b_n, \phi) = \frac{\phi}{a_n} \text{ and } E(a_n, b_n, \phi) = a_n \phi;$$

\* Gauss, *Werke*, III, pp. 361-404.

further if  $\phi = \frac{1}{2}\pi$ , it is seen that

$$F(a_n, b_n) = \frac{\pi}{2a_n} \text{ and } E(a_n, b_n) = \frac{\pi}{2} a_n, \text{ where } a_n = \mu.$$

The equation  $F(a, b, \phi) = \frac{1}{2} F(a_1, b_1, \phi_1)$  gives

$$\begin{aligned} F(a, b, \phi) &= \frac{1}{2} F(a_1, b_1, \phi_1) = \frac{1}{2^2} F(a_2, b_2, \phi_2) \\ &= \dots = \frac{1}{2^n} F(a_n, b_n, \phi_n) = \frac{1}{2^n a_n} \phi_n, \end{aligned}$$

where the  $\phi$ 's are to be calculated from the formula

$$\begin{aligned} \sin \phi_1 &= \frac{a_1 \sin 2\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}, \\ \sin \phi_2 &= \frac{a_2 \sin 2\phi_1}{\sqrt{a_1^2 \cos^2 \phi_1 + b_1^2 \sin^2 \phi_1}}, \dots \end{aligned}$$

**Art. 52.** *The integrals of the second kind.* Note that, since

$$F(a, b, \phi) = \frac{1}{2} F(a_1, b_1, \phi_1),$$

the formula above for the  $E$ -function may be written

$$\begin{aligned} E(a, b, \phi) - a^2 F(a, b, \phi) &= E(a_1, b_1, \phi_1) - a_1^2 F(a_1, b_1, \phi_1) \\ &\quad + F(a_1, b_1, \phi_1)(a_1^2 - \frac{1}{2}a^2 - \frac{1}{2}b_1^2) + c_1 \sin \phi_1; \end{aligned}$$

$$\text{or, since } a_1^2 - \frac{1}{2}a^2 - \frac{1}{2}b_1^2 = -\frac{1}{4}(a^2 - b^2) = -a_1 c_1,$$

the above equation is

$$\begin{aligned} E(a, b, \phi) - a^2 F(a, b, \phi) &= E(a_1, b_1, \phi_1) - a_1^2 F(a_1, b_1, \phi_1) \\ &\quad - a_1 c_1 F(a_1, b_1, \phi_1) + c_1 \sin \phi_1. \end{aligned}$$

Observing that, as  $n$  increases,

$$\lim [E(a_n, b_n, \phi_n) - a_n^2 F(a_n, b_n, \phi_n)] = 0,$$

it is seen that

$$\begin{aligned} E(a, b, \phi) - a^2 F(a, b, \phi) &= -[2a_1 c_1 + 4a_2 c_2 + 8a_3 c_3 + \dots] F(a, b, \phi) \\ &\quad + c_1 \sin \phi_1 + c_2 \sin \phi_2 + c_3 \sin \phi_3 + \dots; \end{aligned}$$

or finally

$$\begin{aligned} E(a, b, \phi) &= [a^2 - 2a_1 c_1 - 4a_2 c_2 - 8a_3 c_3 - \dots] F(a, b, \phi) \\ &\quad + c_1 \sin \phi_1 + c_2 \sin \phi_2 + c_3 \sin \phi_3 + \dots \end{aligned}$$

In particular, if  $\phi = \frac{1}{2}\pi$ , we have Art. 49,  $\phi_1 = \pi$ ,  $\phi_2 = 2\pi$ , . . . , and then

$$E(a, b) = [a^2 - 2a_1c_1 - 4a_2c_2 - \dots] \frac{\pi}{2a_n}.$$

It also follows immediately that

$$E(k, \phi) = \left[ 1 - \frac{2a_1c_1}{a^2} - \frac{4a_2c_2}{a^2} - \dots \right] F(k, \phi) \\ + \frac{c_1}{a} \sin \phi_1 + \frac{c_2}{a} \sin \phi_2 + \frac{c_3}{a} \sin \phi_3 + \dots ;$$

or, noting that

$$\frac{a_1c_1}{a_2} = \frac{1}{4}k^2, \quad \frac{a_2c_2}{a_1c_1} = \frac{1}{4}k_1, \quad \frac{a_3c_3}{a_2c_2} = \frac{1}{4}k_2, \dots, \\ \frac{c_1}{a} = \frac{k_1}{1+k_1}, \\ \frac{c_2}{a_1} = \frac{k_2}{1+k_2}, \quad \frac{a_1}{a} = \frac{1}{1+k_1}, \\ \frac{c_3}{a_2} = \frac{k_3}{1+k_3}, \quad \frac{a_2}{a_1} = \frac{1}{1+k_2}, \quad \frac{a_1}{a} = \frac{1}{1+k_1}, \dots,$$

the equation becomes,

$$E(k, \phi) = \left[ 1 - \frac{1}{2}k^2 \left( 1 + \frac{1}{2}k_1 + \frac{1}{4}k_1k_2 + \frac{1}{8}k_1k_2k_3 + \dots \right) \right] F(k, \phi) \\ + \frac{k_1}{1+k_1} \sin \phi_1 + \frac{k_2}{(1+k_1)(1+k_2)} \sin \phi_2 \\ + \frac{k_3}{(1+k_1)(1+k_2)(1+k_3)} \sin \phi_3 + \dots$$

Further since

$$\frac{1}{1+k_1} = \frac{k}{2\sqrt{k_1}}, \quad \text{or} \quad \frac{1}{1+k_1} = \frac{k}{2\sqrt{k_1}}, \\ \frac{1}{1+k_2} = \frac{k_1}{2\sqrt{k_2}}, \quad \text{or} \quad \frac{1}{(1+k_1)(1+k_2)} = \frac{k\sqrt{k_1}}{4\sqrt{k_2}}, \\ \frac{1}{1+k_3} = \frac{k_2}{2\sqrt{k_3}}, \quad \text{or} \quad \frac{1}{(1+k_1)(1+k_2)(1+k_3)} = \frac{k\sqrt{k_1k_2}}{8\sqrt{k_3}}, \\ \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot$$

the last line of the above expression may be written

$$k[\frac{1}{2}\sqrt{k_1} \sin \phi_1 + \frac{1}{4}\sqrt{k_1 k_2} \sin \phi_2 + \frac{1}{8}\sqrt{k_1 k_2 k_3} \sin \phi_3 + \dots].$$

In particular if  $\phi = \frac{1}{2}\pi$ , we have

$$E_1 = E\left(k, \frac{\pi}{2}\right) = [1 - \frac{1}{2}k^2(1 + \frac{1}{2}k_1 + \frac{1}{4}k_1 k_2 + \frac{1}{8}k_1 k_2 k_3 + \dots)]F_1(k).$$

**Art. 53.** As a numerical example (see Legendre, *Traité* etc., T. I, p. 91), let  $a = 1$ ,  $b = \frac{1}{2}\sqrt{2 - \sqrt{3}} = \cos 75^\circ$ , and let  $\tan \phi = \sqrt{\frac{2}{\sqrt{3}}}$ .

It follows that  $k^2 = 1 - \frac{b^2}{a^2} = \sin 75^\circ$ .

The following table may be at once constructed.

| Index | $a$       | $b$       | $c$       | $k$       | $k'$      | $\phi$    |
|-------|-----------|-----------|-----------|-----------|-----------|-----------|
|       |           |           |           |           |           | ° ' "     |
| (0)   | 1.0000000 | 0.2588190 | .....     | 0.9659258 | 0.2588190 | 47 3 31   |
| (1)   | 0.6294095 | 0.5087426 | 0.3705905 | 0.5887908 | 0.8082856 | 62 36 3   |
| (2)   | 0.5690761 | 0.5658688 | 0.0603334 | 0.1060200 | 0.9943636 | 119 55 48 |
| (3)   | 0.5674724 | 0.5674701 | 0.0016037 | 0.0028260 | 0.9999959 | 240 0 0   |
| (4)   | 0.5674713 | 0.5674713 | 0.0000011 | 0.0000020 | 0.9999999 | 480 0 0   |

(See Cayley, loc. cit., p. 335.)

The complete integral  $F_1 = \frac{\pi}{2} \frac{1}{a_4} = 2.768063 \dots$  and

$$F(75^\circ, 47^\circ 3' 31'') = \frac{\phi_4}{8} \cdot \frac{1}{a_4} = 0.9226877 \dots$$

Note that the first integral is *three times* the second. It is also seen that

$$\begin{aligned} \frac{1}{2} \left( 1 - \frac{E_1}{F_1} \right) &= a_1 c_1 = .2332532 \\ &+ 2a_2 c_2 = .0686686 \\ &+ 4a_3 c_3 = .0036402 \\ &+ 8a_4 c_4 = .0000051 \\ &= .3055671 \end{aligned}$$

and  $E_1 = 1.0764051 \dots$



The computation of  $E(k, \phi)$  is found in the next article.

**Art. 54.** To establish in a somewhat different manner the results that were given in the preceding article, consider \* a function  $G(k, \phi)$  composed of an integral of the first and of an integral of the second kind, such that

$$G(k, \phi) = \int_0^\phi \frac{\alpha + \beta \sin^2 \phi}{\sqrt{1 - k^2 \sin^2 \phi}} d\phi,$$

where  $\alpha$  and  $\beta$  are constants.

Making in this integral the substitutions of Arts. 42 and 48, namely

$$\frac{d\phi}{\Delta\phi} = \frac{1+k_1}{2} \frac{d\phi_1}{\Delta\phi_1}, \quad \sin^2 \phi = \frac{1}{2}(1+k_1 \sin^2 \phi_1 - \Delta\phi_1 \cos \phi_1),$$

it is seen that

$$G(k, \phi) = \frac{1+k_1}{2} [G(k_1, \phi_1) - \frac{1}{2}\beta \sin \phi_1], \quad \dots \quad (1)$$

where

$$G(k_1, \phi_1) = \int_0^{\phi_1} \frac{\alpha_1 + \beta_1 \sin^2 \phi_1}{\Delta\phi_1} d\phi_1,$$

the constants  $\alpha_1$  and  $\beta_1$  being defined by the relations

$$\alpha_1 = \alpha + \frac{1}{2}\beta, \quad \beta_1 = \frac{1}{2}\beta k_1.$$

We saw in Art. 48 that

$$k_1 = \frac{1 - \sqrt{1 - k^2}}{1 + \sqrt{1 - k^2}}, \quad \tan(\phi_1 - \phi) = \sqrt{1 - k^2} \tan \phi,$$

where  $k_1 < k$  and  $\phi_1 > \phi$ .

It follows directly from (1) that

$$\begin{aligned} G(k, \phi) = & \frac{1+k_1}{2} \cdot \frac{1+k_2}{2} \dots \frac{1+k_n}{2} G(k_n, \phi_n) \\ & - \frac{1}{2} \left[ \frac{1+k_1}{2} \beta \sin \phi_1 + \frac{1+k_1}{2} \cdot \frac{1+k_2}{2} \beta_1 \sin \phi_2 + \dots \right. \\ & \left. + \frac{1+k_1}{2} \cdot \frac{1+k_2}{2} \dots \frac{1+k_n}{2} \beta_{n-1} \sin \phi_n \right], \end{aligned}$$

\* See also Legendre, *Traité*, etc., I, p. 108.

where

$$\beta_p = \beta \frac{k_1 k_2 \dots k_p}{2^p},$$

and

$$\alpha_p = \alpha + \frac{1}{2} \beta \left( 1 + \frac{k_1}{2} + \frac{k_1 k_2}{2^2} + \dots + \frac{k_1 k_2 \dots k_{p-1}}{2^{p-1}} \right).$$

Since  $\beta_n$  becomes 0 with  $k_n$ , it is seen that

$$\lim_{n \rightarrow \infty} G(k_n, \phi_n) = \int_0^{\phi_n} \alpha_n d\phi = \alpha_n \phi_n.$$

From Art. 43 we had

$$\frac{1+k_1}{2} \cdot \frac{1+k_2}{2} \dots \frac{1+k_n}{2} \phi_n = F(k, \phi),$$

and, see Art. 42,

$$\frac{1+k_1}{2} = \frac{\sqrt{k_1}}{k}, \quad \frac{1+k_2}{2} = \frac{\sqrt{k_2}}{k_1}, \quad \dots$$

It follows that the above formula becomes

$$G(k, \phi) = F(k, \phi) \left[ \alpha + \frac{1}{2} \beta \left( 1 + \frac{k_1}{2} + \frac{k_1 k_2}{2^2} + \frac{k_1 k_2 k_3}{2^3} + \dots \right) \right] \\ - \frac{\beta}{k} \left( \frac{\sqrt{k_1}}{2} \sin \phi_1 + \frac{\sqrt{k_1 k_2}}{2^2} \sin \phi_2 + \frac{\sqrt{k_1 k_2 k_3}}{2^3} \sin \phi_3 + \dots \right).$$

If in this formula we put  $\alpha = 1$ ,  $\beta = -k^2$ , it becomes

$$E(k, \phi) = F(k, \phi) \left[ 1 - \frac{k^2}{2} \left( 1 + \frac{k_1}{2} + \frac{k_1 k_2}{2^2} + \frac{k_1 k_2 k_3}{2^3} + \dots \right) \right] \\ + k \left[ \frac{\sqrt{k_1}}{2} \sin \phi_1 + \frac{\sqrt{k_1 k_2}}{2^2} \sin \phi_2 + \frac{\sqrt{k_1 k_2 k_3}}{2^3} \sin \phi_3 + \dots \right],$$

where

$$k_p = \frac{1 - \sqrt{1 - k_{p-1}^2}}{1 + \sqrt{1 - k_{p-1}^2}}$$

and

$$\tan (\phi_p - \phi_{p-1}) = \sqrt{1 - k_{p-1}^2} \tan \phi_{p-1}.$$

These results verify those of Art. 52.

With Legendre, *Fonct. Ellip.*, T. I., p. 114, we may find

$$E(k, \phi) \text{ where } k = \sin 75^\circ \text{ and } \tan \phi = \sqrt{\frac{2}{\sqrt{3}}}.$$

Using the results of Art. 53 it is seen that

$$\frac{k\sqrt{k_1}}{2} \sin \phi_1 = .3290186$$

$$\frac{k\sqrt{k_1 k_2}}{4} \sin \phi_2 = .0522872$$

$$\frac{k\sqrt{k_1 k_2 k_3}}{8} \sin \phi_3 = -.0013888$$

$$\frac{k\sqrt{k_1 k_2 k_3 k_4}}{16} \sin \phi_4 = .0000010$$

---


$$\text{sum} = .3799180$$

Writing

$$L = 1 - \frac{k^2}{2} - \frac{k^2 k_1}{4} - \frac{k^2 k_1 k_2}{8} - \frac{k^2 k_1 k_2 k_3}{16},$$

it is found that  $L = .3888658 \dots$

In Art. 53 it was seen that  $F(k, \phi) = .9226877 \dots$

It follows that  $E(k, \phi) = F(k, \phi)L + .3799180 \dots = .7387196 \dots$

Further since

$$E\left(k, \frac{\pi}{2}\right) = F\left(k, \frac{\pi}{2}\right)L,$$

there follows

$$E_1 = 1.0764049 \dots$$

**Art. 55. Inverse order of transformation.** If the modulus  $k$  is nearer unity than zero, the following method is preferable. The equation (1) of the preceding article may be written

$$G(k_1, \phi_1) = \frac{2}{1+k_1} G(k, \phi) + \frac{\beta_1}{k_1} \sin \phi_1, \text{ since } \frac{\beta_1}{k_1} = \frac{\beta}{2}.$$

If in this formula the suffixes be interchanged, then

$$G(k, \phi) = \frac{2}{1+k} G(k_1, \phi_1) + \frac{\beta}{k} \sin \phi,$$

where now

$$\beta_1 = \frac{2\beta}{k}, \quad \alpha_1 = \alpha - \frac{\beta}{k},$$

$$k_1 = \frac{2\sqrt{k}}{1+k}, \quad \sin(2\phi_1 - \phi) = k \sin \phi,$$

$$k_1 > k, \quad \phi_1 < \phi.$$

The continued repetition of (2) gives

$$\begin{aligned} G(k, \phi) &= \frac{\beta}{k} \sin \phi + \frac{\beta_1}{\sqrt{k}} \sin \phi_1 + \frac{\sqrt{k_1}}{\sqrt{k}} \beta_2 \sin \phi_2 \\ &+ \frac{\sqrt{k_1 k_2}}{\sqrt{k}} \beta_3 \sin \phi_3 + \frac{\sqrt{k_1 k_2 \dots k_{n-2}}}{\sqrt{k}} \beta_{n-1} \sin \phi_{n-1} \\ &+ k_n \frac{\sqrt{k_1 k_2 \dots k_{n-1}}}{\sqrt{k}} G(k_n, \phi_n), \end{aligned}$$

where

$$\beta_p = \frac{2^p \beta}{k k_1 \dots k_{p-1}},$$

and

$$\alpha_p = \alpha - \frac{\beta}{k} \left( 1 + \frac{2}{k_1} + \frac{2^2}{k_1 k_2} + \dots + \frac{2^{p-1}}{k_1 k_2 \dots k_{p-1}} \right).$$

Since  $k_n$  approaches unity (rapidly) as  $n$  increases,

$$\begin{aligned} \lim_n G(k_n, \phi_n) &= \int_0^{\phi_n} \frac{\alpha_n + \beta_n \sin^2 \phi}{\cos \phi} d\phi \\ &= (\alpha_n + \beta_n) \log_e \tan \left( \frac{\pi}{4} + \frac{\phi_n}{2} \right) - \beta_n \sin \phi_n. \end{aligned}$$

In Art. 45 it was shown that

$$\lim_n k_n \sqrt{\frac{k_1 k_2 \dots k_{n-1}}{k}} \log \tan \left( \frac{\pi}{4} + \frac{\phi_n}{2} \right) = F(k, \phi).$$

We may consequently write the above formula

$$G(k, \phi) = F(k, \phi) \left[ \alpha - \frac{\beta}{k} \left( 1 + \frac{2}{k_1} + \frac{2^2}{k_1 k_2} + \dots + \frac{2^{n-1} - 2^n}{k_1 k_2 \dots k_{n-1}} \right) \right] \\ + \frac{\beta}{k} \left[ \sin \phi + \frac{2}{\sqrt{k}} \sin \phi_1 + \frac{2^2}{\sqrt{k k_1}} \sin \phi_2 + \frac{2^3}{\sqrt{k k_1 k_2}} \sin \phi_3 + \dots \right. \\ \left. + \frac{2^{n-1}}{\sqrt{k k_1 \dots k_{n-2}}} \sin \phi_{n-1} - \frac{2^n}{\sqrt{k k_1 \dots k_{n-1}}} \sin \phi_n \right].$$

Writing  $\alpha = 1$ ,  $\beta = -k^2$  in this formula, it becomes

$$E(k, \phi) = F(k, \phi) \left[ 1 + k \left( 1 + \frac{2}{k_1} + \frac{2^2}{k_1 k_2} + \dots \right. \right. \\ \left. \left. + \frac{2^{n-1}}{k_1 k_2 \dots k_{n-1}} - \frac{2^n}{k_1 k_2 \dots k_{n-1}} \right) \right] \\ - k \left( \sin \phi + \frac{2}{\sqrt{k}} \sin \phi_1 + \frac{2^2}{\sqrt{k k_1}} \sin \phi_2 + \dots \right. \\ \left. + \frac{2^{n-1}}{\sqrt{k k_1 k_2 \dots k_{n-2}}} \sin \phi_{n-1} - \frac{2^n}{\sqrt{k k_1 \dots k_{n-1}}} \sin \phi_n \right),$$

where

$$k_p = \frac{2\sqrt{k_{p-1}}}{1 + k_{p-1}} \text{ and } \sin(2\phi_p - \phi_{p-1}) = k_{p-1} \sin \phi_{p-1}.$$

Taking the example of the preceding article, and using the values given in Art. 53, it is seen that

$$\begin{aligned} -k \sin \phi &= -0.7071070 \\ -2\sqrt{k} \sin \phi_1 &= -1.4146540 \\ +4 \frac{\sqrt{k}}{\sqrt{k_1}} \sin \phi_2 &= 2.8293085 \end{aligned}$$

$$F(k, \phi) = .9226877$$

and

$$\begin{aligned} F(k, \phi) \left[ 1 + k - \frac{2k}{k_1} \right] &= \underline{0.0311720} \\ E(k, \phi) &= 0.7387195 \dots \end{aligned}$$



**Art. 56.** Two of the principal problems that appear in practice will now be given.

**PROBLEM 1.** *When  $u$  and  $k$  are given, calculate the values of  $sn\,u$ ,  $cn\,u$ ,  $dn\,u$ .*

1. Computation of  $sn\,u$ . In the Table II, p. 96, is found an immediate answer to the problem.

For when  $u$  and  $k=\sin\theta$  are known, the value  $\phi$  may be found in the table and then  $sn\,u$  from the formula  $sn\,u=\sin\phi$ .

If, for example,  $k=\frac{1}{2}=\sin\theta$ , and  $u=.47551$ , it is seen that for  $\theta=30^\circ$ ,  $u=.47551$ , we have  $\phi=27^\circ$ , and  $\sin\phi=.45399=sn\,u$ .

2. The computation of  $cn\,u$  and  $dn\,u$  are had from the formulas

$$cn\,u = \pm \sqrt{(1 - sn\,u)(1 + sn\,u)},$$

$$dn\,u = \pm \sqrt{(1 - ksn\,u)(1 + ksn\,u)}.$$

**PROBLEM 2.** *Having given the elliptic function, calculate the argument.*

1. If  $sn\,u$  is known, find  $u$ . Table II furnishes the solution. Suppose that  $a$  is the given value of  $sn\,u$ , and suppose that  $k=\sin\theta$  is also known. Hence, since  $sn\,u=\sin\phi=a$ , we may determine  $\phi$ . With  $\theta$  and  $\phi$  known, we find the value of  $u$  from the table. Denote this value by  $u_0$ . From the relation  $sn\,u=sn\,u_0$ , we have (Art. 21),

$$u = u_0 + 4mK + 2m'iK'.$$

Further in the formula (Art. 12).

$$sn\,u = -sn(u + 2K),$$

substitute  $u = -u_0$ , and then we have  $-sn\,u_0 = -sn(2K - u_0)$ , so that  $u$  may also have the form

$$u = 2K - u_0 + 4mK + 2m'iK'.$$

2. If  $cn\,u$  and  $dn\,u$  are given,  $sn\,u$  and then  $u$  may be found as above.

## CHAPTER V

### MISCELLANEOUS EXAMPLES AND PROBLEMS

1. *The rectification of the lemniscate.* The equation of the curve is

$$(y^2 + x^2)^2 + a^2(y^2 - x^2) = 0;$$

or, writing  $x = r \cos \theta$ ,  $y = r \sin \theta$ , the equation becomes

$$r^2 = a^2 \cos 2\theta.$$

From the expression  $ds^2 = dr^2 + r^2 d\theta^2$ , the differential of arc is

$$ds = \mp \frac{dr}{\sqrt{1 - \frac{r^4}{a^4}}} = \mp \frac{ad\theta}{\sqrt{1 - 2 \sin^2 \theta}}.$$

Writing, see II of Art. 3,  $r = a \cos \phi$ , so that  $2 \sin^2 \theta = \sin^2 \phi$ , it is seen that

$$s = a \int_0^\theta \frac{d\theta}{\sqrt{1 - 2 \sin^2 \theta}} = \frac{a}{\sqrt{2}} \int_0^\phi \frac{d\phi}{\sqrt{1 - \frac{1}{2} \sin^2 \phi}} = \frac{a}{\sqrt{2}} F\left(\frac{1}{\sqrt{2}}, \phi\right),$$

which may be calculated at once from the tables when  $a$  and  $\theta$  (or  $\phi$ ) are given. A quadrant of the lemniscate is

$$S_q = a \int_0^{\frac{\pi}{4}} \frac{d\theta}{\sqrt{1 - 2 \sin^2 \theta}} = \frac{a}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1 - \frac{1}{2} \sin^2 \phi}} = \frac{a}{\sqrt{2}} K\left(\frac{1}{\sqrt{2}}\right).$$

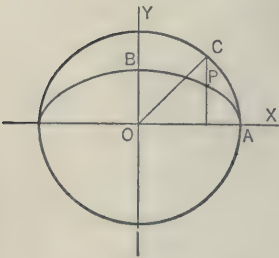


FIG. 18.

2. *The rectification of the ellipse.*

Let the equation be  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ,  $a > b$ .

From the integral

$$s = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

we have, by writing  $k^2 = \frac{a^2 - b^2}{a^2}$ ,  $x = at$ ,

$$s = a \int_0^t \frac{(1 - k^2 t^2) dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}.$$

Finally writing  $t = \sin \phi$  (see Art. 3) and that is  $x = a \sin \phi$ , we have

$$s = \int_0^\phi \Delta \phi d\phi = aE(\phi).$$

Here  $k$  is the *numerical eccentricity* of the ellipse. The angle  $\phi = COY = 90^\circ - COA$ , where in astronomy the angle  $COA$  is known as the *eccentric anomaly* of the point  $P$ . Writing  $\phi = \pi/2$ , it is seen that the quadrant of the ellipse is  $aE$ , where  $E$  is the *complete integral* of the second kind.

If the equation of the ellipse is taken in the form

$$x = a \sin \phi, \quad y = b \cos \phi,$$

it follows at once that

$$ds^2 = a^2(1 - k^2 \sin^2 \phi) d\phi^2, \quad \text{or} \quad s = aE(\phi).$$

3. The major and minor axes of an ellipse are 100 and 50 centimeters respectively. Find the length of the arc between the points (0, 25) and (48, 7). Find also the length of the arc between the points (48, 7) and (50, 0). Determine the length of its quadrant.

4. If  $\lambda$  denotes the *latitude* of a point  $P$  on the earth's surface, the equation of the ellipse through this point as indicated in the figure, may be written in the form

$$x = \frac{a \cos \lambda}{\sqrt{1 - e^2 \sin^2 \lambda}}, \quad y = \frac{a(1 - e^2) \sin \lambda}{\sqrt{1 - e^2 \sin^2 \lambda}}.$$

It follows at once that

$$ds^2 = dx^2 + dy^2 = \frac{a^2(1 - e^2)^2 d\lambda^2}{(1 - e^2 \sin^2 \lambda)^3},$$

so that

$$s = a(1 - e^2) \int_0^\lambda \frac{d\lambda}{(1 - e^2 \sin^2 \lambda)^{3/2}}.$$

This integral may be at once evaluated by the third formula in Art. 41.

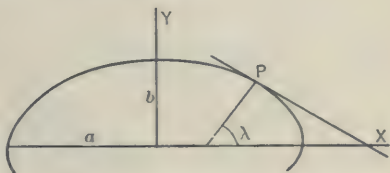


FIG. 19.

Compute the lengths of arc of the ellipse between  $10^\circ$  and  $11^\circ$  and between  $79^\circ$  and  $80^\circ$  where  $a = 6378278$  meters and  $e^2 = 0.0067686$ . Compare these distances with the length of an arc that subtends  $1^\circ$  upon a circle with radius = 6378278 meters.

5. Plot the curves, the *elastic curves*, which are defined through the differential equation

$$d\phi = \pm \frac{y^2 dy}{\sqrt{a^4 - y^4}},$$

for the values  $a = 1, 2, 4, 9$ .

6. The axes of two right cylinders of radii  $a$  and  $b$  respectively ( $a > b$ ) intersect at right angles. Find the volume common to both.

Let the  $z$ -axis be that of the larger cylinder and the  $y$ -axis that of the smaller, so that the equations of the cylinders are

$$x^2 + y^2 = a^2 \quad \text{and} \quad x^2 + z^2 = b^2 \quad \text{respectively.}$$

The volume in question is

$$V = 8 \int_0^b \sqrt{a^2 - x^2} \sqrt{b^2 - x^2} dx.$$

Writing  $t = sn^{-1}\left(\frac{x}{b}, \frac{b}{a}\right)$ , (see formula 5a, Art. 23), then  $x = b \operatorname{sn} t$ ,  $b^2 - x^2 = b^2 \operatorname{cn}^2 t$ ,  $a^2 - x^2 = a^2 \operatorname{dn}^2 t$ ,  $d\phi = b \operatorname{cn} t \operatorname{dn} t dt$ .

It follows that

$$V = 8ab^2 \int_0^K \left[ 1 - \frac{a^2 + b^2}{a^2} \operatorname{sn}^2 t + \frac{b^2}{a^2} \operatorname{sn}^4 t \right] dt. \quad (\text{See Byerly, } \textit{Int. Cal.}, 1902, \text{p. 276.})$$

Noting (see sixth formula of Art. 41, and (ii) of Art. 48) that

$$\int_0^K \operatorname{sn}^2 t dt = \frac{1}{k^2} [K - E] \quad \text{and} \quad 3k^4 \int_0^K \operatorname{sn}^4 t dt = 2K - 2E + k^2 K - 2k^2 E, \quad k^2 = \frac{b^2}{a^2},$$

it follows at once that

$$V = \frac{8}{3} a [(a^2 + b^2)E - (a^2 - b^2)K].$$

Compute  $V$  when  $a = 60$  and  $b = 12$  centimeters respectively; also find the volume common to both when the shortest distance between the axes is 8 centimeters.

7. The differential equation of motion of the simple pendulum is

$$\frac{d^2 s}{dt^2} = -g \frac{dy}{ds};$$

or multiplying by  $\frac{2ds}{dt}$  and integrating,

$$\left( \frac{ds}{dt} \right)^2 = -2gy + C.$$

If the pendulum bob starts from the lowest point of its circular path with the initial velocity that would be acquired by a particle falling freely in a vacuum through the distance  $y_0$ , so that  $v_0^2 = 2gy_0$  (Byerly, loc. cit., p. 215), it is seen that this is the value of  $C$ , and consequently

$$\left( \frac{ds}{dt} \right)^2 = 2g(y_0 - y).$$

Further taking the starting-point as the origin (see figure) the equation of the circular path is  $x^2 + y^2 - 2ay = 0$ , so that

$$\left( \frac{ds}{dt} \right)^2 = \frac{a^2}{2ay - y^2} \left( \frac{dy}{dt} \right)^2,$$

and consequently

$$t = \frac{a}{\sqrt{2g}} \int_0^y \frac{dy}{\sqrt{(y_0 - y)(2ay - y^2)}},$$

which is the time required to reach that point of the path whose ordinate is  $y$ .

Writing  $k^2 = \frac{y_0}{2a}$  and  $\sin^2 \phi = \frac{y}{y_0}$ , this integral becomes at once

$$t = \sqrt{\frac{a}{g}} \int_0^\phi \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \sqrt{\frac{a}{g}} F(k, \phi).$$

Let  $OC=CA=a$  be the length of the pendulum. Let  $A$  be the highest point reached by it in the oscillation so that the ordinate of  $A$  is  $y_0$ . Let the angle  $ACO$  be  $\alpha$ , and let  $\theta$  be the angle  $PCO$ , where  $P$  is the point reached at the expiration of the time  $t$ .

It is seen that

$$\frac{y_0}{a} = 1 - \cos \alpha,$$

so that

$$\sqrt{\frac{y_0}{2a}} = \sqrt{\frac{1}{2}(1 - \cos \alpha)} = \sin \frac{\alpha}{2} = k;$$

and similarly,

$$\sqrt{\frac{y}{2a}} = \sin \frac{\theta}{2}.$$

It follows also that

$$\sin \phi = \sqrt{\frac{g}{y_0}} = \frac{\sin \frac{\theta}{2}}{\sin \frac{\alpha}{2}}.$$

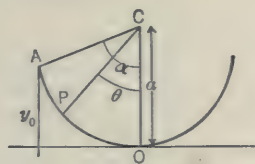


FIG. 20.

When  $\theta = \alpha$ ,  $\sin \phi = 1$ , or  $\phi = \frac{\pi}{2}$ , and consequently, the time of a half-oscillation is  $\sqrt{\frac{a}{g}} F\left(\sin \frac{\alpha}{2}, \frac{\pi}{2}\right)$ .

Show by Table I that when  $a = 36^\circ$ , the time of oscillation is 1.0253 . . .

times greater than that given by the approximate formula  $t = \sqrt{\frac{a}{g}} \pi$ .

The following problems taken from Byerly's Calculus are instructive:

8. A pendulum swings through an angle of  $180^\circ$ ; required, the time of oscillation. *Ans.*  $3.708\sqrt{\frac{a}{g}}$ .

9. The time of vibration of a pendulum swinging in an arc of  $72^\circ$  is observed to be 2 seconds; how long does it take it to fall through an arc of  $5^\circ$ , beginning at a point  $20^\circ$  from the highest point of the arc of swing? *Ans.* 0.095 . . . second.

10. A pendulum for which  $\sqrt{\frac{a}{g}}$  is  $\frac{1}{2}$ , vibrates through an arc of  $180^\circ$ ; through what arc does it rise in the first half second after it has passed its lowest point? In the first  $\frac{1}{8}$  of a second? *Ans.*  $69^\circ$ ;  $20^\circ 6'$ .

11. Show that a pendulum which beats seconds when swinging through an angle of  $6^\circ$ , will lose 11 to 12 seconds a day if made to swing through  $8^\circ$  and 26 seconds a day if made to swing through  $10^\circ$ .

(Simpson's *Fluxions*, § 464.)



## CHAPTER VI

### FIVE-PLACE TABLES

THE following tables of integrals are given in Levy's *Théorie des fonctions elliptiques*. As stated by Professor Levy, he was assisted by Professor G. Humbert in compiling these tables from the ten-place tables that are found in the second volume of Legendre's Treatise.

Table I gives values of the integrals

$$K = \int_0^{\frac{1}{2}\pi} \frac{d\phi}{\sqrt{1 - \sin^2 \theta \sin^2 \phi}} \quad \text{and} \quad E = \int_0^{\frac{1}{2}\pi} d\phi \sqrt{1 - \sin^2 \theta \sin^2 \phi}.$$

For example, if  $\theta = 78^\circ 30'$ , then  $K = 3.01918$  and  $E = 1.05024$ .

Table II gives values of the integral

$$F(k, \phi) = \int_0^\phi \frac{d\phi}{\sqrt{1 - \sin^2 \theta \sin^2 \phi}}.$$

For example, if  $\theta = 65^\circ$  and  $\phi = 81^\circ$ , then  $F(k, \phi) = 1.94377$ .

Table III gives values of the integral

$$E(k, \phi) = \int_0^\phi d\phi \sqrt{1 - \sin^2 \theta \sin^2 \phi}.$$

For example, if  $\theta = 40^\circ$  and  $\phi = 34^\circ$ , then  $E(k, \phi) = 0.57972$ .

## I.—THE COMPLETE ELLIPTIC INTEGRALS OF THE FIRST AND SECOND KINDS

| $\theta$ | K       | E       | $\theta$ | K       | E       | $\theta$ | K        | E       |
|----------|---------|---------|----------|---------|---------|----------|----------|---------|
| 0°       | 1.57080 | 1.57080 | 50°      | 1.93558 | 1.30554 | 82° 0'   | 3.36987  | 1.02784 |
| 1        | 092     | 068     | 51       | 5386    | 29628   | 12       | 9457     | 670     |
| 2        | 127     | 032     | 52       | 7288    | 28695   | 24       | 3.41994  | 558     |
| 3        | 187     | 1.56972 | 53       | 9267    | 27757   | 36       | 4601     | 447     |
| 4        | 271     | 888     | 54       | 2.01327 | 26815   | 48       | 7282     | 338     |
| 5        | 379     | 781     | 55       | 3472    | 25868   | 83 0     | 3.50042  | 231     |
| 6        | 511     | 650     | 56       | 5706    | 24918   | 12       | 2884     | 126     |
| 7        | 668     | 495     | 57       | 8036    | 23966   | 24       | 5814     | 023     |
| 8        | 849     | 296     | 58       | 2.10466 | 23013   | 36       | 8837     | 1921    |
| 9        | 1.58054 | 114     | 59       | 3002    | 22059   | 48       | 3.61959  | 821     |
| 10       | 284     | 1.55889 | 60       | 2.15652 | 21106   | 84 0     | 3.65186  | 1.01724 |
| 11       | 539     | 640     | 61       | 8421    | 20154   | 12       | 8525     | 628     |
| 12       | 820     | 368     | 62       | 2.21319 | 19205   | 24       | 3.71984  | 534     |
| 13       | 1.59125 | 073     | 63       | 4355    | 18259   | 36       | 5572     | 443     |
| 14       | 457     | 1.54755 | 64       | 7538    | 17318   | 48       | 9298     | 354     |
| 15       | 814     | 415     | 65       | 2.30879 | 16383   | 85 0     | 3.83174  | 266     |
| 16       | 1.60198 | 052     | 66       | 4390    | 15455   | 12       | 7211     | 181     |
| 17       | 608     | 1.53667 | 67       | 8087    | 14535   | 24       | 3.91423  | 099     |
| 18       | 1.61045 | 260     | 68       | 2.41984 | 13624   | 36       | 5827     | 018     |
| 19       | 510     | 1.52831 | 69       | 6100    | 12725   | 48       | 4.00437  | 0940    |
| 20       | 1.62003 | 380     | 70° 0'   | 2.50455 | 11838   | 86 0     | 5276     | 865     |
| 21       | 523     | 1.51908 | 30       | 2729    | 11399   | 12       | 4.10366  | 792     |
| 22       | 1.63073 | 415     | 71 0     | 5073    | 10964   | 24       | 5736     | 721     |
| 23       | 632     | 1.50901 | 30       | 7490    | 10533   | 36       | 4.21416  | 653     |
| 24       | 1.64260 | 366     | 72 0     | 9982    | 106     | 48       | 7444     | 588     |
| 25       | 900     | 1.49811 | 30       | 2.62555 | 09683   | 87 0     | 4.33865  | 526     |
| 26       | 1.65570 | 237     | 73 0     | 5214    | 265     | 12       | 40733    | 466     |
| 27       | 6272    | 1.48643 | 30       | 7962    | 8851    | 24       | 8115     | 410     |
| 28       | 7006    | 7006    | 74 0     | 2.70807 | 443     | 36       | 56190    | 356     |
| 29       | 7773    | 1.47397 | 30       | 3752    | 039     | 48       | 64765    | 306     |
| 30       | 1.68575 | 1.46746 | 75 0     | 6806    | 7641    | 88 0     | 74272    | 258     |
| 31       | 9411    | 077     | 30       | 9975    | 248     | 12       | 84785    | 215     |
| 32       | 1.70284 | 1.45391 | 76 0     | 2.83267 | 6861    | 24       | 96542    | 174     |
| 33       | 1192    | 44687   | 30       | 6691    | 480     | 36       | 5.09876  | 137     |
| 34       | 2139    | 43966   | 77 0     | 2.90256 | 106     | 48       | 25274    | 104     |
| 35       | 3125    | 229     | 30       | 3974    | 5738    | 89 0     | 43491    | 075     |
| 36       | 4150    | 42476   | 78 0     | 7857    | 378     | 6        | 54020    | 062     |
| 37       | 5217    | 41707   | 30       | 3.01918 | 024     | 12       | 65792    | 050     |
| 38       | 6326    | 40924   | 79 0     | 6173    | 4679    | 18       | 79140    | 049     |
| 39       | 7479    | 126     | 30       | 3.10640 | 4341    | 24       | 94550    | 030     |
| 40       | 1.78677 | 1.39314 | 80 0     | 5339    | 011     | 30       | 6.12778  | 021     |
| 41       | 9922    | 38489   | 12       | 7288    | 3882    | 36       | 35038    | 014     |
| 42       | 1.81216 | 37650   | 24       | 9280    | 754     | 42       | 63854    | 008     |
| 43       | 2560    | 36800   | 36       | 3.21317 | 628     | 48       | 7.04398  | 004     |
| 44       | 3957    | 35938   | 48       | 3400    | 503     | 54       | 73711    | 011     |
| 45       | 5407    | 35064   | 81 0     | 5530    | 379     | 90       | $\infty$ | 000     |
| 46       | 6915    | 34181   | 12       | 7711    | 257     |          |          |         |
| 47       | 8481    | 33287   | 24       | 9945    | 126     |          |          |         |
| 48       | 1.90108 | 32384   | 36       | 3.32234 | 017     |          |          |         |
| 49       | 1800    | 31473   | 48       | 4580    | 2900    |          |          |         |

## II.—ELLIPTIC INTEGRALS OF THE FIRST KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 0°       | 5°      | 10°     | 15°     | 20°     | 25°     | 30°     | 35°     | 40°     | 45°     |
| 1°     | 0.01745  | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 |
| 2      | 03491    | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   |
| 3      | 05236    | 05236   | 05236   | 05236   | 05236   | 05236   | 05237   | 05237   | 05237   | 05237   |
| 4      | 06981    | 06981   | 06981   | 06982   | 06982   | 06982   | 06983   | 06983   | 06984   | 06984   |
| 5      | 08727    | 08727   | 08727   | 08727   | 08728   | 08729   | 08729   | 08730   | 08731   | 08732   |
| 6      | 10472    | 10472   | 10473   | 10473   | 10474   | 10475   | 10477   | 10478   | 10480   | 10482   |
| 7      | 12217    | 12218   | 12218   | 12219   | 12221   | 12223   | 12225   | 12227   | 12230   | 12233   |
| 8      | 13963    | 13963   | 13964   | 13966   | 13968   | 13971   | 13974   | 13978   | 13981   | 13985   |
| 9      | 15708    | 15708   | 15710   | 15712   | 15715   | 15719   | 15724   | 15729   | 15735   | 15740   |
| 10     | 17453    | 17454   | 17456   | 17459   | 17464   | 17469   | 17475   | 17482   | 17490   | 17498   |
| 11     | 19199    | 19200   | 19202   | 19206   | 19212   | 19220   | 19228   | 19237   | 19247   | 19258   |
| 12     | 20944    | 20945   | 20949   | 20954   | 20962   | 20971   | 20982   | 20994   | 21007   | 21021   |
| 13     | 22689    | 22691   | 22695   | 22702   | 22712   | 22724   | 22738   | 22753   | 22770   | 22787   |
| 14     | 24435    | 24436   | 24442   | 24451   | 24463   | 24478   | 24495   | 24514   | 24535   | 24556   |
| 15     | 26180    | 26182   | 26189   | 26200   | 26215   | 26233   | 26254   | 26278   | 26303   | 26330   |
| 16     | 27925    | 27928   | 27936   | 27949   | 27967   | 27989   | 28015   | 28044   | 28075   | 28107   |
| 17     | 29671    | 29674   | 29684   | 29699   | 29721   | 29748   | 29779   | 29813   | 29850   | 29889   |
| 18     | 31416    | 31420   | 31431   | 31450   | 31475   | 31507   | 31544   | 31585   | 31629   | 31675   |
| 19     | 33161    | 33166   | 33179   | 33201   | 33231   | 33268   | 33312   | 33360   | 33412   | 33466   |
| 20     | 34907    | 34912   | 34927   | 34953   | 34988   | 35031   | 35082   | 35138   | 35199   | 35262   |
| 21     | 36652    | 36658   | 36676   | 36706   | 36746   | 36796   | 36855   | 36920   | 36990   | 37063   |
| 22     | 38397    | 38404   | 38425   | 38459   | 38505   | 38563   | 38630   | 38705   | 38786   | 38871   |
| 23     | 40143    | 40151   | 40174   | 40213   | 40266   | 40331   | 40408   | 40494   | 40587   | 40683   |
| 24     | 41888    | 41897   | 41924   | 41968   | 42027   | 42102   | 42189   | 42287   | 42392   | 42503   |
| 25     | 43633    | 43643   | 43674   | 43723   | 43791   | 43875   | 43973   | 44084   | 44203   | 44328   |
| 26     | 45379    | 45390   | 45424   | 45479   | 45555   | 45650   | 45761   | 45885   | 46020   | 46161   |
| 27     | 47124    | 47137   | 47174   | 47236   | 47321   | 47427   | 47551   | 47690   | 47841   | 48000   |
| 28     | 48869    | 48883   | 48925   | 48994   | 49080   | 49207   | 49345   | 49500   | 49669   | 49846   |
| 29     | 50615    | 50630   | 50677   | 50753   | 50858   | 50988   | 51142   | 51315   | 51503   | 51700   |
| 30     | 52360    | 52377   | 52428   | 52513   | 52628   | 52773   | 52943   | 53134   | 53343   | 53562   |
| 31     | 54105    | 54124   | 54181   | 54273   | 54401   | 54560   | 54747   | 54959   | 55189   | 55432   |
| 32     | 55851    | 55871   | 55933   | 56035   | 56175   | 56349   | 56555   | 56788   | 57042   | 57310   |
| 33     | 57596    | 57619   | 57686   | 57797   | 57950   | 58141   | 58367   | 58623   | 58902   | 59197   |
| 34     | 59341    | 59366   | 59439   | 59561   | 59727   | 59936   | 60183   | 60463   | 60769   | 61093   |
| 35     | 61087    | 61113   | 61193   | 61325   | 61506   | 61734   | 62003   | 62308   | 62643   | 62998   |
| 36     | 62832    | 62861   | 62948   | 63090   | 63287   | 63534   | 63827   | 64159   | 64524   | 64912   |
| 37     | 64577    | 64609   | 64702   | 64857   | 65070   | 65337   | 65655   | 66016   | 66413   | 66836   |
| 38     | 66323    | 66356   | 66457   | 66624   | 66854   | 67144   | 67487   | 67879   | 68309   | 68769   |
| 39     | 68068    | 68104   | 68213   | 68393   | 68641   | 68953   | 69324   | 69747   | 70214   | 70713   |
| 40     | 69813    | 69852   | 69969   | 70162   | 70429   | 70765   | 71165   | 71622   | 72126   | 72667   |
| 41     | 71558    | 71600   | 71726   | 71933   | 72219   | 72580   | 73010   | 73502   | 74047   | 74632   |
| 42     | 73304    | 73349   | 73483   | 73704   | 74011   | 74398   | 74860   | 75380   | 75976   | 76608   |
| 43     | 75049    | 75097   | 75240   | 75477   | 75805   | 76219   | 76714   | 77282   | 77914   | 78594   |
| 44     | 76794    | 76846   | 76998   | 77251   | 77600   | 78043   | 78573   | 79182   | 79860   | 80594   |
| 45     | 0.78540  | 0.78594 | 0.78756 | 0.79025 | 0.79398 | 0.79871 | 0.80437 | 0.81088 | 0.81815 | 0.82602 |

## II.—ELLIPTIC INTEGRALS OF THE FIRST KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 50°      | 55°     | 60°     | 65°     | 70°     | 75°     | 80°     | 85°     | 90°     |
| 1°     | 0.01745  | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 |
| 2      | 03491    | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   | 03491   |
| 3      | 05237    | 05238   | 05238   | 05238   | 05238   | 05238   | 05238   | 05238   | 05238   |
| 4      | 06985    | 06985   | 06986   | 06986   | 06986   | 06987   | 06987   | 06987   | 06987   |
| 5      | 08733    | 08734   | 08735   | 08736   | 08736   | 08737   | 08737   | 08738   | 08738   |
| 6      | 10483    | 10485   | 10486   | 10488   | 10489   | 10490   | 10491   | 10491   | 10491   |
| 7      | 12235    | 12238   | 12240   | 12242   | 12244   | 12246   | 12247   | 12248   | 12248   |
| 8      | 13989    | 13993   | 13997   | 14000   | 14003   | 14005   | 14007   | 14008   | 14008   |
| 9      | 15746    | 15751   | 15757   | 15761   | 15765   | 15769   | 15771   | 15772   | 15773   |
| 10     | 17505    | 17513   | 17520   | 17526   | 17532   | 17536   | 17540   | 17542   | 17543   |
| 11     | 19268    | 19278   | 19288   | 19296   | 19304   | 19310   | 19314   | 19317   | 19318   |
| 12     | 21034    | 21047   | 21059   | 21071   | 21080   | 21088   | 21094   | 21098   | 21099   |
| 13     | 22804    | 22821   | 22836   | 22851   | 22863   | 22873   | 22880   | 22885   | 22886   |
| 14     | 24578    | 24599   | 24618   | 24636   | 24652   | 24664   | 24674   | 24680   | 24681   |
| 15     | 26356    | 26382   | 26406   | 26428   | 26448   | 26463   | 26475   | 26482   | 26484   |
| 16     | 28139    | 28171   | 28200   | 28227   | 28251   | 28270   | 28284   | 28293   | 28295   |
| 17     | 29927    | 29965   | 30001   | 30034   | 30062   | 30085   | 30102   | 30112   | 30116   |
| 18     | 31721    | 31766   | 31809   | 31848   | 31881   | 31909   | 31929   | 31942   | 31946   |
| 19     | 33520    | 33574   | 33624   | 33670   | 33710   | 33742   | 33766   | 33781   | 33786   |
| 20     | 35326    | 35388   | 35447   | 35501   | 35548   | 35586   | 35615   | 35632   | 35638   |
| 21     | 37137    | 37210   | 37279   | 37342   | 37396   | 37441   | 37474   | 37494   | 37501   |
| 22     | 38956    | 39040   | 39119   | 39192   | 39255   | 39307   | 39346   | 39369   | 39377   |
| 23     | 40782    | 40878   | 40969   | 41053   | 41126   | 41186   | 41230   | 41257   | 41266   |
| 24     | 42614    | 42724   | 42829   | 42925   | 43008   | 43077   | 43128   | 43159   | 43169   |
| 25     | 44455    | 44580   | 44699   | 44808   | 44904   | 44982   | 45040   | 45075   | 45088   |
| 26     | 46304    | 46445   | 46580   | 46704   | 46812   | 46901   | 46967   | 47008   | 47021   |
| 27     | 48161    | 48320   | 48472   | 48612   | 48735   | 48835   | 48910   | 48956   | 48972   |
| 28     | 50027    | 50206   | 50377   | 50534   | 50672   | 50785   | 50870   | 50922   | 50939   |
| 29     | 51902    | 52102   | 52293   | 52470   | 52624   | 52752   | 52847   | 52905   | 52925   |
| 30     | 53787    | 54009   | 54223   | 54420   | 54593   | 54736   | 54843   | 54908   | 54931   |
| 31     | 55681    | 55928   | 56166   | 56386   | 56579   | 56739   | 56858   | 56931   | 56956   |
| 32     | 57586    | 57860   | 58123   | 58367   | 58582   | 58760   | 58893   | 58975   | 59003   |
| 33     | 59501    | 59803   | 60095   | 60365   | 60604   | 60802   | 60950   | 61042   | 61073   |
| 34     | 61427    | 61760   | 62082   | 62381   | 62646   | 62865   | 63029   | 63131   | 63166   |
| 35     | 63364    | 63730   | 64085   | 64415   | 64707   | 64950   | 65132   | 65245   | 65284   |
| 36     | 65313    | 65715   | 66104   | 66468   | 66790   | 67058   | 67260   | 67385   | 67428   |
| 37     | 67273    | 67713   | 68141   | 68540   | 68895   | 69131   | 69414   | 69552   | 69599   |
| 38     | 69246    | 69727   | 70195   | 70633   | 71023   | 71349   | 71594   | 71747   | 71799   |
| 39     | 71232    | 71756   | 72267   | 72746   | 73175   | 73533   | 73804   | 73972   | 74029   |
| 40     | 73231    | 73801   | 74358   | 74882   | 75352   | 75745   | 76043   | 76228   | 76291   |
| 41     | 75243    | 75862   | 76469   | 77041   | 77555   | 77987   | 78313   | 78517   | 78586   |
| 42     | 77269    | 77940   | 78600   | 79224   | 79786   | 80258   | 80617   | 80841   | 80917   |
| 43     | 79308    | 80035   | 80752   | 81432   | 82045   | 82562   | 82954   | 83200   | 83284   |
| 44     | 81362    | 82149   | 82926   | 83665   | 84333   | 84898   | 85329   | 85598   | 85690   |
| 45     | 0.83431  | 0.84281 | 0.85122 | 0.85925 | 0.86653 | 0.87270 | 0.87741 | 0.88037 | 0.88137 |



## II.—ELLIPTIC INTEGRALS OF THE FIRST KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 0°       | 5°      | 10°     | 15°     | 20°     | 25°     | 30°     | 35°     | 40°     | 45°     |
| 46°    | 0.80285  | 0.80343 | 0.80515 | 0.80801 | 0.81198 | 0.81701 | 0.82305 | 0.83001 | 0.83779 | 0.84623 |
| 47     | 82030    | 82092   | 82275   | 82578   | 82999   | 83535   | 84178   | 84920   | 85752   | 86656   |
| 48     | 83776    | 83841   | 84035   | 84356   | 84803   | 85371   | 86055   | 86846   | 87734   | 88701   |
| 49     | 85521    | 85590   | 85795   | 86135   | 86609   | 87211   | 87937   | 88779   | 89725   | 90759   |
| 50     | 87266    | 87339   | 87556   | 87915   | 88416   | 89054   | 89825   | 90719   | 91725   | 92829   |
| 51     | 89012    | 89088   | 89317   | 89697   | 90226   | 90901   | 91716   | 92665   | 93735   | 94912   |
| 52     | 90757    | 90838   | 91078   | 91479   | 92037   | 92750   | 93613   | 94618   | 95755   | 97007   |
| 53     | 92502    | 92587   | 92841   | 93262   | 93850   | 94603   | 95514   | 96578   | 97784   | 0.99115 |
| 54     | 94248    | 94337   | 94603   | 95047   | 95666   | 96458   | 97420   | 0.98545 | 0.99822 | 1.01237 |
| 55     | 95993    | 96086   | 96366   | 96832   | 97483   | 0.98317 | 0.99331 | 1.00519 | 1.01871 | 0.9371  |
| 56     | 97738    | 97836   | 98130   | 0.98618 | 0.99302 | 1.00179 | 1.01247 | 0.92499 | 0.93928 | 0.95519 |
| 57     | 0.99484  | 0.99586 | 0.99894 | 1.00406 | 1.01123 | 0.92044 | 0.93167 | 0.94487 | 0.95996 | 0.97680 |
| 58     | 1.01229  | 1.01336 | 1.01658 | 0.92194 | 0.92946 | 0.93912 | 0.95092 | 0.96481 | 0.98073 | 0.99854 |
| 59     | 0.92974  | 0.93086 | 0.93423 | 0.93984 | 0.94770 | 0.95783 | 0.97021 | 0.98482 | 1.00159 | 1.02042 |
| 60     | 0.94720  | 0.94837 | 0.95188 | 0.95774 | 0.96597 | 0.97657 | 0.98955 | 1.00490 | 1.02256 | 1.04243 |
| 61     | 0.96465  | 0.96587 | 0.96954 | 0.97566 | 0.98425 | 0.99534 | 1.00894 | 1.02504 | 1.04361 | 1.06457 |
| 62     | 0.98210  | 0.98338 | 0.98720 | 0.99358 | 1.00255 | 1.01414 | 1.02837 | 1.04525 | 1.06476 | 1.08685 |
| 63     | 0.99956  | 1.00088 | 1.00486 | 1.01151 | 1.02087 | 1.03296 | 1.04784 | 1.06552 | 1.08601 | 2.0926  |
| 64     | 1.01701  | 1.01839 | 1.02253 | 1.02945 | 1.03920 | 1.05182 | 1.06735 | 1.08586 | 2.09735 | 2.1380  |
| 65     | 1.03446  | 1.03590 | 1.04020 | 1.04740 | 1.05755 | 1.07070 | 1.08691 | 2.0626  | 2.2877  | 2.5447  |
| 66     | 1.05192  | 1.05340 | 1.05787 | 1.06536 | 1.07592 | 1.08961 | 2.0651  | 2.2672  | 2.5029  | 2.7727  |
| 67     | 1.06937  | 1.07091 | 1.07555 | 1.08333 | 1.09430 | 2.0854  | 2.2615  | 2.4724  | 2.7190  | 3.0020  |
| 68     | 1.08682  | 1.08842 | 1.09324 | 2.0130  | 2.1269  | 2.2750  | 2.4583  | 2.6782  | 2.9359  | 3.2325  |
| 69     | 2.0428   | 2.0593  | 2.1092  | 2.1928  | 2.3110  | 2.4648  | 2.6555  | 2.8846  | 3.1537  | 3.4642  |
| 70     | 2.2173   | 2.2345  | 2.2861  | 2.3727  | 2.4953  | 2.6548  | 2.8530  | 3.0915  | 3.3723  | 3.6972  |
| 71     | 2.3918   | 2.4096  | 2.4630  | 2.5527  | 2.6796  | 2.8451  | 3.0509  | 3.2990  | 3.5917  | 3.9313  |
| 72     | 2.5664   | 2.5847  | 2.6400  | 2.7328  | 2.8641  | 3.0356  | 3.2491  | 3.5070  | 3.8118  | 4.1666  |
| 73     | 2.7409   | 2.7599  | 2.8169  | 2.9129  | 3.0488  | 3.2263  | 3.4477  | 3.7155  | 4.0328  | 4.4030  |
| 74     | 2.9154   | 2.9350  | 2.9939  | 3.0930  | 3.2335  | 3.4172  | 3.6466  | 3.9244  | 4.2544  | 4.6404  |
| 75     | 3.0900   | 3.1102  | 3.1710  | 3.2733  | 3.4184  | 3.6083  | 3.8457  | 4.1339  | 4.4767  | 4.8788  |
| 76     | 3.2645   | 3.2853  | 3.3480  | 3.4535  | 3.6034  | 3.7996  | 4.0452  | 4.3437  | 4.6997  | 5.1183  |
| 77     | 3.4390   | 3.4605  | 3.5251  | 3.6339  | 3.7884  | 3.9911  | 4.2449  | 4.5540  | 4.9232  | 5.3586  |
| 78     | 3.6136   | 3.6356  | 3.7022  | 3.8143  | 3.9736  | 4.1827  | 4.4449  | 4.7647  | 5.1474  | 5.5999  |
| 79     | 3.7881   | 3.8108  | 3.8793  | 3.9947  | 4.1588  | 4.3744  | 4.6451  | 4.9757  | 5.3721  | 5.8419  |
| 80     | 3.9626   | 3.9860  | 4.0565  | 4.1752  | 4.3442  | 4.5663  | 4.8455  | 5.1870  | 5.5973  | 6.0848  |
| 81     | 4.1372   | 4.1612  | 4.2336  | 4.3557  | 4.5296  | 4.7583  | 5.0462  | 5.3987  | 5.8230  | 6.3283  |
| 82     | 4.3117   | 4.3364  | 4.4108  | 4.5362  | 4.7150  | 4.9504  | 5.2470  | 5.6106  | 6.0491  | 6.5725  |
| 83     | 4.4862   | 4.5115  | 4.5879  | 4.7168  | 4.9005  | 5.1426  | 5.4479  | 5.8228  | 6.2756  | 6.8172  |
| 84     | 4.6608   | 4.6867  | 4.7651  | 4.8974  | 5.0861  | 5.3350  | 5.6490  | 6.0352  | 6.5024  | 7.0625  |
| 85     | 4.8353   | 4.8619  | 4.9423  | 5.0781  | 5.2717  | 5.5273  | 5.8503  | 6.2478  | 6.7295  | 7.3082  |
| 86     | 5.0098   | 5.0371  | 5.1195  | 5.2587  | 5.4574  | 5.7198  | 6.0516  | 6.4605  | 6.9569  | 7.5542  |
| 87     | 5.1844   | 5.2123  | 5.2968  | 5.4394  | 5.6431  | 5.9123  | 6.2530  | 6.6734  | 7.1844  | 7.8006  |
| 88     | 5.3589   | 5.3875  | 5.4740  | 5.6200  | 5.8288  | 6.1048  | 6.4545  | 6.8864  | 7.4121  | 8.0472  |
| 89     | 5.5334   | 5.5627  | 5.6512  | 5.8007  | 6.0145  | 6.2974  | 6.6560  | 7.0994  | 7.6399  | 8.2939  |
| 90     | 1.57080  | 1.57379 | 1.58284 | 1.59814 | 1.62003 | 1.64900 | 1.68575 | 1.73125 | 1.78677 | 1.85407 |



## II.—ELLIPTIC INTEGRALS OF THE FIRST KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |          |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|----------|
|        | 50°      | 55°     | 60°     | 65°     | 70°     | 75°     | 80°     | 85°     | 90°      |
| 46°    | 0.85515  | 0.86431 | 0.87342 | 0.88213 | 0.89005 | 0.89678 | 0.90193 | 0.90517 | 0.90628  |
| 47     | 87614    | 88601   | 89585   | 90529   | 91390   | 92224   | 92687   | 93042   | 93163    |
| 48     | 89729    | 90791   | 91853   | 92875   | 93811   | 94610   | 95226   | 95614   | 95747    |
| 49     | 91860    | 93001   | 94146   | 95252   | 96267   | 97139   | 0.97810 | 0.98235 | 0.98381  |
| 50     | 94008    | 95232   | 96465   | 0.97660 | 0.98762 | 0.99711 | 1.00444 | 1.00909 | 1.01068  |
| 51     | 96171    | 97484   | 0.98811 | 1.00102 | 1.01297 | 1.02329 | 03129   | 03638   | 03812    |
| 52     | 0.98352  | 0.99759 | 1.01185 | 02578   | 03872   | 04995   | 05868   | 06425   | 06616    |
| 53     | 1.00550  | 1.02055 | 03587   | 05089   | 06491   | 07711   | 08665   | 09274   | 09483    |
| 54     | 02765    | 04374   | 06018   | 07637   | 09155   | 10481   | 11521   | 12188   | 12418    |
| 55     | 04998    | 06716   | 08479   | 10223   | 11865   | 13307   | 14442   | 15171   | 15423    |
| 56     | 07248    | 09082   | 10971   | 12848   | 14624   | 16190   | 17430   | 18229   | 18505    |
| 57     | 09517    | 11472   | 13494   | 15513   | 17433   | 19136   | 20488   | 21364   | 21667    |
| 58     | 11803    | 13886   | 16050   | 18220   | 20295   | 22145   | 23623   | 24582   | 24916    |
| 59     | 14108    | 16325   | 18638   | 20970   | 23212   | 25223   | 26837   | 27890   | 28257    |
| 60     | 16432    | 18788   | 21254   | 23764   | 26186   | 28371   | 30135   | 31292   | 31696    |
| 61     | 18773    | 21277   | 23916   | 26604   | 29219   | 31594   | 33524   | 34795   | 35240    |
| 62     | 21134    | 23792   | 26606   | 29490   | 32314   | 34897   | 37008   | 38407   | 38899    |
| 63     | 23513    | 26332   | 29332   | 32425   | 35473   | 38281   | 40594   | 42135   | 42679    |
| 64     | 25910    | 28898   | 32094   | 35409   | 38699   | 41753   | 44288   | 45989   | 46591    |
| 65     | 28326    | 31491   | 34893   | 38443   | 41994   | 45316   | 48098   | 49977   | 50645    |
| 66     | 30760    | 34109   | 37728   | 41529   | 45360   | 48976   | 52031   | 54112   | 54855    |
| 67     | 33212    | 36753   | 40600   | 44668   | 48800   | 52738   | 56096   | 58404   | 59232    |
| 68     | 35683    | 39423   | 43510   | 47860   | 52317   | 56606   | 60303   | 62868   | 63794    |
| 69     | 38171    | 42119   | 46457   | 51107   | 55913   | 60586   | 64661   | 67518   | 68557    |
| 70     | 40677    | 44840   | 49441   | 54410   | 59591   | 64684   | 69181   | 72372   | 73542    |
| 71     | 43200    | 47587   | 52463   | 57768   | 63352   | 68905   | 73877   | 77450   | 78771    |
| 72     | 45739    | 50359   | 55522   | 61182   | 67198   | 73256   | 78759   | 82774   | 84273    |
| 73     | 48296    | 53155   | 58618   | 64653   | 71132   | 77743   | 83844   | 88370   | 90079    |
| 74     | 50867    | 55974   | 61750   | 68180   | 75155   | 82371   | 89146   | 1.94267 | 1.96226  |
| 75     | 53455    | 58817   | 64918   | 71763   | 79269   | 87145   | 1.94682 | 2.00499 | 2.02759  |
| 76     | 56056    | 61682   | 68120   | 75401   | 83473   | 92073   | 2.00470 | 07106   | 09732    |
| 77     | 58672    | 64560   | 71356   | 79094   | 87768   | 1.97157 | 06529   | 14136   | 17212    |
| 78     | 61302    | 67476   | 74625   | 82840   | 92154   | 2.02403 | 12878   | 21644   | 25280    |
| 79     | 63943    | 70403   | 77924   | 86637   | 1.96630 | 07813   | 19538   | 29694   | 34040    |
| 80     | 66597    | 73347   | 81253   | 90484   | 2.01193 | 13390   | 26527   | 38365   | 43625    |
| 81     | 69261    | 76309   | 84609   | 94377   | 05840   | 19131   | 33866   | 47748   | 54209    |
| 82     | 71935    | 79286   | 87991   | 1.98313 | 10568   | 25035   | 41569   | 57954   | 66031    |
| 83     | 74618    | 82278   | 91395   | 2.02290 | 15371   | 31097   | 49648   | 69109   | 79422    |
| 84     | 77309    | 85281   | 94821   | 06303   | 20244   | 37309   | 58105   | 81362   | 2.94870  |
| 85     | 80006    | 88296   | 1.98264 | 10348   | 25178   | 43658   | 66935   | 2.94869 | 3.13130  |
| 86     | 82710    | 91320   | 2.01723 | 14421   | 30166   | 50129   | 76116   | 3.09782 | 35467    |
| 87     | 85418    | 94351   | 05194   | 18515   | 35108   | 56703   | 85612   | 26198   | 3.64253  |
| 88     | 88129    | 1.97388 | 08674   | 22627   | 40265   | 63357   | 2.95366 | 44116   | 4.04813  |
| 89     | 90843    | 2.00429 | 12161   | 26750   | 45354   | 70068   | 3.05304 | 63279   | 4.74135  |
| 90     | 1.93558  | 2.03472 | 2.15652 | 2.30879 | 2.50455 | 2.76806 | 3.15339 | 3.83174 | $\infty$ |

## III.—ELLIPTIC INTEGRALS OF THE SECOND KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 0°       | 5°      | 10°     | 15°     | 20°     | 25°     | 30°     | 35°     | 40°     | 45°     |
| 1      | 0.01745  | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 |
| 2      | 03491    | 03491   | 03491   | 03491   | 03491   | 03491   | 03490   | 03490   | 03490   | 03490   |
| 3      | 05236    | 05236   | 05236   | 05236   | 05236   | 05236   | 05235   | 05235   | 05235   | 05235   |
| 4      | 06981    | 06981   | 06981   | 06981   | 06981   | 06980   | 06980   | 06979   | 06979   | 06978   |
| 5      | 08727    | 08727   | 08726   | 08726   | 08725   | 08725   | 08744   | 08723   | 08722   | 08721   |
| 6      | 10472    | 10472   | 10471   | 10471   | 10470   | 10469   | 10467   | 10466   | 10464   | 10462   |
| 7      | 12217    | 12217   | 12216   | 12215   | 12214   | 12212   | 12210   | 12207   | 12205   | 12202   |
| 8      | 13963    | 13962   | 13961   | 13960   | 13957   | 13955   | 13951   | 13948   | 13944   | 13940   |
| 9      | 15708    | 15707   | 15706   | 15704   | 15700   | 15696   | 15692   | 15687   | 15681   | 15676   |
| 10     | 17453    | 17453   | 17451   | 17447   | 17443   | 17438   | 17431   | 17427   | 17417   | 17409   |
| 11     | 19199    | 19198   | 19195   | 19191   | 19185   | 19178   | 19169   | 19160   | 19150   | 19140   |
| 12     | 20944    | 20943   | 20939   | 20934   | 20926   | 20917   | 20906   | 20894   | 20881   | 20868   |
| 13     | 22689    | 22688   | 22683   | 22676   | 22667   | 22655   | 22641   | 22626   | 22609   | 22593   |
| 14     | 24435    | 24433   | 24427   | 24419   | 24406   | 24392   | 24374   | 24355   | 24335   | 24314   |
| 15     | 26180    | 26178   | 26171   | 26160   | 26145   | 26127   | 26106   | 26083   | 26058   | 26032   |
| 16     | 27925    | 27923   | 27914   | 27901   | 27883   | 27861   | 27836   | 27807   | 27777   | 27746   |
| 17     | 29671    | 29667   | 29658   | 29642   | 29620   | 29594   | 29563   | 29529   | 29493   | 29455   |
| 18     | 31416    | 31412   | 31401   | 31382   | 31357   | 31325   | 31289   | 31248   | 31205   | 31161   |
| 19     | 33161    | 33157   | 33143   | 33121   | 33092   | 33055   | 33012   | 32965   | 32914   | 32862   |
| 20     | 34907    | 34901   | 34886   | 34860   | 34825   | 34783   | 34733   | 34678   | 34619   | 34558   |
| 21     | 36652    | 36646   | 36628   | 36598   | 36558   | 36509   | 36451   | 36387   | 36319   | 36249   |
| 22     | 38397    | 38390   | 38370   | 38336   | 38290   | 38233   | 38167   | 38094   | 38015   | 37934   |
| 23     | 40143    | 40135   | 40111   | 40073   | 40020   | 39955   | 39880   | 39796   | 39707   | 39614   |
| 24     | 41888    | 41879   | 41852   | 41809   | 41749   | 41676   | 41590   | 41496   | 41394   | 41289   |
| 25     | 43633    | 43623   | 43593   | 43544   | 43477   | 43394   | 43298   | 43191   | 43076   | 42958   |
| 26     | 45379    | 45367   | 45333   | 45278   | 45203   | 45110   | 45002   | 44882   | 44753   | 44620   |
| 27     | 47124    | 47111   | 47074   | 47012   | 46928   | 46824   | 46703   | 46569   | 46425   | 46276   |
| 28     | 48869    | 48855   | 48813   | 48745   | 48651   | 48536   | 48402   | 48252   | 48092   | 47926   |
| 29     | 50615    | 50599   | 50553   | 50477   | 50373   | 50245   | 50097   | 49931   | 49753   | 49569   |
| 30     | 52360    | 52343   | 52292   | 52208   | 52094   | 51953   | 51788   | 51605   | 51409   | 51205   |
| 31     | 54105    | 54086   | 54030   | 53938   | 53813   | 53657   | 53476   | 53275   | 53059   | 52834   |
| 32     | 55851    | 55830   | 55768   | 55667   | 55530   | 55360   | 55161   | 54940   | 54703   | 54456   |
| 33     | 57596    | 57573   | 57506   | 57396   | 57245   | 57059   | 56842   | 56600   | 56341   | 56070   |
| 34     | 59341    | 59317   | 59243   | 59123   | 58959   | 58756   | 58520   | 58256   | 57972   | 57677   |
| 35     | 61087    | 61060   | 60980   | 60850   | 60672   | 60451   | 60194   | 59907   | 59598   | 59276   |
| 36     | 62832    | 62803   | 62716   | 62575   | 62382   | 62143   | 61864   | 61552   | 61217   | 60868   |
| 37     | 64577    | 64546   | 64452   | 64300   | 64091   | 63832   | 63530   | 63193   | 62830   | 62451   |
| 38     | 66323    | 66289   | 66188   | 66023   | 65798   | 65519   | 65193   | 64828   | 64436   | 64027   |
| 39     | 68068    | 68031   | 67923   | 67746   | 67503   | 67203   | 66851   | 66459   | 66035   | 65594   |
| 40     | 69813    | 69774   | 69658   | 69467   | 69207   | 68884   | 68506   | 68084   | 67628   | 67153   |
| 41     | 71558    | 71517   | 71392   | 71188   | 70909   | 70562   | 70157   | 69703   | 69214   | 68703   |
| 42     | 73304    | 73259   | 73126   | 72907   | 72609   | 72238   | 71804   | 71318   | 70793   | 70245   |
| 43     | 75049    | 75001   | 74859   | 74626   | 74307   | 73910   | 73446   | 72927   | 72365   | 71778   |
| 44     | 76794    | 76744   | 76592   | 76343   | 76003   | 75580   | 75085   | 74530   | 73931   | 73303   |
| 45     | 0.78540  | 0.78486 | 0.78324 | 0.78059 | 0.77697 | 0.77247 | 0.76720 | 0.76128 | 0.75489 | 0.74819 |

## III.—ELLIPTIC INTEGRALS OF THE SECOND KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 50°      | 55°     | 60°     | 65°     | 70°     | 75°     | 80°     | 85°     | 90°     |
| 1°     | 0.01745  | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 | 0.01745 |
| 2      | 03490    | 03490   | 03490   | 03490   | 03490   | 03490   | 03490   | 03490   | 03490   |
| 3      | 05235    | 05234   | 05234   | 05234   | 05234   | 05234   | 05234   | 05234   | 05234   |
| 4      | 06978    | 06978   | 06977   | 06977   | 06976   | 06976   | 06976   | 06976   | 06976   |
| 5      | 08720    | 08719   | 08718   | 08718   | 08717   | 08716   | 08716   | 08716   | 08716   |
| 6      | 10461    | 10459   | 10458   | 10456   | 10455   | 10454   | 10453   | 10453   | 10453   |
| 7      | 12199    | 12197   | 12195   | 12192   | 12190   | 12189   | 12188   | 12187   | 12187   |
| 8      | 13936    | 13932   | 13929   | 13925   | 13923   | 13920   | 13919   | 13918   | 13917   |
| 9      | 15670    | 15665   | 15660   | 15655   | 15651   | 15648   | 15645   | 15644   | 15643   |
| 10     | 17401    | 17394   | 17387   | 17381   | 17375   | 17371   | 17367   | 17365   | 17365   |
| 11     | 19130    | 19120   | 19110   | 19102   | 19095   | 19089   | 19084   | 19082   | 19081   |
| 12     | 20855    | 20842   | 20830   | 20819   | 20809   | 20801   | 20796   | 20792   | 20791   |
| 13     | 22576    | 22559   | 22544   | 22530   | 22518   | 22508   | 22501   | 22497   | 22495   |
| 14     | 24293    | 24272   | 24253   | 24236   | 24221   | 24209   | 24200   | 24194   | 24192   |
| 15     | 26006    | 25981   | 25957   | 25936   | 25917   | 25902   | 25891   | 25884   | 25882   |
| 16     | 27714    | 27684   | 27655   | 27629   | 27606   | 27588   | 27575   | 27567   | 27564   |
| 17     | 29418    | 29381   | 29347   | 29315   | 29288   | 29267   | 29250   | 29241   | 29237   |
| 18     | 31116    | 31073   | 31032   | 30995   | 30963   | 30937   | 30917   | 30906   | 30902   |
| 19     | 32809    | 32758   | 32710   | 32666   | 32629   | 32598   | 32575   | 32561   | 32557   |
| 20     | 34496    | 34437   | 34381   | 34330   | 34286   | 34250   | 34224   | 34207   | 34202   |
| 21     | 36178    | 36109   | 36044   | 35985   | 35934   | 35892   | 35862   | 35843   | 35837   |
| 22     | 37853    | 37773   | 37699   | 37631   | 37572   | 37525   | 37490   | 37468   | 37461   |
| 23     | 39521    | 39431   | 39345   | 39268   | 39201   | 39146   | 39106   | 39081   | 39073   |
| 24     | 41183    | 41080   | 40983   | 40895   | 40819   | 40757   | 40711   | 40683   | 40674   |
| 25     | 42838    | 42722   | 42612   | 42513   | 42426   | 42356   | 42304   | 42273   | 42262   |
| 26     | 44486    | 44355   | 44232   | 44120   | 44023   | 43944   | 43885   | 43849   | 43837   |
| 27     | 46126    | 45980   | 45842   | 45716   | 45607   | 45518   | 45453   | 45413   | 45399   |
| 28     | 47759    | 47595   | 47441   | 47301   | 47180   | 47081   | 47007   | 46962   | 46947   |
| 29     | 49383    | 49202   | 49031   | 48875   | 48740   | 48629   | 48548   | 48498   | 48481   |
| 30     | 51000    | 50799   | 50609   | 50437   | 50287   | 50165   | 50074   | 50019   | 50000   |
| 31     | 52608    | 52386   | 52177   | 51986   | 51821   | 51686   | 51586   | 51525   | 51504   |
| 32     | 54207    | 53964   | 53733   | 53524   | 53341   | 53193   | 53082   | 53015   | 52992   |
| 33     | 55798    | 55531   | 55278   | 55048   | 54848   | 54684   | 54563   | 54489   | 54464   |
| 34     | 57379    | 57087   | 56811   | 56559   | 56340   | 56161   | 56028   | 55947   | 55919   |
| 35     | 58952    | 58634   | 58332   | 58057   | 57818   | 57622   | 57477   | 57388   | 57358   |
| 36     | 60515    | 60169   | 59841   | 59541   | 59280   | 59067   | 58909   | 58811   | 58779   |
| 37     | 62068    | 61693   | 61337   | 61011   | 60727   | 60495   | 60323   | 60217   | 60182   |
| 38     | 63612    | 63206   | 62820   | 62467   | 62159   | 61907   | 61720   | 61605   | 61566   |
| 39     | 65146    | 64707   | 64290   | 63908   | 63574   | 63302   | 63099   | 62974   | 62932   |
| 40     | 66671    | 66197   | 65746   | 65334   | 64974   | 64679   | 64459   | 64324   | 64279   |
| 41     | 68185    | 67675   | 67189   | 66745   | 66356   | 66038   | 65801   | 65655   | 65606   |
| 42     | 69688    | 69140   | 68619   | 68140   | 67722   | 67379   | 67124   | 66966   | 66913   |
| 43     | 71182    | 70594   | 70034   | 69520   | 69070   | 68701   | 68426   | 68257   | 68200   |
| 44     | 72665    | 72036   | 71435   | 70884   | 70401   | 70005   | 69710   | 69527   | 69466   |
| 45     | 0.74137  | 0.73465 | 0.72822 | 0.72232 | 0.71715 | 0.71289 | 0.70972 | 0.70777 | 0.70711 |



## III.—ELLIPTIC INTEGRALS OF THE SECOND KIND

| $\phi$ | $\theta$  |           |            |            |            |            |            |            |            |            |
|--------|-----------|-----------|------------|------------|------------|------------|------------|------------|------------|------------|
|        | $0^\circ$ | $5^\circ$ | $10^\circ$ | $15^\circ$ | $20^\circ$ | $25^\circ$ | $30^\circ$ | $35^\circ$ | $40^\circ$ | $45^\circ$ |
| 46°    | 0.80285   | 0.80228   | 0.80056    | 0.79775    | 0.79390    | 0.78911    | 0.78350    | 0.77721    | 0.77040    | 0.76326    |
| 47     | 82030     | 81969     | 81787      | 81489      | 81081      | 80573      | 79977      | 79308      | 78584      | 77824      |
| 48     | 83776     | 83711     | 83518      | 83202      | 82770      | 82231      | 81599      | 80890      | 80121      | 79313      |
| 49     | 85521     | 85453     | 85249      | 84914      | 84457      | 83887      | 83217      | 82466      | 81651      | 80794      |
| 50     | 87266     | 87194     | 86979      | 86626      | 86142      | 85539      | 84832      | 84036      | 83173      | 82265      |
| 51     | 89012     | 88936     | 88709      | 88336      | 87826      | 87189      | 86442      | 85601      | 84689      | 83728      |
| 52     | 90757     | 90677     | 90438      | 90045      | 89507      | 88836      | 88048      | 87161      | 86197      | 85182      |
| 53     | 92502     | 92418     | 92166      | 91753      | 91187      | 90481      | 89650      | 88715      | 87698      | 86627      |
| 54     | 94248     | 94159     | 93895      | 93450      | 92865      | 92122      | 91248      | 90264      | 89193      | 88063      |
| 55     | 95993     | 95900     | 95622      | 95166      | 94541      | 93761      | 92843      | 91807      | 90680      | 89490      |
| 56     | 97738     | 97641     | 97350      | 96872      | 96216      | 95397      | 94433      | 93345      | 92160      | 90908      |
| 57     | 0.99484   | 0.99381   | 0.99077    | 0.98576    | 97889      | 97030      | 96019      | 94878      | 93634      | 92318      |
| 58     | 1.01229   | 1.01122   | 1.00803    | 1.00279    | 0.99560    | 0.98661    | 97602      | 96405      | 95100      | 93719      |
| 59     | 02974     | 02863     | 02529      | 01981      | 1.01229    | 1.00289    | 0.99180    | 97928      | 96560      | 95111      |
| 60     | 04720     | 04603     | 04255      | 03683      | 02897      | 01915      | 1.00756    | 0.99445    | 98013      | 96495      |
| 61     | 06465     | 06343     | 05980      | 05383      | 04563      | 03538      | 02327      | 1.00957    | 0.99460    | 97871      |
| 62     | 08210     | 08084     | 07705      | 07083      | 06228      | 05158      | 03895      | 02405      | 1.00900    | 0.99238    |
| 63     | 09956     | 09824     | 09430      | 08781      | 07891      | 06776      | 05459      | 03967      | 02334      | 1.00598    |
| 64     | 11701     | 11564     | 11154      | 10479      | 09553      | 08392      | 07020      | 05465      | 03762      | 01949      |
| 65     | 13446     | 13304     | 12878      | 12176      | 11213      | 10005      | 08577      | 06958      | 05183      | 03293      |
| 66     | 15192     | 15043     | 14601      | 13873      | 12871      | 11616      | 10132      | 08447      | 06599      | 04629      |
| 67     | 16937     | 16783     | 16324      | 15568      | 14529      | 13225      | 11683      | 09932      | 08009      | 05957      |
| 68     | 18682     | 18523     | 18047      | 17263      | 16185      | 14832      | 13231      | 11412      | 09413      | 07279      |
| 69     | 20428     | 20262     | 19769      | 18957      | 17839      | 16437      | 14776      | 12888      | 10812      | 08593      |
| 70     | 22173     | 22002     | 21491      | 20650      | 19493      | 18040      | 16318      | 14360      | 12205      | 09901      |
| 71     | 23918     | 23741     | 23213      | 22343      | 21145      | 19640      | 17857      | 15828      | 13594      | 11202      |
| 72     | 25664     | 25481     | 24935      | 24034      | 22796      | 21239      | 19394      | 17293      | 14977      | 12497      |
| 73     | 27409     | 27220     | 26656      | 25726      | 24446      | 22837      | 20928      | 18754      | 16356      | 13786      |
| 74     | 29154     | 28959     | 28377      | 27417      | 26094      | 24432      | 22459      | 20211      | 17731      | 15068      |
| 75     | 30900     | 30698     | 30097      | 29107      | 27742      | 26026      | 23989      | 21666      | 19101      | 16346      |
| 76     | 32645     | 32437     | 31818      | 30796      | 29389      | 27619      | 25516      | 23117      | 20467      | 17618      |
| 77     | 34390     | 34176     | 33538      | 32486      | 31035      | 29210      | 27041      | 24566      | 21830      | 18885      |
| 78     | 36136     | 35915     | 35258      | 34174      | 32680      | 30800      | 28565      | 26012      | 23189      | 20148      |
| 79     | 37881     | 37654     | 36978      | 35862      | 34325      | 32389      | 30086      | 27456      | 24544      | 21407      |
| 80     | 39626     | 39393     | 38698      | 37550      | 35968      | 33976      | 31606      | 28897      | 25897      | 22661      |
| 81     | 41372     | 41132     | 40417      | 39238      | 37611      | 35563      | 33124      | 30336      | 27246      | 23912      |
| 82     | 43117     | 42871     | 42137      | 40925      | 39254      | 37148      | 34641      | 31773      | 28594      | 25159      |
| 83     | 44862     | 44610     | 43856      | 42612      | 40896      | 38733      | 36157      | 33209      | 29939      | 26404      |
| 84     | 46608     | 46349     | 45575      | 44299      | 42537      | 40317      | 37672      | 34643      | 31282      | 27646      |
| 85     | 48353     | 48087     | 47294      | 45985      | 44178      | 41900      | 39186      | 36076      | 32623      | 28886      |
| 86     | 50098     | 49826     | 49013      | 47671      | 45819      | 43483      | 40699      | 37508      | 33963      | 30124      |
| 87     | 51844     | 51565     | 50732      | 49357      | 47459      | 45066      | 42211      | 38939      | 35302      | 31360      |
| 88     | 53589     | 53304     | 52451      | 51043      | 49100      | 46648      | 43723      | 40369      | 36640      | 32596      |
| 89     | 55334     | 55042     | 54170      | 52729      | 50740      | 48230      | 45235      | 41799      | 37977      | 33830      |
| 90     | 1.57080   | 1.56781   | 1.55889    | 1.54415    | 1.52380    | 1.49811    | 1.46746    | 1.43229    | 1.39314    | 1.35064    |

## III.—ELLIPTIC INTEGRALS OF THE SECOND KIND

| $\phi$ | $\theta$ |         |         |         |         |         |         |         |         |
|--------|----------|---------|---------|---------|---------|---------|---------|---------|---------|
|        | 50°      | 55°     | 60°     | 65°     | 70°     | 75°     | 80°     | 85°     | 90°     |
| 46°    | 0.75599  | 0.74881 | 0.74195 | 0.73564 | 0.73010 | 0.72554 | 0.72215 | 0.72005 | 0.71934 |
| 47     | 77050    | 76285   | 75553   | 74879   | 74287   | 73800   | 73436   | 73211   | 73135   |
| 48     | 78490    | 77676   | 76896   | 76177   | 75546   | 75025   | 74636   | 74396   | 74314   |
| 49     | 79920    | 79054   | 78225   | 77459   | 76786   | 76230   | 75815   | 75558   | 75471   |
| 50     | 81338    | 80419   | 79538   | 78724   | 78007   | 77414   | 76971   | 76697   | 76604   |
| 51     | 82746    | 81772   | 80836   | 79971   | 79208   | 78578   | 78106   | 77814   | 77715   |
| 52     | 84143    | 83111   | 82120   | 81202   | 80391   | 79720   | 79218   | 78907   | 78801   |
| 53     | 85529    | 84438   | 83388   | 82415   | 81554   | 80842   | 80307   | 79976   | 79864   |
| 54     | 86904    | 85752   | 84641   | 83610   | 82698   | 81941   | 81374   | 81021   | 80902   |
| 55     | 88269    | 87052   | 85879   | 84788   | 83822   | 83020   | 82417   | 82042   | 81915   |
| 56     | 89622    | 88340   | 87101   | 85949   | 84926   | 84076   | 83436   | 83039   | 82904   |
| 57     | 90965    | 89614   | 88308   | 87092   | 86011   | 85110   | 84432   | 84010   | 83867   |
| 58     | 92297    | 90876   | 89500   | 88217   | 87075   | 86122   | 85404   | 84957   | 84805   |
| 59     | 93619    | 92125   | 90677   | 89325   | 88119   | 87112   | 86352   | 85878   | 85717   |
| 60     | 94930    | 93362   | 91839   | 90415   | 89144   | 88080   | 87276   | 86773   | 86603   |
| 61     | 96231    | 94586   | 92986   | 91488   | 90148   | 89025   | 88175   | 87643   | 87462   |
| 62     | 97521    | 95797   | 94118   | 92543   | 91132   | 89948   | 89049   | 88486   | 88295   |
| 63     | 0.98802  | 0.96996 | 0.95236 | 0.93581 | 0.92096 | 0.90848 | 0.89898 | 0.89303 | 0.89101 |
| 64     | I.00072  | 0.98183 | 0.96339 | 0.94602 | 0.93041 | 0.91725 | 0.90723 | 0.90094 | 0.89879 |
| 65     | 01333    | 0.99358 | 0.97427 | 0.95606 | 0.93965 | 0.92580 | 0.91523 | 0.90858 | 0.90631 |
| 66     | 02585    | I.00522 | 0.98502 | 0.96593 | 0.94870 | 0.93412 | 0.92297 | 0.91595 | 0.91355 |
| 67     | 03827    | 01674   | 0.99562 | 0.97564 | 0.95756 | 0.94222 | 0.93047 | 0.92305 | 0.92050 |
| 68     | 05060    | 02815   | I.00609 | 0.98518 | 0.96622 | 0.95010 | 0.93771 | 0.92987 | 0.92718 |
| 69     | 06284    | 03945   | 01643   | 0.99456 | 0.97469 | 0.95775 | 0.94470 | 0.93642 | 0.93358 |
| 70     | 07500    | 05064   | 02664   | I.00379 | 0.98298 | 0.96519 | 0.95144 | 0.94270 | 0.93969 |
| 71     | 08707    | 06173   | 03672   | 01286   | 0.99108 | 0.97240 | 0.95793 | 0.94870 | 0.94552 |
| 72     | 09907    | 07272   | 04668   | 02178   | 0.99900 | 0.97940 | 0.96417 | 0.95442 | 0.95106 |
| 73     | 11098    | 08362   | 05651   | 03056   | I.00674 | 0.98619 | 0.97016 | 0.95987 | 0.95630 |
| 74     | 12283    | 09442   | 06624   | 03919   | 01431   | 0.99278 | 0.97590 | 0.96503 | 0.96126 |
| 75     | 13460    | 10513   | 07586   | 04769   | 02172   | 0.99916 | 0.98141 | 0.96992 | 0.96593 |
| 76     | 14631    | 11577   | 08537   | 05607   | 02896   | I.00534 | 0.98667 | 0.97453 | 0.97030 |
| 77     | 15795    | 12632   | 09478   | 06432   | 03605   | 01133   | 0.99170 | 0.97887 | 0.97437 |
| 78     | 16954    | 13680   | 10410   | 07245   | 04300   | 01714   | 0.99650 | 0.98293 | 0.97815 |
| 79     | 18107    | 14721   | 11333   | 08047   | 04981   | 02277   | I.00107 | 0.98671 | 0.98163 |
| 80     | 19255    | 15755   | 12249   | 08839   | 05648   | 02823   | 00543   | 0.99023 | 0.98481 |
| 81     | 20399    | 16784   | 13156   | 09621   | 06304   | 03354   | 00958   | 0.99348 | 0.98769 |
| 82     | 21538    | 17807   | 14057   | 10395   | 06948   | 03870   | 01354   | 0.99646 | 0.99027 |
| 83     | 22673    | 18825   | 14952   | 11161   | 07582   | 04372   | 01731   | 0.99920 | 0.99255 |
| 84     | 23805    | 19839   | 15841   | 11920   | 08207   | 04863   | 02091   | I.00168 | 0.99452 |
| 85     | 24934    | 20850   | 16726   | 12673   | 08825   | 05343   | 02436   | 00394   | 0.99619 |
| 86     | 26061    | 21857   | 17606   | 13421   | 09435   | 05813   | 02768   | 00598   | 0.99256 |
| 87     | 27186    | 22862   | 18484   | 14165   | 10041   | 06277   | 03089   | 00784   | 0.99863 |
| 88     | 28310    | 23865   | 19359   | 14906   | 10642   | 06735   | 03401   | 00954   | 0.99939 |
| 89     | 29432    | 24867   | 20233   | 15645   | 11241   | 07188   | 03708   | 01113   | 0.99985 |
| 90     | I.30554  | I.25868 | I.21106 | I.16383 | I.11838 | I.07641 | I.04011 | I.01266 | I.00000 |





## INDEX

- Abel, Niels, 6, 7, 24, 29, 31  
 Addition-Theorem, 7  
 d'Alembert, Jean, 6  
 Amplitude, 24  
 Andoyer, H., 6  
 Appell and Lacour, *Fonctions Elliptiques*, 7, 14, 21  
 Arithmetico-Geometrical Mean, 78  
  
 Bernoulli, Jacob, 6, 7  
 Burnside, W. S., 54  
 Byerly, *Integral Calculus*, 48, 63, 90, 91  
  
 Calculation of elliptic functions, 33  
 Calculation of elliptic function, when modulus and argument are given, 87  
 Calculation of argument, when  $k$  and the function are given, 87  
 Cayley, Arthur, 6, 14, 16, 28, 65, 68, 73, 76  
 Clifford, W. K., 41  
 Co-amplitude, 35  
 Computation, see Numerical computation  
 Cosine-amplitude =  $cn$ , 24  
  
 Delta-amplitude =  $dn$ , 24  
 Durège, *Elliptische Funktionen*, 59  
  
 $E(k, \phi)$ , see Integral  $E(k, \phi)$   
 $E_1$ , the complete integral, 16, 17, 81  
     Tables for  $E_1$ , 93  
*Elastic curves*, 6, 89  
 Enneper, *Elliptische Funktionen*, 8, 10, 12  
 Euler, Leonhard, 7  
  
 $F(k, h)$ , see Integral  $F(k, h)$   
 $F_1$ , the complete integral, 16, 17, 81  
     Tables for  $F_1$ , 93  
 Fagnano, G. C., 6, 7  
 Fricke, *Elliptische Funktionen*, 8  
 Functions, elliptic, 22-39  
  
 Gauss, C. F., 78  
 Graphs of the integrals of the first and second kinds, 16, 17, 18, 19, 20  
 Graphs of the functions  $sn$ ,  $cn$  and  $dn$ , 36, 37, 38  
 Greenhill, A. G., 6, 52  
 Gudermann, M., 24, 28  
 Gudermannian, The, 28, 29  
  
 Hancock, Harris, 6, 10, 15  
 Hermite, Chas., 6  
 Humbert, G., 92  
 Hyperbola, rectification of, 6  
  
 Infinities, 35, 36, 37, 38  
 Integrals, the  $F(k, \phi)$  and  $E(k, \phi)$ , their graphs, 15-20  
 Integrals, tables for  $F(k, \phi)$  and  $E(k, \phi)$ , 94-101  
 Integral, the  $\Pi(n, k, \phi)$ , 15  
 Integrals, the general elliptic, 9, 14  
 Integrals of certain elliptic functions, 58-63  
 Integrals of first kind reduced to normal form, 41-63  
 Integrals, normal, 10, 15  
 Integrals, of the first, second and third kinds, 9, 60

- Jacobi, C. G. J., 5, 6, 7, 24, 25, 26, 29, 31, 34, 65  
 Jacobi's imaginary transformation, 25, 34  
 $K, K'$ , tables for, 93  
 $K, K'$ , the transcendents, 26  
 Lacour, E., see Paul Appell  
 Lagrange, Joseph L., 7, 68  
 Landen, John, 68  
 Landen's transformations, 65-73  
 Latitude, 89  
 Legendre, Adrien Marie, 5, 6, 7, 9, 10, 11, 12, 15, 21, 22, 23, 24, 61, 68, 71, 81, 82, 84, 92  
 Legendre's transformation, 10-13  
 Lemniscate, 7, 88  
 Levy, *Fonctions elliptiques*, 21, 92  
 Maclaurin, Colin, 6  
 Modulus, 12  
 Molk, Jules, see Jules Tannery  
 Numerical computation of integrals of 1st and 2d kinds, 69-87  
 Pendulum, simple, 90, 91  
 Periods, 27, 31, 35-38  
 Periodic functions, property of, 7, 31  
 Period-parallelogram, 32  
 Period-parallelogram of  $sn u$ , p. 35, of  $cn u$ , p. 37, of  $dn u$ , p. 38.  
 Period-strip, 31  
 Quadrant of lemniscate, 88; of ellipse, 89  
 Rectification of ellipse, 88; of lemniscate, 88  
 Reduction formulas, 59-61  
 Richelot, F. S., 10  
 Serret, J. A., 7  
 Simpson's Fluxions, 91  
 Sine-amplitude =  $sn$ , 24  
 Tables of complete integrals of 1st and 2d kinds, 93  
 Tables of elliptic integrals of 1st kind, 94-97  
 Tables of elliptic integrals of 2d kind, 98-101  
 Tables of useful integrals, 61-63  
 Tannery and Molk, *Fonctions Elliptiques*, 14  
 Volume common to two cylinders, 89  
 Wallis's formula, 26, 60  
 Weierstrass, Karl, 24, 28  
 Zeroes, 35-38

128  
 110  
 865







**PLEASE DO NOT REMOVE  
CARDS OR SLIPS FROM THIS POCKE**

---

**UNIVERSITY OF TORONTO LIBRAR**

---

